

МДМА

3,4-метилendioксиметамфетамин (МДМА), широко известный как **экстази** в форме таблеток и **молли** в форме кристаллов,^{[20][21]} является **энтактогеном** со **стимулирующими** и слабыми **психоделическими** свойствами.^{[18][22][23]}

MDMA was first synthesized in 1912 by Merck chemist Anton Köllisch.^[24] It was used to enhance psychotherapy beginning in the 1970s and became popular as a street drug in the 1980s.^{[25][26]} MDMA is commonly associated with **dance parties**, **raves**, and **electronic dance music**.^[27] Tablets sold as ecstasy may be **mixed** with other substances such as **ephedrine**, **amphetamine**, and **methamphetamine**.^[25] In 2016, about 21 million people between the ages of 15 and 64 used ecstasy (0.3% of the world population).^[28] In the United States, as of 2017, about 7% of people have used MDMA at some point in their lives and 0.9% have used it in the last year.^[29] The lethal risk from one dose of MDMA is estimated to be from 1 death in 20,000 instances to 1 death in 50,000 instances.^[30]

The purported pharmacological effects that may be **prosocial** include altered sensations, increased energy, empathy, and pleasure.^{[23][25]} When taken by mouth, effects begin in 30 to 45 minutes and last three to six hours.^{[13][26]} Short-term adverse effects include **grinding of the teeth**, **blurred vision**, **sweating**, and a **rapid heartbeat**,^[25] and extended use can also lead to addiction, **memory problems**, **paranoia**, and **difficulty sleeping**. Deaths have been reported due to increased body temperature and dehydration. MDMA acts primarily by increasing the release of the **neurotransmitters serotonin**, **dopamine**, and **norepinephrine** in parts of the brain.^{[25][26]} It belongs to the **substituted amphetamine classes of drugs**.^{[10][31]} MDMA is structurally similar to **mescaline** (a psychedelic), methamphetamine (a stimulant), as well as **endogenous monoamine neurotransmitters** such as serotonin, norepinephrine, and dopamine.^[32]

MDMA has limited approved medical uses in a small number of countries,^[33] but is illegal in most jurisdictions.^[34] **MDMA-assisted psychotherapy** may substantially improve PTSD symptoms, response, and remission rates compared to psychotherapy alone, but the evidence is low to very low certainty and safety data are limited, with some increased transient adverse events.^[35] In the United States, the **Food and Drug Administration** (FDA) has given MDMA **breakthrough therapy** status (though there are no current clinical indications in the US).^[36] Canada has allowed limited distribution of MDMA upon application to and approval by **Health Canada**.^[37] In Australia, it may be prescribed in the treatment of PTSD by specifically authorised psychiatrists.^[38]

Uses

Recreational

MDMA is often considered the drug of choice within the [rave](#) culture and is also used at clubs, festivals, and [house parties](#).^[15] In the rave environment, the sensory effects of music and lighting are often highly [synergistic](#) with the drug. The psychedelic [amphetamine](#) quality of MDMA offers multiple appealing aspects to users in the rave setting. Some users enjoy the feeling of mass communion from the inhibition-reducing effects of the drug, while others use it as party fuel because of the drug's stimulatory effects.^[39] MDMA is used less often than other stimulants, typically less than once per week.^[40]

MDMA is sometimes taken in conjunction with other psychoactive drugs such as [LSD](#),^[41] [psilocybin mushrooms](#), [2C-B](#), and [ketamine](#). The combination with LSD is called "candy-flipping".^[41] The combination with 2C-B is called "nexus flipping". For this combination, most people take the MDMA first, wait until the peak is over, and then take the 2C-B.^[42]

MDMA is often co-administered with [alcohol](#), [methamphetamine](#), and prescription drugs such as [SSRIs](#) with which MDMA has several drug-drug interactions.^{[43][44][45]} Three life-threatening reports of MDMA co-administration with [ritonavir](#) have been reported;^[46] with ritonavir having severe and dangerous drug-drug interactions with a wide range of both psychoactive, anti-psychotic, and non-psychoactive drugs.^[47]

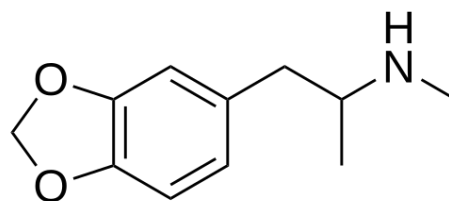
Medical

МДМА

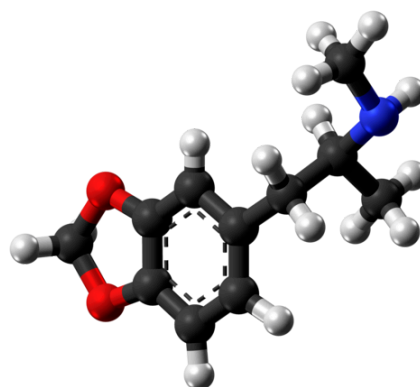
МНН:

Мидомафетамин

[1]



Скелетная структура рацемического МДМА



Шарообразная модель МДМА

Клинические данные

Произношение

метилендиокси-
метамфетамин:
[/mɛθɪliːndaɪˈbksi/](#)
[/mɛθæmˈfɛtəmiːn/](#)

Другие названия

[3,4-МДМА](#); экстази (Е, Х, ХТС);
мидомафетамин;
«Молли»;
«Мэнди»;^{[2][3]}
«Фингерс»/«Пингас»^[4]

AHFS/Drugs.com

МДМА (<https://www.drugs.com/MDMA.html>)

Зависимость ответственность

Физическое:
нетипично^[5]
Психологический:
умеренный^[6]

Before it was widely banned, it saw limited use in psychotherapy.^{[8][10][48]} In 2017 the [United States Food and Drug Administration](#) (FDA) granted breakthrough therapy designation for MDMA-assisted psychotherapy for [post-traumatic stress disorder](#) (PTSD).^{[49][50]}

MDMA has limited approved medical uses in a small number of countries,^[51] but is illegal in most jurisdictions.^[52] Canada has allowed limited distribution of MDMA upon application to and approval by [Health Canada](#).^[53] In Australia, it may be prescribed in the treatment of PTSD by specifically authorised psychiatrists.^[54]

Fourteen systematic reviews of up to 353 participants suggest MDMA-assisted psychotherapy may substantially improve PTSD symptoms, response, and remission rates compared to psychotherapy alone, but the evidence is low to very low certainty and safety data are limited, with some increased transient adverse events.^[35]

Some researchers have proposed that psychedelics in general may act as [active](#) "super [placebos](#)" used for therapeutic purposes.^{[55][56]}

Others

Small doses of MDMA are used by some religious practitioners as an [entheogen](#) to enhance prayer or meditation.^[57] MDMA has been used as an adjunct to [New Age](#) spiritual practices.^[58]

Зависимость ответственность	Низкий– средний ^{[7][8][9]}
Способы введения	Обычно: в рот ^[10] Нечасто встречается: инсуффляция , ^[10] ингаляция , ^[10] инъекция , ^{[10][11]} ректальное введение
Класс лекарств	Энтактоген ; Стимулятор ; Психоделик ; Вещество , высвобождающее серотонин , норадреналин и дофамин ; 5-НТ₂ рецептор агонист
Код УВД	Нет
Правовой статус	
Правовой статус	AU : S8 (посттравматическое стрессовое расстройство) S9 (все остальные значения) Класс F2BR : (Запрещенные психотропные вещества) ^[12] CA : Список I DE : Приложение I (только для разрешенного научного использования) NZ : Класс B UK : Класс A

Forms



1 gram of pure MDMA hydrochloride crystals

MDMA has become widely known as ecstasy (shortened "E", "X", or "XTC"), usually referring to its tablet form, although this term may also include the presence of possible [adulterants](#) or diluents. The UK term "mandy" and the US term "molly" colloquially refer to MDMA in a crystalline powder form that is thought to be free of adulterants.^{[2][3][59]} MDMA is also sold in the form of the hydrochloride salt, either as loose crystals or in [gelcaps](#).^{[60][61]} MDMA tablets can sometimes be found in a shaped form that may depict characters from [popular culture](#). These are sometimes collectively referred to as "fun tablets".^{[62][63]}

Partly due to the global supply shortage of [sassafras oil](#)—a problem largely assuaged by use of improved or alternative modern methods of synthesis—the purity of substances sold as molly have been found to vary widely. Some of these substances contain [methyldone](#), [ethylone](#), [MDPV](#), [mephedrone](#), or any other of the group of compounds commonly known as [bath salts](#), in addition to, or in place of, MDMA.^{[3][59][60][61]} Powdered MDMA ranges from pure MDMA to crushed tablets with 30–40% purity.^[10] MDMA tablets typically have low purity due to bulking agents that are added to dilute the drug and increase

US: Список I ООН: Психотропные вещества Списка I ЕС: Полностью запрещены SE: Список I	
Фармакокинетические данные	
Биодоступность	Устно: Неизвестно ^[13]
Связывание с белками	Неизвестно ^[14]
Метаболизм	Печень, CYP450 принимают активное участие в процессе, в том числе CYP2D6
Метаболиты	MDA, HMMA, HMA, HHMA, HHA, THMA, THA, MDP2P, MDOH ^[15] 3-ММОТИ ^[16]
Начало действия	Оральный: 30–45 мин ^[13]
Период выведения период полувыведения	- МДМА: 8,7 (диапазон 4,6–16) часов^{[17][18]} (S)-МДМА: 5,1 (диапазон 3,5–7,4) часов^[17] (R)-МДМА: 11 (диапазон 5,1–24) часов^[17]
Продолжительность действия	3–6 часов ^{[19][8][13]}
Выведение	Почка
Идентификаторы	
Название IUPAC	[показать]

profits (notably lactose) and binding agents.^[10] Tablets sold as ecstasy sometimes contain [3,4-methylenedioxyamphetamine](#) (MDA), [3,4-methylenedioxyethylamphetamine](#) (MDEA), other amphetamine derivatives, caffeine, opiates, or painkillers.^[8] Some tablets contain little or no MDMA.^{[8][10][64]} The proportion of seized ecstasy tablets with MDMA-like impurities has varied annually and by country.^[10] The average content of MDMA in a preparation is 70 to 120 mg with the purity having increased since the 1990s.^[8]

MDMA is usually consumed by mouth. It is also sometimes [snorted](#).^[25]

Номер CAS	42542-10-9 (https://commonchemistry.org/detail?cas_rn=42542-10-9) ✓ ^[TOXNET] 64057-70-1 (https://commonchemistry.org/detail?cas_rn=64057-70-1&title=) (HCl)
PubChem CID	1615 (https://pubchem.ncbi.nlm.nih.gov/compound/1615)
IUPHAR/BPS	4574 (https://www.guidetopharmacology.org/GRAC/LigandDisplayForward?ligandId=4574)
АптеЧный банк	DB01454 (https://www.drugbank.ca/drugs/DB01454) ✓
Химспидер	1556 (https://www.chemspider.com/Chemical-Structure.1556.html) ✓
UNII	KE1SEN21RM (https://precision.fda.gov/uniisearch/srs/unii/KE1SEN21RM)
KEGG	D11172 (https://www.kegg.jp/entry/D11172) ✓
ChEBI	CHEBI:1391 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=CHEBI:1391) ✓
ChEMBL	ChEMBL43048 (https://www.ebi.ac.uk/chembl/explore/compo

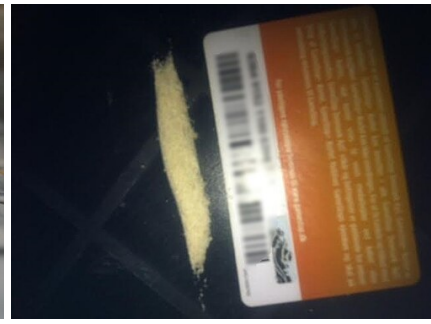
	und/ChEMBL43048) ✓
PDB ligand	B41 (PDBe (https://www.ebi.ac.uk/pdbe-srv/PDBeXplore/ligand/?ligand=B41) , RCSB PDB (https://www.rcsb.org/ligand/B41))
CompTox Dashboard (EPA)	DTXSID90860791 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID90860791)
Chemical and physical data	
Formula	C ₁₁ H ₁₅ NO ₂
Molar mass	193.246 g·mol ⁻¹
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=CC%28NC%29CC1%3DCC%3DC%28OCO2%29C2%3DC1)
Chirality	Racemic mixture
Density	1.1 g/cm ³
Boiling point	105 °C (221 °F) at 0.4 mmHg (experimental)
SMILES	[show]
InChI	[show]
(verify) (https://en.wikipedia.org/w/index.php?title=Special:ComparePages&rev1=6321640&page2=MDMA)	



Ecstasy tablets which may contain MDMA



A salt of MDMA (typically white) with impurities, resulting in a tan discoloration



Crushed MDMA (1 gram) crystals

Effects

In general, MDMA users report feeling the onset of subjective effects within 30 to 60 minutes of oral consumption and reaching peak effect at 75 to 120 minutes, which then plateaus for about 3.5 hours.^[65] The desired short-term psychoactive effects of MDMA have been reported to include:

- **Euphoria** – a sense of general **well-being** and happiness^{[23][66]}
- Increased self-confidence, sociability, and perception of facilitated communication^{[8][23][66]}
- **Entactogenic** effects—increased **empathy** or feelings of closeness with others^{[23][66]} and oneself^[8]
- **Dilated pupils**^[8]
- Relaxation and reduced anxiety^[8]
- Increased emotionality^[8]
- A sense of inner peace^[66]
- Mild hallucination^[66]
- Enhanced sensation, perception, or sexuality^{[8][23][66]}
- Altered sense of time^[26]

The experience elicited by MDMA depends on the dose, setting, and user.^[8] The variability of the induced altered state is lower compared to other psychedelics. For example, MDMA used at parties is associated with high motor activity, reduced sense of identity, and poor awareness of surroundings. Use of MDMA individually or in small groups in a quiet environment and when concentrating, is associated with increased lucidity, concentration, sensitivity to aesthetic aspects of the environment, enhanced awareness of emotions, and improved capability of communication.^{[15][67]} In psychotherapeutic settings, MDMA effects have been characterized by infantile ideas, mood lability, and memories and moods connected with childhood experiences.^{[67][68]}

MDMA has been described as an "[empathogenic](#)" drug because of its empathy-producing effects.^{[69][70]} Results of several studies show the effects of increased empathy with others.^[69] When testing MDMA for medium and high doses, it showed increased hedonic and arousal continuum.^{[71][72]} The effect of MDMA increasing sociability is consistent, while its effects on empathy have been more mixed.^[73]

Side effects

Short-term

Acute adverse effects are usually the result of high or multiple doses, although single dose toxicity can occur in susceptible individuals.^[23] The most serious short-term physical health risks of MDMA are [hyperthermia](#) and [dehydration](#).^{[66][74]} Cases of life-threatening or fatal [hyponatremia](#) (excessively low sodium concentration in the blood) have developed in MDMA users attempting to prevent dehydration by [consuming excessive amounts of water](#) without replenishing [electrolytes](#).^{[66][74][30]}

The immediate adverse effects of MDMA use can include:

- [Bruxism](#) (grinding and clenching of the teeth)^{[8][15][23]}
- [Dehydration](#)^{[15][66][74]}
- [Diarrhea](#)^[66]
- [Erectile dysfunction](#)^{[8][75]}
- [Hyperthermia](#)^{[8][15][74]}
- Increased wakefulness or [insomnia](#)^{[8][66]}
- Increased perspiration and sweating^{[66][74]}
- Increased [heart rate](#) and [blood pressure](#)^{[8][15][74]}
- Increased [psychomotor](#) activity^[8]
- Loss of [appetite](#)^{[8][64]}
- Nausea and vomiting^[23]
- Visual and auditory hallucinations (rarely)^[8]

Other adverse effects that may occur or persist for up to a week following cessation of moderate MDMA use include:^{[64][23]}

Physiological

- Insomnia^[64]
- Loss of appetite^[64]

- Tiredness or lethargy^{[76][77]}

- Trismus (lockjaw)^[23]

Psychological

- Anhedonia^[64]
- Anxiety or paranoia^[64]
- Depression^{[64][23]}
- Impulsiveness^[64]
- Irritability^[64]
- Memory impairment^[23]
- Restlessness^[64]

Long-term

As of 2015, the long-term effects of MDMA on human brain structure and function have not been fully determined.^[78] However, there is consistent evidence of structural and functional deficits in MDMA users with high lifetime exposure.^[78] These structural or functional changes appear to be dose dependent and may be less prominent in MDMA users with only a moderate (typically <50 doses used and <100 tablets consumed) lifetime exposure. Nonetheless, moderate MDMA use may still be neurotoxic and what constitutes moderate use is not clearly established.^[79]

Furthermore, it is not clear yet whether "typical" recreational users of MDMA (1 to 2 pills of 75 to 125 mg MDMA or analogue every 1 to 4 weeks) will develop neurotoxic brain lesions.^[80] Long-term exposure to MDMA in humans has been shown to produce marked neurodegeneration in striatal, hippocampal, prefrontal, and occipital serotonergic axon terminals.^{[78][81]} Neurotoxic damage to serotonergic axon terminals has been shown to persist for more than two years.^[81] Elevations in brain temperature from MDMA use are positively correlated with MDMA-induced neurotoxicity.^{[15][78][79]} However, most studies on MDMA and serotonergic neurotoxicity in humans focus more on heavy users who consume as much as seven times or more the amount that most users report taking. The evidence for the presence of serotonergic neurotoxicity in casual users who take lower doses less frequently is not conclusive.^[82]

However, adverse neuroplastic changes to brain microvasculature and white matter have been observed to occur in humans using low doses of MDMA.^{[15][78]} Reduced gray matter density in certain brain structures has also been noted in human MDMA users.^{[15][78]} Global reductions in gray matter volume, thinning of the parietal and orbitofrontal cortices, and decreased hippocampal activity have been observed in long term users.^[8] The effects established so far for recreational use of ecstasy lie in the range of moderate to severe effects for serotonin transporter reduction.^[83]

Impairments in multiple aspects of cognition, including attention, learning, memory, visual processing, and sleep, have been found in regular MDMA users.^{[8][23][84][78]} The magnitude of these impairments is correlated with lifetime MDMA usage^{[23][84][78]} and are partially reversible with abstinence.^[8] Several forms of memory are impaired by chronic ecstasy use,^{[23][84]} however,

the effects for memory impairments in ecstasy users are generally small overall.^{[85][86]} MDMA use is also associated with increased impulsivity and depression.^[8]

Serotonin depletion following MDMA use can cause depression in subsequent days. In some cases, depressive symptoms persist for longer periods.^[8] Some studies indicate repeated recreational use of ecstasy is associated with depression and anxiety, even after quitting the drug.^[87] Depression is one of the main reasons for cessation of use.^[8]

At high doses, MDMA induces a [neuroimmune response](#) that, through several mechanisms, increases the permeability of the [blood–brain barrier](#), thereby making the brain more susceptible to environmental toxins and [pathogens](#).^{[88][89]} In addition, MDMA has [immunosuppressive](#) effects in the [peripheral nervous system](#) and pro-inflammatory effects in the [central nervous system](#).^[90]

MDMA may increase the risk of [cardiac valvulopathy](#) in heavy or long-term users due to activation of serotonin [5-HT_{2B} receptors](#).^{[91][92]} MDMA induces cardiac [epigenetic changes](#) in [DNA methylation](#), particularly hypermethylation changes.^[93]

Reinforcement disorders

Approximately 60% of MDMA users experience [withdrawal](#) symptoms when they stop taking MDMA.^[64] Some of these symptoms include fatigue, loss of appetite, depression, and trouble concentrating.^[64] [Tolerance](#) to some of the desired and adverse effects of MDMA is expected to occur with consistent MDMA use.^[64] A 2007 [delphic analysis](#) of a panel of experts in pharmacology, psychiatry, law, policing and others estimated MDMA to have a psychological dependence and physical dependence potential roughly three-fourths to four-fifths that of cannabis.^[94]

MDMA has been shown to induce [ΔFosB](#) in the [nucleus accumbens](#).^[95] Because MDMA releases dopamine in the [striatum](#), the mechanisms by which it induces ΔFosB in the nucleus accumbens are analogous to other dopaminergic psychostimulants.^{[95][96]} Therefore, chronic use of MDMA at high doses can result in [altered brain structure](#) and [drug addiction](#) that occur as a consequence of ΔFosB overexpression in the nucleus accumbens.^[96] MDMA is less addictive than other stimulants such as methamphetamine and cocaine.^{[97][98]} Compared with amphetamine, MDMA and its metabolite MDA are less reinforcing.^[99]

One study found approximately 15% of chronic MDMA users met the [DSM-IV](#) diagnostic criteria for [substance dependence](#).^[100] However, there is little evidence for a specific diagnosable MDMA dependence syndrome because MDMA is typically used relatively infrequently.^[40]

There are currently no medications to treat MDMA addiction.^[101]

During pregnancy

MDMA is a moderately [teratogenic drug](#) (i.e., it is toxic to the fetus).^{[102][103]} [In utero](#) exposure to MDMA is associated with a [neuro-](#) and [cardiotoxicity](#)^[103] and impaired motor functioning. Motor delays may be temporary during infancy or long-term. The severity of these developmental delays increases with heavier MDMA use.^{[84][104]} MDMA has been shown to promote the survival of fetal dopaminergic neurons in culture.^[105]

Overdose

MDMA [overdose](#) symptoms vary widely due to the involvement of multiple [organ systems](#). Some of the more overt overdose symptoms are listed in the table below. The number of instances of fatal MDMA intoxication is low relative to its usage rates. In most fatalities, MDMA was not the only drug involved. Acute toxicity is mainly caused by [serotonin syndrome](#) and [sympathomimetic](#) effects.^[100] MDMA's toxicity in overdose may be exacerbated by [caffeine](#), with which it is frequently cut in order to increase volume.^[106] The estimated fatal dose of MDMA in humans is around 15 or 16 times a typical recreational dose.^[107] Sympathomimetic side effects can be managed with [carvedilol](#).^{[108][109]} A scheme for management of acute MDMA toxicity has been published focusing on treatment of hyperthermia, hyponatraemia, serotonin syndrome, and multiple organ failure.^[110]

Symptoms of overdose

System	Minor or moderate overdose ^[111]	Severe overdose ^[111]
Cardiovascular		- Disseminated intravascular coagulation^[66] - Intracranial hemorrhage^[66] - Severe hypertension^{[66][112]} or hypotension^[66] - Hypotensive bleeding^[8]
Central nervous system	- Hyperreflexia^[111] - Agitation^{[66][112]} - Mental confusion^[66] - Paranoia^{[66][112]} - Stimulant psychosis^{[8][15]}	- Cognitive and memory impairment^[66] potentially to the point of retrograde or anterograde amnesia^[113] - Coma^{[15][112]} - Convulsions^{[66][112]} - Hallucinations^{[66][112]} - Loss of consciousness^[15] - Serotonin syndrome^{[15][66][30]}
Musculoskeletal		- Muscle rigidity^[66] - Rhabdomyolysis (i.e., rapid muscle breakdown)^{[66][30]}
Respiratory		Acute respiratory distress syndrome^[66]
Urinary		Acute kidney injury^{[66][114]}
Other		- Cerebral edema^[15] - Hepatitis^{[66][30]} - Hyperpyrexia (a life-threatening elevation of body temperature greater than or equal to 40.0 or 41.5 °C (104.0 or 106.7 °F))^{[66][30]} - Hyponatremia (syndrome of inappropriate antidiuretic hormone)^{[66][74][30]}

Interactions

A number of [drug interactions](#) can occur between MDMA and other drugs, including [serotonergic](#) drugs.^{[64][115]} MDMA also interacts with drugs which inhibit [CYP450](#) enzymes, like [ritonavir](#) (Norvir), particularly [CYP2D6](#) inhibitors.^[64] Life-threatening reactions and death have occurred in people who took MDMA while on ritonavir.^[116] [Bupropion](#), a strong CYP2D6 inhibitor, has been found to increase MDMA exposure with administration of MDMA.^{[117][118]} Concurrent use of MDMA with certain other serotonergic drugs can result in a life-threatening condition called [serotonin syndrome](#).^{[8][64]} Severe overdose resulting in death has also been reported in people who took MDMA in combination with certain [monoamine oxidase inhibitors](#) (MAOIs),^{[8][64]} such as [phenelzine](#) (Nardil), [tranylcypromine](#) (Parnate), or [moclobemide](#) (Aurorix, Manerix).^[119] [Serotonin reuptake inhibitors](#) (SRIs) such as [citalopram](#) (Celexa), [duloxetine](#) (Cymbalta),

[fluoxetine](#) (Prozac), and [paroxetine](#) (Paxil) have been shown to block most of the subjective effects of MDMA.^[120] [Norepinephrine reuptake inhibitors](#) (NRIs) such as [reboxetine](#) (Edronax) have been found to reduce [emotional excitation](#) and feelings of [stimulation](#) with MDMA but do not appear to influence its [entactogenic](#) or [mood-elevating](#) effects.^[120]

MDMA [induces the release of monoamine neurotransmitters](#) and thereby acts as an indirectly acting [sympathomimetic](#) and produces a variety of [cardiostimulant](#) effects.^[117] It [dose-dependently](#) increases [heart rate](#), [blood pressure](#), and [cardiac output](#).^{[117][121]} SRIs like citalopram and [paroxetine](#), as well as the [serotonin 5-HT_{2A} receptor antagonist](#) [ketanserin](#), have been found to partially block the increases in heart rate and blood pressure with MDMA.^{[117][122]} It is notable in this regard that [serotonergic psychedelics](#) such as [psilocybin](#), which act as [serotonin 5-HT_{2A} receptor agonists](#), likewise have sympathomimetic effects.^{[123][124][125]} The NRI [reboxetine](#) and the [serotonin–norepinephrine reuptake inhibitor](#) (SNRI) [duloxetine](#) block MDMA-induced increases in heart rate and blood pressure.^[117] Conversely, [bupropion](#), a [norepinephrine–dopamine reuptake inhibitor](#) (NDRI) with only weak [dopaminergic](#) activity,^{[126][127]} reduced MDMA-induced heart rate and circulating [norepinephrine](#) increases but did not affect MDMA-induced blood pressure increases.^{[117][118]} On the other hand, the robust NDRI [methylphenidate](#), which has sympathomimetic effects of its own, has been found to augment the cardiovascular effects and increases in circulating [norepinephrine](#) and [epinephrine](#) levels induced by MDMA.^{[117][128]}

The [non-selective beta blocker](#) [pindolol](#) blocked MDMA-induced increases in heart rate but not blood pressure.^{[117][108][129]} The [α₂-adrenergic receptor agonist](#) [clonidine](#) did not affect the cardiovascular effects of MDMA, though it reduced blood pressure.^{[117][108][130]} The [α₁-adrenergic receptor](#) antagonists [doxazosin](#) and [prazosin](#) blocked or reduced MDMA-induced blood pressure increases but augmented MDMA-induced heart rate and cardiac output increases.^{[117][108][131][121]} The dual α₁- and [β-adrenergic receptor](#) blocker [carvedilol](#) reduced MDMA-induced heart rate and blood pressure increases.^{[117][108][109]} In contrast to the cases of serotonergic and noradrenergic agents, the [dopamine D₂ receptor](#) antagonist [haloperidol](#) did not affect the cardiovascular responses to MDMA.^{[117][132]} Due to the theoretical risk of "unopposed α-stimulation" and possible consequences like [coronary vasospasm](#), it has been suggested that dual α₁- and β-adrenergic receptor antagonists like [carvedilol](#) and [labetalol](#), rather than selective beta blockers, should be used in the management of stimulant-induced [sympathomimetic toxicity](#), for instance in the context of [overdose](#).^{[108][133]}

Pharmacology

Pharmacodynamics

MDMA is an [entactogen](#) or [empathogen](#), as well as a [stimulant](#), [euphoriant](#), and weak [psychedelic](#).^{[18][146]} It is a [substrate](#) of the [monoamine transporters](#) (MATs) and acts as a [monoamine releasing agent](#) (MRA).^{[18][147][148][149]} The drug is specifically a well-balanced [serotonin–norepinephrine–dopamine releasing agent](#) (SNDRA).^{[18][147][148][149]} To a lesser extent, MDMA also acts as a [serotonin–norepinephrine–dopamine reuptake inhibitor](#) (SNDRI).^{[18][147][148]} MDMA enters [monoaminergic neurons](#) via the MATs and then, via poorly understood [mechanisms](#), reverses the direction of these transporters to produce [efflux](#) of the [monoamine neurotransmitters](#) rather than the usual [reuptake](#).^{[18][150][151][152][153]} Induction of monoamine efflux by [amphetamines](#) in general may involve [intracellular Na⁺](#) and [Ca²⁺](#) elevation and [PKC](#) and [CaMKII \$\alpha\$](#) activation.^{[150][151][152]} MDMA also acts on the [vesicular monoamine transporter 2](#) (VMAT2) on [synaptic vesicles](#) to increase the [cytosolic](#) concentrations of the monoamine neurotransmitters available for efflux.^{[18][147]} By inducing release and reuptake inhibition of [serotonin](#), [norepinephrine](#), and [dopamine](#), MDMA increases levels of these neurotransmitters in the [brain](#) and [periphery](#) and thereby indirectly activates the [receptors](#) of these neurotransmitters.^{[18][147][153]} There are species differences in the balance of MAT activities of MDMA, with it showing greater influence on serotonin in rodents than in humans.^[153]

In addition to its actions as an SNDRA, MDMA directly interacts with a number of [monoamine](#) and other [receptors](#).^{[18][134][135][137][136]} With varying [potencies](#) ranging from high to low, it is a [partial agonist](#) of several of the serotonin [5-HT₁](#) [receptors](#), including the serotonin [5-HT_{1A}](#), [5-HT_{1B}](#), and [5-HT_{1D}](#) [receptors](#), and is a partial agonist of the serotonin [5-HT₂](#) [receptors](#), including the serotonin [5-HT_{2A}](#), [5-HT_{2B}](#), and [5-HT_{2C}](#) [receptors](#).^{[18][136][154][155][156]} The drug also interacts with [\$\alpha_2\$ -adrenergic receptors](#), with the [sigma \$\sigma_1\$](#) and [\$\sigma_2\$ receptors](#), with the [imidazoline I₁ receptor](#), and with other [targets](#).^{[18][134][135][136][137]} Along with the preceding receptor interactions, MDMA is a potent partial agonist of the rodent [trace amine-associated receptor 1](#) (TAAR1).^{[143][144]} Conversely however, due to species differences, it is far weaker in terms of potency as an agonist of the human TAAR1.^{[18][143][144][157]} The drug appears to act as a weak partial agonist of the human TAAR1 rather than as an [efficacious](#)

Activities of MDMA^[18]

Target	Affinity (K _i , nM)
SERT	0.73–13,300 (K _i) 380–2,500 (IC ₅₀) 50–72 (EC ₅₀) (rat)
NET	27,000–30,500 (K _i) 360–405 (IC ₅₀) 54–110 (EC ₅₀) (rat)
DAT	6,500–>10,000 (K _i) 1,440–21,000 (IC ₅₀) 51–278 (EC ₅₀) (rat)
5-HT_{1A}	6,300–12,200 (K _i) 126–36,000 nM (EC ₅₀) 21–64% (E _{max})
5-HT_{1B}	>10,000 (K _i) 513–1,700 (EC ₅₀) 46–91% (E _{max})
5-HT_{1D}	>10,000 (K _i) 51–2,630 (EC ₅₀) 28–51% (E _{max})
5-HT_{1E}	>10,000
5-HT_{1F}	ND
5-HT_{2A}	4,600–>10,000 (K _i) 1,260–12,500 (EC ₅₀) 21–77% (E _{max})
5-HT_{2B}	269–2,000 (K _i) 126–>20,000 (EC ₅₀) 32–84% (E _{max})
5-HT_{2C}	1,550–>13,000 (K _i) 575–9,100 (EC ₅₀) 44–92% (E _{max})
5-HT₃	>10,000
5-HT₄	ND
5-HT_{5A}	>10,000
5-HT₆	>10,000
5-HT₇	>10,000
α_{1A}	6,900–>10,000
α_{1B}	>10,000
α_{1D}	ND
α_{2A}	2,532–15,000
α_{2B}	1,785

agonist.^{[143][144]} In relation to the preceding findings, MDMA has been said to be essentially inactive as a human TAAR1 agonist.^[18] TAAR1 activation is thought to auto-inhibit and constrain the effects of amphetamines that possess TAAR1 agonism, for instance MDMA in rodents.^{[147][158][159][138][160]}

Elevation of serotonin, norepinephrine, and dopamine levels by MDMA is believed to mediate most of the drug's effects, including its entactogenic, stimulant, euphoriant, [hyperthermic](#), and [sympathomimetic](#) effects.^{[18][147][161][73]} The entactogenic effects of MDMA, including increased [sociability](#), [empathy](#), [feelings of closeness](#), and [reduced aggression](#), are thought to be mainly due to induction of serotonin release.^{[73][120][19]} The exact [serotonin receptors](#) responsible for MDMA's entactogenic effects are unclear, but may include the serotonin 5-HT_{1A} receptor,^[162] 5-HT_{1B} receptor,^[163] and 5-HT_{2A} receptor,^[164] as well as 5-HT_{1A} receptor-mediated [oxytocin](#) release and consequent activation of the [oxytocin receptor](#).^{[18][73][165][166][146][153]} Induction of dopamine release is thought to be importantly involved in the stimulant and euphoriant effects of MDMA,^{[153][18][154][167]} while induction of norepinephrine release and serotonin 5-HT_{2A} receptor stimulation are believed to mediate its sympathomimetic effects.^{[117][147]} Activation of serotonin 5-HT_{1B} and 5-HT_{2A} receptors is also thought to be involved in the stimulant and euphoriant effects of MDMA, while serotonin 5-HT_{2C} receptor activation is thought to constrain these effects and limit MDMA's [reinforcing potential](#).^{[153][168][169][170][171][172][173]} Serotonin 5-HT_{2B} receptor signaling appears to be required for MDMA-induced serotonin release and effects.^{[153][174][175][176][177][178]} MDMA has been associated with a unique subjective "magic" or [euphoria](#) that few or no other known entactogens are said to fully reproduce.^{[179][180]} The mechanisms underlying this property of MDMA are unknown, but it has been theorized to be due to a specific mixture and balance of pharmacological activities, including combined serotonin, norepinephrine, and dopamine release and direct serotonin receptor agonism.^{[181][179][180][182]}

MDMA is often said to have mild or weak [psychedelic](#) effects.^{[183][120][19][184]} These effects are said to be [dose-dependent](#), such that greater [hallucinogenic](#) effects are produced at higher doses.^{[183][185]} The mild hallucinogenic effects of MDMA include [perceptual changes](#) like intensification of [visual](#), [auditory](#), and [tactile perception](#) (e.g., brightened colors), a state of

α_{2C}	1,123–1,346
β_1, β_2	>10,000
D ₁	>13,600
D ₂	25,200
D ₃	>17,700
D ₄	>10,000
D ₅	>10,000
H ₁	2,138–>14,400
H ₂	>10,000
H ₃ , H ₄	ND
M ₁	>10,000
M ₂	>10,000
M ₃	1,850–>10,000
M ₄	8,250–>10,000
M ₅	6,340–>10,000
nACh	>10,000
TAAR1	250–370 (K _i) (rat) 1,000–1,700 (EC ₅₀) (rat) 56% (E _{max}) (rat) 2,400–3,100 (K _i) (mouse) 4,000 (EC ₅₀) (mouse) 71% (E _{max}) (mouse) 35,000 (EC ₅₀) (human) 26% (E _{max}) (human)
I ₁	220
σ_1, σ_2	ND
Notes: The smaller the value, the more avidly the drug binds to the site. Proteins are human unless otherwise specified. Refs: [134][135][18][136][137][138][139] [140][141][142][143][144][145]	

dissociation with feelings of [depersonalization](#) and [derealization](#) (e.g., "oceanic boundlessness"), and [thinking disturbances](#).^{[183][186][120][184][122][187][185]} Conversely, overt [hallucinations](#) do not occur, MDMA's hallucinogenic effects are described as "non-problematic" for users, and are said to be much less than those of [3,4-methylenedioxyamphetamine](#) (MDA) or especially those of fully effective [serotonergic psychedelics](#) like [psilocybin](#).^{[186][122][19]} The hallucinogenic effects of MDMA have been theorized to be mediated by serotonin 5-HT_{2A} receptor activation analogously to the case of classical psychedelics.^{[183][122][185][188][186][17]} Accordingly, the serotonin 5-HT_{2A} receptor [antagonist](#) ketanserin has been reported to reduce MDMA-induced perceptual changes in humans.^{[183][120][122][185]} Conversely however, it failed to affect MDMA-induced feelings of dissociation and oceanic boundlessness.^{[183][120][185]} In contrast, the serotonin reuptake inhibitor [citalopram](#), which blocks MDMA-induced serotonin release, diminished all of the psychoactive and hallucinogenic effects of MDMA.^{[183][120][187][122]} It has been noted that *N*-[methylation](#) of psychedelic [phenethylamines](#), as in the [structural](#) difference between MDA and MDMA, has invariably greatly reduced or abolished their psychedelic activity.^{[189][190]} Whereas MDA and psychedelics like psilocybin induce the [head-twitch response](#) in rodents, a behavioral proxy of psychedelic effects, findings on MDMA and the head-twitch response are mixed and conflicting.^{[191][192][120]} In addition, whereas MDA fully substitutes for psychedelics like [LSD](#) and [DOM](#) in rodent [drug discrimination](#) tests, MDMA does not do so, nor do psychedelics generally fully substitute for MDMA.^{[193][120][194][195]}

Long-term repeated activation of serotonin 5-HT_{2B} receptors by MDMA is thought to result in increased risk of organ complications such as [valvular heart disease](#) (VHD) and [primary pulmonary hypertension](#) (PPH).^{[196][197][123][198][181][199]} MDMA has been associated with [serotonergic neurotoxicity](#).^{[200][19][201]} This may be due to formation of toxic MDMA [metabolites](#) and/or induction of [simultaneous serotonin and dopamine release](#), with consequent uptake of dopamine into serotonergic neurons and breakdown into [toxic species](#).^{[200][19][201]} Serotonin 5-HT₂ receptor agonists or serotonergic psychedelics may potentiate the neurotoxicity of MDMA.^{[202][203][204][205]}

MDMA is a [racemic mixture](#) of two [enantiomers](#), (S)-MDMA and (*R*)-MDMA.^{[154][17]} (S)-MDMA is much more potent as an SNDR *in vitro* and in producing MDMA-like subjective effects in humans than (*R*)-MDMA.^{[154][149][17][206]} By contrast, (*R*)-MDMA acts as a lower-potency [serotonin–norepinephrine releasing agent](#) (SNRA) with weak or negligible effects on dopamine.^{[154][149][207]} Relatedly, (*R*)-MDMA shows weak or negligible stimulant-like and [rewarding](#) effects in animals.^{[154][208]} Both (S)-MDMA and (*R*)-MDMA produce entactogen-type effects in animals and humans.^{[154][17]} In addition, both (S)-MDMA and (*R*)-MDMA are weak agonists of the serotonin 5-HT₂ receptors.^{[154][167][17][155][156]} (*R*)-MDMA is more potent and efficacious as a serotonin 5-HT_{2A} and 5-HT_{2B} receptor agonist than (S)-MDMA, whereas (S)-MDMA is somewhat more potent as an agonist of the serotonin 5-HT_{2C} receptor.^{[154][167][17]} Due to it being a more potent serotonin 5-HT_{2A} receptor agonist than (S)-MDMA, (*R*)-MDMA has been

hypothesized to have greater psychedelic effects than (S)-MDMA or racemic MDMA.^{[209][17]} However, this proved not to be the case in a direct clinical comparison of (R)-MDMA, (S)-MDMA, and racemic MDMA, with equivalent hallucinogen-like effects instead found between the three interventions.^{[209][17]}

MDMA produces MDA as a minor [active metabolite](#).^[111] [Peak levels](#) of MDA are about 5 to 10% of those of MDMA and [total exposure](#) to MDA is almost 10% of that of MDMA with [oral](#) MDMA administration.^{[111][197]} As a result, MDA may contribute to some extent to the effects of MDMA.^{[111][188]} MDA is an entactogen, stimulant, and weak psychedelic similarly to MDMA.^[19] Like MDMA, it acts as a potent and well-balanced SNDR and as a weak serotonin 5-HT₂ receptor agonist.^{[149][155][156]} However, MDA shows much more potent and efficacious serotonin 5-HT_{2A}, 5-HT_{2B}, and 5-HT_{2C} receptor agonism than MDMA.^{[167][188][156][155]} Accordingly, MDA produces greater psychedelic effects than MDMA in humans^[19] and might particularly contribute to the mild psychedelic-like effects of MDMA.^[188] On the other hand, MDA may also be importantly involved in [toxicity](#) of MDMA, such as [cardiac valvulopathy](#).^{[210][197][155]}

The [duration of action](#) of MDMA (3–6 hours) is much shorter than its [elimination half-life](#) (8–9 hours) would imply.^[211] In relation to this, MDMA's duration and the offset of its effects appear to be determined more by [rapid acute tolerance](#) rather than by circulating drug concentrations.^[43] Similar findings have been made for [amphetamine](#) and [methamphetamine](#).^{[212][213][214][215]} One mechanism by which [tolerance](#) to MDMA may occur is [internalization](#) of the [serotonin transporter](#) (SERT).^{[216][217][218][219][220]} Although MDMA and serotonin are not significant TAAR1 agonists in humans, TAAR1 activation by MDMA may result in SERT internalization, for instance in rodents in whom MDMA is a potent TAAR1 agonist.^{[219][220][221][143]} It is thought that brain serotonin levels are depleted after MDMA administration but that levels typically return to normal within 24 to 48 hours.^[8]

Monoamine release by MDMA and related agents (EC_{50} , nM)

Compound	Serotonin	Norepinephrine	Dopamine
Amphetamine	ND	ND	ND
(S)-Amphetamine (<i>d</i>)	698–1,765	6.6–7.2	5.8–24.8
(R)-Amphetamine (<i>l</i>)	ND	9.5	27.7
Methamphetamine	ND	ND	ND
(S)-Methamphetamine (<i>d</i>)	736–1,292	12.3–13.8	8.5–24.5
(R)-Methamphetamine (<i>l</i>)	4,640	28.5	416
MDA	160	108	190
MDMA	49.6–72	54.1–110	51.2–278
(S)-MDMA (<i>d</i>)	74	136	142
(R)-MDMA (<i>l</i>)	340	560	3,700
MDEA	47	2,608	622
MBDB	540	3,300	>100,000
MDAI	114	117	1,334

Notes: The smaller the value, the more strongly the drug releases the neurotransmitter. The assays were done in rat brain synaptosomes and human potencies may be different. See also Monoamine releasing agent § Activity profiles for a larger table with more compounds. **Refs:** [149][155][222][223][224][225][226][227][18]

MDMA, MDA, and enantiomers at serotonin 5-HT₂ receptors

Compound	5-HT _{2A}		5-HT _{2B}		5-HT _{2C}	
	EC_{50} (nM)	E_{max}	EC_{50} (nM)	E_{max}	EC_{50} (nM)	E_{max}
Serotonin	53	92%	1.0	100%	22	91%
MDA	1,700	57%	190	80%	ND	ND
(S)-MDA (<i>d</i>)	18,200	89%	100	81%	7,400	73%
(R)-MDA (<i>l</i>)	5,600	95%	150	76%	7,400	76%
MDMA	6,100	55%	2,000–>20,000	32%	ND	ND
(S)-MDMA (<i>d</i>)	10,300	9%	6,000	38%	2,600	53%
(R)-MDMA (<i>l</i>)	3,100	21%	900	27%	5,400	27%

Notes: The smaller the K_{act} or EC_{50} value, the more strongly the compound produces the effect. **Refs:** [156][155][228]

Pharmacokinetics

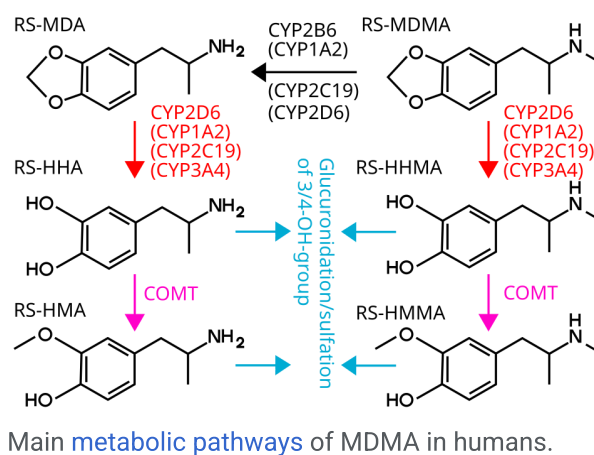
Absorption

The MDMA concentration in the bloodstream starts to rise after about 30 minutes,^[229] and reaches its maximal concentration between 1.5 and 3 hours after oral administration.^[230] It is then slowly metabolized and excreted, with levels of MDMA and its metabolites decreasing to half their peak concentration over the next several hours.^[231] The duration of action of MDMA is about 3 to 6 hours.^[19]

Distribution

The plasma protein binding of MDMA is unknown.^[14]

Metabolism



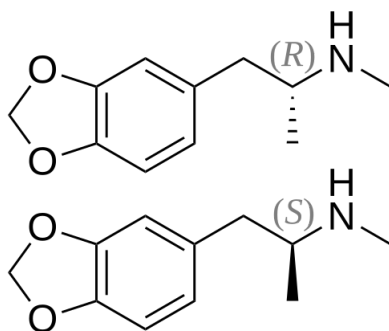
Metabolites of MDMA that have been identified in humans include 3,4-methylenedioxyamphetamine (MDA), 4-hydroxy-3-methoxymethamphetamine (HMMA), 4-hydroxy-3-methoxyamphetamine (HMA), 3,4-dihydroxyamphetamine (DHA) (also called alpha-methyldopamine (α -Me-DA)), 3,4-dihydroxymethamphetamine (HHMA), 3,4-methylenedioxyphenylacetone (MDP2P), and 3,4-methylenedioxy-N-hydroxyamphetamine (MDOH). The contributions of these metabolites to the psychoactive and toxic effects of MDMA are an area of active research. 80% of MDMA is metabolised in the liver, and about 20% is excreted unchanged in the urine.^[15]

MDMA is known to be metabolized by two main metabolic pathways: (1) O-demethylenation followed by catechol-O-methyltransferase (COMT)-catalyzed methylation or glucuronide/sulfate conjugation; and (2) N-dealkylation, deamination, and oxidation to the corresponding benzoic acid derivatives conjugated with glycine.^[111] The metabolism may be primarily by cytochrome P450 (CYP450) enzymes CYP2D6 and CYP3A4 and COMT. Complex, nonlinear pharmacokinetics arise via autoinhibition of CYP2D6 and CYP2D8, resulting in zeroth order kinetics at higher doses. It is thought that this can result in sustained and higher concentrations of MDMA if the user takes consecutive doses of the drug.^[232]

Elimination

MDMA and metabolites are primarily excreted as conjugates, such as sulfates and glucuronides.^[233] MDMA is a **chiral** compound and has been almost exclusively administered as a **racemate**. However, the two enantiomers have been shown to exhibit different kinetics. The disposition of MDMA may also be stereoselective, with the S-enantiomer having a shorter elimination half-life and greater excretion than the R-enantiomer. Evidence suggests^[234] that the area under the **blood plasma** concentration versus time curve (AUC) was two to four times higher for the (R)-enantiomer than the (S)-enantiomer after a 40 mg oral dose in human volunteers. Likewise, the plasma half-life of (*R*)-MDMA was significantly longer than that of the (S)-enantiomer (5.8 ± 2.2 hours vs 3.6 ± 0.9 hours).^[64] However, because MDMA excretion and metabolism have nonlinear kinetics,^[235] the half-lives would be higher at more typical doses (100 mg is sometimes considered a typical dose).^[230]

Chemistry



MDMA is a **racemic mixture** and exists as two **enantiomers**: (*R*)-MDMA (top) and (S)-MDMA (bottom). Depicted in this figure are their skeletal structures.



A powdered salt of MDMA



Reactors used to synthesize MDMA on an industrial scale in a clandestine chemical factory in Cikande, Indonesia

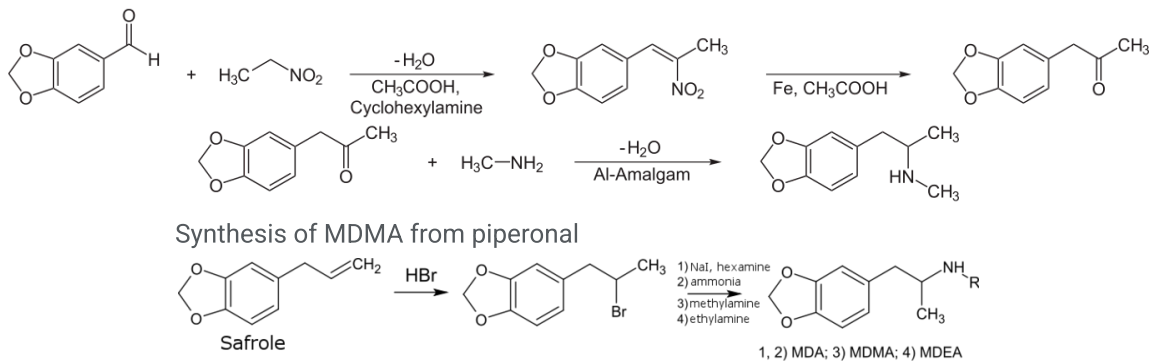
MDMA, also known as 3,4-methylenedioxy-*N*-methylamphetamine or as 1-(1,3-benzodioxol-5-yl)-*N*-methylpropan-2-amine, is a chemical compound of the phenethylamine, amphetamine, and methylenedioxyphenethylamine (MDxx) families.^[236] It is a derivative of various related compounds including phenethylamine, amphetamine, methamphetamine, homopiperonylamine (MDPEA), and 3,4-methylenedioxyamphetamine (MDA).^[236] MDMA is a racemic mixture of two enantiomers, (*R*)-MDMA and (*S*)-MDMA.^{[236][154][17]}

Properties

The chemical properties of MDMA have been described.^{[236][10]} As a free base, MDMA is a colorless oil insoluble in water.^[10] The most common salt of MDMA is the hydrochloride salt.^[10] Pure MDMA hydrochloride is water-soluble and appears as a white or off-white powder or crystal.^[10]

Synthesis

The chemical synthesis of MDMA has been described.^{[236][18]} There are numerous methods available to synthesize MDMA via different intermediates.^{[237][238][239][240]} The original MDMA synthesis described in Merck's patent involves brominating safrole to 1-(3,4-methylenedioxyphenyl)-2-bromopropane and then reacting this adduct with methylamine.^{[241][242]} Most illicit MDMA is synthesized using MDP2P (3,4-methylenedioxyphenyl-2-propanone) as a precursor. MDP2P in turn is generally synthesized from piperonal, safrole or isosafrole.^[243] One method is to isomerize safrole to isosafrole in the presence of a strong base, and then oxidize isosafrole to MDP2P. Another method uses the Wacker process to oxidize safrole directly to the MDP2P intermediate with a palladium catalyst. Once the MDP2P intermediate has been prepared, a reductive amination leads to racemic MDMA (an equal parts mixture of (*R*)-MDMA and (*S*)-MDMA). Relatively small quantities of essential oil are required to make large amounts of MDMA. The essential oil of *Ocotea cymbarum*, for example, typically contains between 80 and 94% safrole. This allows 500 mL of the oil to produce between 150 and 340 grams of MDMA.^[244]



Detection in body fluids

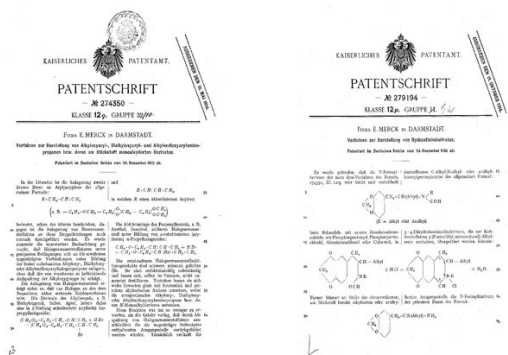
MDMA and MDA may be quantitated in blood, plasma or urine to monitor for use, confirm a diagnosis of poisoning, or assist in the forensic investigation of a traffic or other criminal violation or a sudden death. Some drug abuse screening programs rely on hair, saliva, or sweat as specimens. Most commercial amphetamine immunoassay screening tests cross-react significantly with MDMA or its major metabolites, but chromatographic techniques can easily distinguish and separately measure each of these substances. The concentrations of MDA in the blood or urine of a person who has taken only MDMA are, in general, less than 10% those of the parent drug. ^{[232][245][246]}

Analogues

Analogues of MDMA include [amphetamine](#), [methamphetamine](#), [homopiperonylamine](#) (MDPEA), [homarylamine](#) (MDMPEA), [lophophine](#) (MMDPEA), [MDA](#), [MMDA](#), [MMDA-2](#), [DMMDA](#), [DMMDA-2](#), [MDEA](#), [MDOH](#), [MDMOH](#) (FLEA), [N-t-BOC-MDMA](#), [methylone](#) (MDMC), [MBDB](#), [5-MAPB](#), [6-MAPB](#), [5-MAPBT](#), [6-MAPBT](#), [SDMA](#), [ODMA](#), [SeDMA](#), [TDMA](#), [MDAI](#), and [MDAT](#), among many others. ^{[236][19]} Deuterated analogues of MDMA such as [d2-MDMA](#) have been described as well. ^{[247][248][249][250][251]}

History

Early research and use



German patents for MDMA synthesis and the subsequent methylhydrastinine synthesis filed by Merck on 24 December 1912 and issued in 1914

MDMA was first [synthesized](#) and [patented](#) in 1912 by [Merck](#) chemist [Anton Köllisch](#).^{[252][253]} At the time, Merck was interested in developing substances that stopped abnormal bleeding. Merck wanted to avoid an existing patent held by [Bayer](#) for one such compound: [hydrastinine](#). Köllisch developed a preparation of a hydrastinine [analogue](#), methylhydrastinine, at the request of fellow lab members, Walther Beckh and Otto Wolfes. MDMA (called methylsafrylamin, safrylmethylamin or N-Methyl-a-Methylhomopiperonylamin in Merck laboratory reports) was an [intermediate compound](#) in the synthesis of methylhydrastinine. Merck was not interested in MDMA itself at the time.^[253] On 24 December 1912, Merck filed two patent applications that described the synthesis and some chemical properties of MDMA^[254] and its subsequent conversion to methylhydrastinine.^[255] Merck records indicate its researchers returned to the compound sporadically. A 1920 Merck patent describes a chemical modification to MDMA.^{[252][256]}

MDMA's [analogue 3,4-methylenedioxyamphetamine](#) (MDA) was first synthesized in 1910 as a [derivative](#) of [adrenaline](#).^[252] [Gordon A. Alles](#), the discoverer of the [psychoactive](#) effects of [amphetamine](#), also discovered the psychoactive effects of MDA in 1930 in a [self-experiment](#) in which he administered a high dose (126 mg) to himself.^{[252][257][258]} However, he did not subsequently describe these effects until 1959.^{[259][257][258]} MDA was later tested as an [appetite suppressant](#) by [Smith, Kline & French](#) and for other uses by other groups in the 1950s.^[252] In relation to the preceding, the psychoactive effects of MDA were discovered well before those of MDMA.^{[252][259]}

In 1927, Max Oberlin studied the pharmacology of MDMA while searching for substances with effects similar to [adrenaline](#) or [ephedrine](#), the latter being structurally similar to MDMA. Compared to ephedrine, Oberlin observed that it had similar effects on [vascular smooth muscle](#) tissue, stronger effects at the uterus, and no "local effect at the eye". MDMA was also found to have effects on [blood sugar](#) levels comparable to high doses of ephedrine. Oberlin concluded that the effects of MDMA were not limited to the [sympathetic nervous system](#). Research was stopped "particularly due to a strong price increase of safrylmethylamine", which was still used as an intermediate in methylhydrastinine synthesis. Albert van Schoor performed simple toxicological tests with the drug in 1952, most likely while researching new stimulants or circulatory medications. After pharmacological studies, research on MDMA was not continued. In 1959, Wolfgang Fruhstorfer synthesized MDMA for pharmacological testing while researching stimulants. It is unclear if Fruhstorfer investigated the effects of MDMA in humans.^[253]

Outside of Merck, other researchers began to investigate MDMA. In 1953 and 1954, the [United States Army](#) commissioned a study of [toxicity](#) and behavioral effects in animals injected with [mescaline](#) and several analogues, including MDMA. Conducted at the [University of Michigan](#) in [Ann Arbor](#), these investigations were declassified in October 1969 and published in 1973.^{[260][259]} A 1960 Polish paper by Biniecki and Krajewski describing the synthesis of MDMA as an intermediate was the first published scientific paper on the substance.^{[253][259][261]}

The first administration of MDMA to humans is unknown.^[153] MDA appeared as a [recreational drug](#) in the mid-1960s.^[252] MDMA may have been in non-medical use in the western United States in 1968.^{[252][262]} An August 1970 report at a meeting of crime laboratory chemists indicates MDMA was being used recreationally in the Chicago area by 1970.^{[153][259][263]} MDMA likely emerged as a substitute for MDA,^[264] a drug at the time popular among users of psychedelics^[265] which was made a [Schedule 1 controlled substance](#) in the United States in 1970.^{[266][267]}

Shulgin's research



Alexander and Ann Shulgin in December 2011

American chemist and [psychopharmacologist Alexander Shulgin](#) reported he synthesized MDMA in 1965 while researching methylenedioxy compounds at [Dow Chemical Company](#), but did not test the psychoactivity of the compound at this time. Around 1970, Shulgin sent instructions for N-methylated MDA (MDMA) synthesis to the founder of a Los Angeles chemical company who had requested them. This individual later provided these instructions to a client in the Midwest. Shulgin may have suspected he played a role in the emergence of MDMA in Chicago.^[259]

Shulgin first heard of the psychoactive effects of N-methylated MDA around 1975 from a young student who reported "amphetamine-like content".^[259] Around 30 May 1976, Shulgin again heard about the effects of N-methylated MDA,^[259] this time from a graduate student in a medicinal chemistry group he advised at [San Francisco State University](#)^{[265][268]} who directed him to the University of Michigan study.^[236] She and two close friends had consumed 100 mg of MDMA and reported positive emotional experiences.^[259] Following the self-trials of a colleague at the [University of San Francisco](#), Shulgin synthesized MDMA and tried it himself in September and October 1976.^{[259][265]} Shulgin and [David E. Nichols](#) first reported on MDMA and its psychoactive effects in humans in a presentation at a conference in Bethesda, Maryland in December 1976.^{[259][269]} This conference paper was subsequently formally published in 1978.^{[252][259][269]} They described MDMA as inducing "an easily controlled altered state of consciousness with

emotional and sensual overtones" comparable "to marijuana, to [psilocybin](#) devoid of the hallucinatory component, or to low levels of MDA".^[269]

While not finding his own experiences with MDMA particularly powerful,^{[236][270]} Shulgin was impressed with the drug's disinhibiting effects and thought it could be useful in therapy.^[270]

Believing MDMA allowed users to strip away habits and perceive the world clearly, Shulgin called the drug *window*.^{[236][271]} Shulgin occasionally used MDMA for relaxation, referring to it as "my low-calorie martini", and gave the drug to friends, researchers, and others who he thought could benefit from it.^[236] One such person was [Leo Zeff](#), a psychotherapist who had been known to use psychedelic substances in his practice. When he tried the drug in 1977, Zeff was impressed with the effects of MDMA and came out of his semi-retirement to promote its use in therapy. Over the following years, Zeff traveled around the United States and occasionally to Europe, eventually training an estimated four thousand psychotherapists in the therapeutic use of MDMA.^{[270][272]} Zeff named the drug *Adam*, believing it put users in a state of primordial innocence.^[265]

Psychotherapists who used MDMA believed the drug eliminated the typical fear response and increased communication. Sessions were usually held in the home of the patient or the therapist. The role of the therapist was minimized in favor of patient self-discovery accompanied by MDMA induced feelings of empathy. Depression, substance use disorders, relationship problems, premenstrual syndrome, and autism were among several psychiatric disorders MDMA assisted therapy was reported to treat.^[267] According to psychiatrist George Greer, therapists who used MDMA in their practice were impressed by the results. Anecdotally, MDMA was said to greatly accelerate therapy.^[270] According to [David Nutt](#), MDMA was widely used in the western US in couples counseling, and was called *empathy*. Only later was the term *ecstasy* used for it, coinciding with rising opposition to its use.^{[273][274]}

Rising recreational use

In the late 1970s and early 1980s, "Adam" spread through personal networks of psychotherapists, psychiatrists, users of psychedelics, and [yuppies](#). Hoping MDMA could avoid criminalization like LSD and mescaline, psychotherapists and experimenters attempted to limit the spread of MDMA and information about it while conducting informal research.^{[267][275]} Early MDMA distributors were deterred from large scale operations by the threat of possible legislation.^[276] Between the 1970s and the mid-1980s, this network of MDMA users consumed an estimated 500,000 doses.^{[23][277]}

A small recreational market for MDMA developed by the late 1970s,^[278] consuming perhaps 10,000 doses in 1976.^[266] By the early 1980s MDMA was being used in Boston and New York City nightclubs such as [Studio 54](#) and [Paradise Garage](#).^{[279][280]} Into the early 1980s, as the recreational market slowly expanded, production of MDMA was dominated by a small group of

therapeutically minded [Boston](#) chemists. Having commenced production in 1976, this "Boston Group" did not keep up with growing demand and shortages frequently occurred.^[276]

Perceiving a business opportunity, Michael Clegg, the Southwest distributor for the Boston Group, started his own "Texas Group" backed financially by Texas friends.^{[276][281]} In 1981,^[276] Clegg had coined "Ecstasy" as a slang term for MDMA to increase its marketability.^{[271][275]} Starting in 1983,^[276] the Texas Group mass-produced MDMA in a Texas lab^[275] or imported it from California^[271] and marketed tablets using pyramid sales structures and toll-free numbers.^[277] MDMA could be purchased via credit card and taxes were paid on sales.^[276] Under the brand name "Sassyfras", MDMA tablets were sold in brown bottles.^[275] The Texas Group advertised "Ecstasy parties" at bars and discos, describing MDMA as a "fun drug" and "good to dance to".^[276] MDMA was openly distributed in [Austin](#) and [Dallas–Fort Worth](#) area bars and nightclubs, becoming popular with yuppies, college students, and gays.^{[264][276][277]}

Recreational use also increased after several cocaine dealers switched to distributing MDMA following experiences with the drug.^[277] A California laboratory that analyzed confidentially submitted drug samples first detected MDMA in 1975. Over the following years the number of MDMA samples increased, eventually exceeding the number of MDA samples in the early 1980s.^{[282][283]} By the mid-1980s, MDMA use had spread to colleges around the United States.^{[276]:33}

Media attention and scheduling

United States

30210	Federal Register / Vol. 49, No. 149 / Friday, July 27, 1984 / Proposed Rules	
and to comment on the current methodology. Comments on other aspects of the temporary alien labor certification program are outside the scope of this invitation for comments.	On March 13, 1984, the Administrator, Drug Enforcement Administration, submitted information relevant to the above period and that trafficking of 3,4-methylenedioxymethamphetamine (MDMA) to the Assistant Secretary for Health, Department of Health and Human Services, thereby, the information documented that 3,4-methylenedioxymethamphetamine, trafficked on the street as MDMA or ecstasy, (1) is an analog of the Schedule I substance, 3,4-methylenedioxyamphetamine (MDA), (2) has no legitimate medical use or medicinal use in the United States, (3) has been classified as a Schedule I substance, (4) produces stimulant and psychotropic effects as human similar to those produced by MDA, and (5) has been associated with medical emergencies as reported by the Drug Abuse Warning Network (DAWN).	Department of Justice regulations (28 CFR 0.100), the Administrator hereby proposes that (1) 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
Reynold J. Thomas, Secretary of Labor. FR 1984-07-27; 49 FR 30210-10; 49 FR 30210-11.	In accordance with the provisions of 21 U.S.C. 812(b), the DEA Administrator requested a scientific and medical evaluation of the relevant information and a scheduling recommendation for 3,4-methylenedioxy-N-methylphenylisopropylamine from the Assistant Secretary for Health.	1. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
DEPARTMENT OF JUSTICE Drug Enforcement Administration 21 CFR Part 1308 Schedules of Controlled Substances Proposed Placement of 3,4-Methylenedioxymethamphetamine into Schedule I	On June 8, 1984, the Administrator of the Drug Enforcement Administration received a letter from the Assistant Secretary for Health, acting on behalf of the Secretary of the Department of Health and Human Services, stating that 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) has a high potential for abuse and presents a significant risk to the public health, and recommending that it should be placed into Schedule I of the Controlled Substances Act.	2. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
ACTION: Notice of proposed rulemaking.	Based upon the investigations and review of the Drug Enforcement Administration and upon the scientific and medical evaluation and the recommendations of the Secretary of Health and Human Services in accordance with 21 U.S.C. 812(b), the Administrator of the Drug Enforcement Administration, pursuant to the provisions of 21 U.S.C. 811(a), finds that:	3. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
SUMMARY: This notice of proposed rulemaking is issued by the Administrator of the Drug Enforcement Administration (DEA) to place the substance, 3,4-methylenedioxy-N-methylphenylisopropylamine, into Schedule I of the Controlled Substances Act (CSA) (21 U.S.C. 811, et seq.). This proposed action follows DEA's review of the above and illicit trafficking of 3,4-methylenedioxy-N-methylphenylisopropylamine, which was found by the Assistant Secretary for Health, Department of Health and Human Services, to support DEA's position that this substance be placed into Schedule I of the CSA. The effect of this proposed action would be to impose the regulatory control mechanisms and criminal sanctions of Schedule I on the manufacturing, distribution and possession of 3,4-methylenedioxy-N-methylphenylisopropylamine.	Under the authority vested in the Attorney General by section 10102 of the CSA (21 U.S.C. 811(a)), and adopted to the Administrator by	4. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
DATE: Comments must be submitted on or before August 27, 1984.	DEA, 1401 I Street, NW, Washington, D.C. 20537. Attention: DEA Federal Register Representative.	5. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
ADDRESS: Comments and objections should be submitted in quadruplicate to the Administrator, Drug Enforcement Administration, 1401 I Street, NW, Washington, D.C. 20537. Attention: DEA Federal Register Representative.	Under the authority vested in the Attorney General by section 10102 of the CSA (21 U.S.C. 811(a)), and adopted to the Administrator by	6. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
FOR FURTHER INFORMATION CONTACT: Howard McGhee, Jr., Chief, Drug Control Section, Drug Enforcement Administration, Washington, D.C. 20537, Telephone (202) 616-1000.	DEA, 1401 I Street, NW, Washington, D.C. 20537. Attention: DEA Federal Register Representative.	7. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:
SUPPLEMENTARY INFORMATION: List of Subjects in 21 CFR Part 1308: Administrative practice and procedure, Drug traffic control, Narcotics, Prescription Drugs.		8. 3,4-methylenedioxy-N-methylphenylisopropylamine (MDMA) be redesignated as (4) (D)-(2), respectively, and that a new (4)(D) be added to read as follows:

27 July 1984 Federal Register notice of the proposed MDMA scheduling

In an early media report on MDMA published in 1982, a [Drug Enforcement Administration](#) (DEA) spokesman stated the agency would ban the drug if enough evidence for abuse could be

found.^[276] By mid-1984, MDMA use was becoming more noticed. Bill Mandel reported on "Adam" in a 10 June [San Francisco Chronicle](#) article, but misidentified the drug as [methyloxymethylenedioxyamphetamine](#) (MMDA). In the next month, the World Health Organization identified MDMA as the only substance out of twenty phenethylamines to be seized a significant number of times.^[275]

After a year of planning and data collection, MDMA was proposed for [scheduling](#) by the DEA on 27 July 1984, with a request for comments and objections.^{[275][284]} The DEA was surprised when a number of psychiatrists, psychotherapists, and researchers objected to the proposed scheduling and requested a hearing.^[267] In a [Newsweek](#) article published the next year, a DEA pharmacologist stated that the agency had been unaware of its use among psychiatrists.^[285] An initial hearing was held on 1 February 1985 at the DEA offices in Washington, D.C., with administrative law judge Francis L. Young presiding.^[275] It was decided there to hold three more hearings that year: Los Angeles on 10 June, Kansas City, Missouri on 10–11 July, and Washington, D.C., on 8–11 October.^{[267][275]}

Sensational media attention was given to the proposed criminalization and the reaction of MDMA proponents, effectively advertising the drug.^[267] In response to the proposed scheduling, the Texas Group increased production from 1985 estimates of 30,000 tablets a month to as many as 8,000 per day, potentially making two million ecstasy tablets in the months before MDMA was made illegal.^[286] By some estimates the Texas Group distributed 500,000 tablets per month in Dallas alone.^[271] According to one participant in an [ethnographic](#) study, the Texas Group produced more MDMA in eighteen months than all other distribution networks combined across their entire histories.^[276] By May 1985, MDMA use was widespread in California, Texas, southern Florida, and the northeastern United States.^{[262][287]} According to the DEA there was evidence of use in twenty-eight states^[288] and Canada.^[262] Urged by Senator [Lloyd Bentsen](#), the DEA announced an [emergency Schedule I classification](#) of MDMA on 31 May 1985. The agency cited increased distribution in Texas, escalating street use, and new evidence of MDA (an analog of MDMA) neurotoxicity as reasons for the emergency measure.^{[287][289][290]} The ban took effect one month later on 1 July 1985^[286] in the midst of [Nancy Reagan's "Just Say No"](#) campaign.^{[291][292]}

As a result of several expert witnesses testifying that MDMA had an accepted medical usage, the administrative law judge presiding over the hearings recommended that MDMA be classified as a [Schedule III](#) substance. Despite this, DEA administrator [John C. Lawn](#) overruled and classified the drug as Schedule I.^{[267][293]} Harvard psychiatrist [Lester Grinspoon](#) then sued the DEA, claiming that the DEA had ignored the medical uses of MDMA, and the federal court sided with Grinspoon, calling Lawn's argument "strained" and "unpersuasive", and vacated MDMA's Schedule I status.^[294] Despite this, less than a month later Lawn reviewed the evidence and reclassified MDMA as Schedule I again, claiming that the expert testimony of several

psychiatrists claiming over 200 cases where MDMA had been used in a therapeutic context with positive results could be dismissed because they were not published in medical journals.^[267] In 2017, the FDA [granted breakthrough therapy](#) designation for its use with psychotherapy for PTSD. However, this designation has been questioned and problematized.^[295]

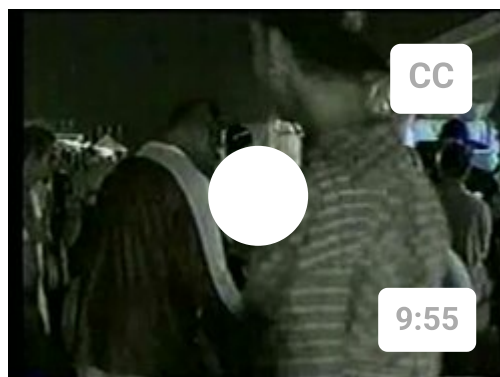
United Nations

While engaged in scheduling debates in the United States, the DEA also pushed for international scheduling.^[286] In 1985, the [World Health Organization's](#) Expert Committee on Drug Dependence recommended that MDMA be placed in Schedule I of the 1971 United Nations [Convention on Psychotropic Substances](#). The committee made this recommendation on the basis of the pharmacological similarity of MDMA to previously scheduled drugs, reports of illicit trafficking in Canada, drug seizures in the United States, and lack of well-defined therapeutic use. While intrigued by reports of psychotherapeutic uses for the drug, the committee viewed the studies as lacking appropriate methodological design and encouraged further research. Committee chairman [Paul Grof](#) dissented, believing international control was not warranted at the time and a recommendation should await further therapeutic data.^[296] The [Commission on Narcotic Drugs](#) added MDMA to Schedule I of the convention on 11 February 1986.^[297]

Post-scheduling



A 1995 Vibe Tribe rave in [Erskineville, New South Wales](#), Australia being broken up by police. MDMA use spread globally along with rave culture.



A 2000 [United States Air Force](#) video dramatizing the dangers of MDMA misuse

The use of MDMA in Texas clubs declined rapidly after criminalization, but by 1991, the drug became popular among young middle-class whites and in nightclubs.^[276] In 1985, MDMA use became associated with [acid house](#) on the Spanish island of [Ibiza](#).^{[276]:50}^[298] Thereafter, in the late 1980s, the drug spread alongside [rave culture](#) to the United Kingdom and then to other European and American cities.^{[276]:50} Illicit MDMA use became increasingly widespread among young adults in universities and later, in high schools. Since the mid-1990s, MDMA has become the most widely used amphetamine-type drug by college students and teenagers.^{[299]:1080} MDMA became one of the four most widely used illicit drugs in the US, along with [cocaine](#), [heroin](#), and [cannabis](#).^[271] According to some estimates as of 2004, only marijuana attracts more first time users in the United States.^[271]

After MDMA was criminalized, most medical use stopped, although some therapists continued to prescribe the drug illegally. Later, Charles Grob initiated an ascending-dose safety study in healthy volunteers. Subsequent FDA-approved MDMA studies in humans have taken place in the United States in Detroit ([Wayne State University](#)), Chicago ([University of Chicago](#)), San Francisco (UCSF and [California Pacific Medical Center](#)), Baltimore ([NIDA–NIH Intramural Program](#)), and [South Carolina](#). Studies have also been conducted in Switzerland (University Hospital of Psychiatry, [Zürich](#)), the Netherlands ([Maastricht University](#)), and Spain ([Universitat Autònoma de Barcelona](#)).^[300]

"Molly", short for 'molecule', was recognized as a slang term for crystalline or powder MDMA in the 2000s.^[301]^[302]

In 2010, the BBC reported that use of MDMA had decreased in the UK in previous years. This may be due to increased seizures during use and decreased production of the precursor chemicals used to manufacture MDMA. Unwitting substitution with other drugs, such as [mephedrone](#) and [methamphetamine](#),^[303] as well as legal alternatives to MDMA, such as [BZP](#), [MDPV](#), and [methyldone](#), are also thought to have contributed to its decrease in popularity.^[304]

In 2017, it was found that some pills being sold as MDMA contained [pentylone](#), which can cause very unpleasant agitation and paranoia.^[305]

According to [David Nutt](#), when [safrole](#) was restricted by the United Nations in order to reduce the supply of MDMA, producers in China began using [anethole](#) instead, but this gives [para-methoxyamphetamine](#) (PMA, also known as "Dr Death"), which is much more toxic than MDMA and can cause overheating, muscle spasms, seizures, unconsciousness, and death. People wanting MDMA are sometimes sold PMA instead.^[273]

In 2025, the BBC reported on a study of 650 survivors from the [Nova music festival massacre](#). Two-thirds were under the influence of recreational drugs (MDMA, LSD, marijuana or psilocybin) when [Hamas](#) attacked the festival on [October 7, 2023](#). MDMA appeared to have a protective effect against later problems with sleeping and emotional distress.^[306]^[307]

Society and culture

Legal status

MDMA is legally controlled in most of the world under the UN [Convention on Psychotropic Substances](#) and other international agreements, although exceptions exist for research and limited medical use. In general, the unlicensed use, sale or manufacture of MDMA are all criminal offences.

Australia

In Australia, MDMA was rescheduled on 1 July 2023 as a schedule 8 substance (available on prescription) when used in the treatment of PTSD, while remaining a schedule 9 substance (prohibited) for all other uses. For the treatment of PTSD, MDMA can only be prescribed by psychiatrists with specific training and authorisation.^[309] In 1986, MDMA was declared an illegal substance because of its allegedly harmful effects and potential for misuse.^[310] Any non-authorized sale, use or manufacture is strictly prohibited by law. Permits for research uses on humans must be approved by a recognized [ethics committee](#) on human research.

In [Western Australia](#) under the Misuse of Drugs Act 1981 4.0g of MDMA is the amount required determining a court of trial, 2.0g is considered a presumption with intent to sell or supply and 28.0g is considered trafficking under Australian law.^[311]

The [Australian Capital Territory](#) passed legislation to decriminalise the possession of small amounts of MDMA, which took effect in October 2023.^{[312][313]}

Canada

In Canada, MDMA is listed as a [Schedule 1](#)^[314] as it is an analogue of amphetamine.^[315] The [Controlled Drugs and Substances Act](#) was updated as a result of the [Safe Streets and Communities Act](#) changing amphetamines from [Schedule III](#) to Schedule I in March 2012. In 2022, the federal government granted [British Columbia](#) a 3-year exemption, legalizing the possession of up to 2.5 grams (0.088 oz) of MDMA in the province from February 2023 until February 2026.^{[316][317]}

Finland

Scheduled in the "government decree on substances, preparations and plants considered to be narcotic drugs".^[318] Ecstasy is considered a very dangerous illegal drug.^[319]

Global estimates of drug users in 2016
(in millions of users)^[308]

Substance	Best estimate	Low estimate	High estimate
Amphetamine-type stimulants	34.16	13.42	55.24
Cannabis	192.15	165.76	234.06
Cocaine	18.20	13.87	22.85
Ecstasy	20.57	8.99	32.34
Opiates	19.38	13.80	26.15
Opioids	34.26	27.01	44.54

Netherlands

In 2024, a Dutch state commission issued a report advocating for MDMA to be made available to patients with PTSD.^[320]

In June 2011, the Expert Committee on the List (Expertcommissie Lijstensystematiek Opiumwet) issued a report which discussed the evidence for harm and the legal status of MDMA, arguing in favor of maintaining it on List I.^{[321][322][323]}

United Kingdom

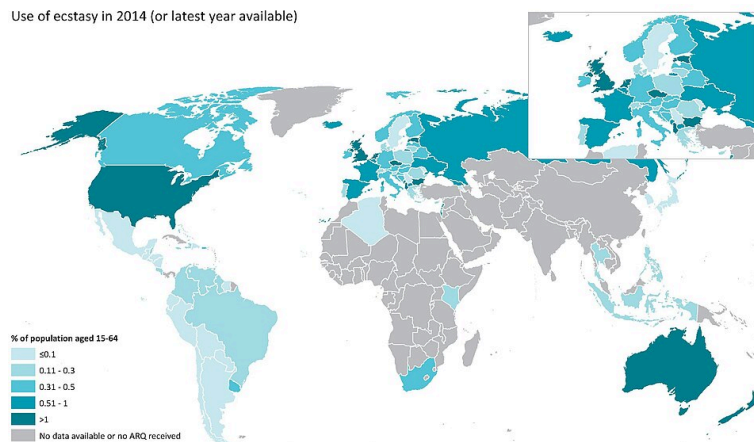
In the United Kingdom, MDMA was made illegal in 1977 by a modification order to the existing [Misuse of Drugs Act 1971](#). Although MDMA was not named explicitly in this legislation, the order extended the definition of Class A drugs to include various ring-substituted phenethylamines.^{[324][325]} The drug is therefore illegal to sell, buy, or possess without a licence in the UK. Penalties include a maximum of seven years and/or unlimited fine for possession; life and/or unlimited fine for production or trafficking.

Some researchers such as [David Nutt](#) have criticized the scheduling of MDMA, which he determined to be a relatively harmless drug.^{[326][327]} An editorial he wrote in the *[Journal of Psychopharmacology](#)*, where he compared the risk of harm for [horse riding](#) (1 adverse event in 350) to that of ecstasy (1 in 10,000) resulted in his dismissal, leading to the resignation of several of his colleagues from the [ACMD](#).^[328]

United States

In the United States, MDMA is listed in [Schedule I](#) of the [Controlled Substances Act](#).^[329] In a 2011 federal court hearing, the [American Civil Liberties Union](#) successfully argued that the sentencing guideline for MDMA/ecstasy is based on outdated science, leading to excessive prison sentences.^[330] Other courts have upheld the sentencing guidelines. The [United States District Court for the Eastern District of Tennessee](#) explained its ruling by noting that "an individual federal district court judge simply cannot marshal resources akin to those available to the Commission for tackling the manifold issues involved with determining a proper drug equivalency."^[321]

Demographics



UNODC map showing the use of ecstasy by country in 2014 for the global population aged 15–64

In 2014, 3.5% of 18-to-25-year-olds had used MDMA in the United States.^[8] In the European Union as of 2018, 4.1% of adults (15–64 years old) have used MDMA at least once in their life, and 0.8% had used it in the last year.^[331] Among young adults, 1.8% had used MDMA in the last year.^[331]

In Europe, an estimated 37% of regular club-goers aged 14 to 35 used MDMA in the past year according to the 2015 European Drug report.^[8] The highest one-year prevalence of MDMA use in Germany in 2012 was 1.7% among people aged 25 to 29 compared with a population average of 0.4%.^[8] Among adolescent users in the United States between 1999 and 2008, girls were more likely to use MDMA than boys.^[332]

Economics

Europe

In 2008 the [European Monitoring Centre for Drugs and Drug Addiction](#) noted that although there were some reports of tablets being sold for as little as €1, most countries in Europe then reported typical retail prices in the range of €3 to €9 per tablet, typically containing 25–65 mg of MDMA.^[333] By 2014 the EMCDDA reported that the range was more usually between €5 and €10 per tablet, typically containing 57–102 mg of MDMA, although MDMA in powder form was becoming more common.^[334]

North America

The [United Nations Office on Drugs and Crime](#) stated in its 2014 World Drug Report that US ecstasy retail prices range from US\$1 to \$70 per pill, or from \$15,000 to \$32,000 per kilogram.^[335] A new research area named Drug Intelligence aims to automatically monitor distribution networks based on image processing and machine learning techniques, in which an Ecstasy pill picture is analyzed to detect correlations among different production batches.^[336] These novel techniques allow police scientists to facilitate the monitoring of illicit distribution networks.

As of October 2015, most of the MDMA in the United States is produced in British Columbia, Canada and imported by Canada-based Asian [transnational criminal organizations](#).^[59] The market for MDMA in the United States is relatively small compared to [methamphetamine](#), [cocaine](#), and [heroin](#).^[59] In the United States, about 0.9 million people used ecstasy in 2010.^[25]

Australia

MDMA is particularly expensive in Australia, costing [A\\$15–A\\$30](#) per tablet. In terms of purity data for Australian MDMA, the average is around 34%, ranging from less than 1% to about 85%. The majority of tablets contain 70–85 mg of MDMA. Most MDMA enters Australia from the Netherlands, the UK, Asia, and the US.^[337]

Corporate logos on pills

A number of ecstasy manufacturers brand their pills with a [logo](#), often that of an unrelated corporation.^[338] Some pills depict logos of products or media popular with children, such as [Shaun the Sheep](#).^[339]

Research

MDMA is being investigated as a potential treatment for social impairments in [autism](#).^[340]

The [British critical psychiatrist Joanna Moncrieff](#) has critiqued the use and study of MDMA and related drugs like [psychedelics](#) for treatment of psychiatric disorders, highlighting concerns including excessive hype around these drugs, blurred lines between medical and recreational use, flawed clinical trial findings, financial conflicts of interest, strong expectancy effects and large placebo responses, short-term benefits over placebo, and their potential for adverse effects, among others.^[341]

See also

- [Substituted methylenedioxyphenethylamine](#)
- *[I Feel Love: MDMA and the Quest for Connection in a Fractured World](#)*
- [Lykos Therapeutics](#)
- [Multidisciplinary Association for Psychedelic Studies](#) (MAPS)

References

1. "FDA Substance Registration System" (<https://precision.fda.gov/uniisearch/srs/unii/KE1SEN21RM>) . United States National Library of Medicine. Archived (<https://web.archive.org/web/20170831220013/https://fdasis.nlm.nih.gov/srs/unii/KE1SEN21RM>) from the original on 31 August 2017. Retrieved 31 August 2017.

2. Luciano RL, Perazella MA (June 2014). "Nephrotoxic effects of designer drugs: synthetic is not better!". *Nature Reviews. Nephrology*. **10** (6): 314–324. doi:10.1038/nrneph.2014.44 (<https://doi.org/10.1038/nrneph.2014.44>) . ISSN 1759-5061 (<https://search.worldcat.org/issn/1759-5061>) . PMID 24662435 (<https://pubmed.ncbi.nlm.nih.gov/24662435>) . S2CID 9817771 (<https://api.semanticscholar.org/CorpusID:9817771>) .
3. "DrugFacts: MDMA (Ecstasy or Molly)" (<https://web.archive.org/web/20141203115531/http://www.drugabuse.gov/publications/drugfacts/mdma-ecstasy-or-molly>) . National Institute on Drug Abuse. Archived from the original (<http://www.drugabuse.gov/publications/drugfacts/mdma-ecstasy-or-molly>) on 3 December 2014. Retrieved 2 December 2014.
4. "Pingers, pingas, pingaz: how drug slang affects the way we use and understand drugs" (<https://theconversation.com/pingers-pingas-pingaz-how-drug-slang-affects-the-way-we-use-and-understand-drugs-129452>) . The Conversation. 8 January 2020. Archived (<https://web.archive.org/web/20210115205246/https://theconversation.com/pingers-pingas-pingaz-how-drug-slang-affects-the-way-we-use-and-understand-drugs-129452>) from the original on 15 January 2021.
5. Palmer RB (2012). *Medical toxicology of drug abuse: synthesized chemicals and psychoactive plants* (<https://books.google.com/books?id=OWFiVaDZnkQC&pg=PA139>) . Hoboken, N.J.: John Wiley & Sons. p. 139. ISBN 978-0-471-72760-6. Archived (<https://web.archive.org/web/20230113000533/https://books.google.com/books?id=OWFiVaDZnkQC&pg=PA139>) from the original on 13 January 2023. Retrieved 4 September 2017.
6. Upfal J (2022). *Australian Drug Guide: The Plain Language Guide to Drugs and Medicines of All Kinds* (9th ed.). Melbourne: Black Inc. p. 319. ISBN 978-1-76064-319-5. "Habit-forming potential moderate. Ecstasy may induce psychological dependence and tolerance to its effect when used frequently."
7. Malenka RC, Nestler EJ, Hyman SE (2009). "Chapter 15: Reinforcement and Addictive Disorders". In Sydor A, Brown RY (eds.). *Molecular Neuropharmacology: A Foundation for Clinical Neuroscience* (2nd ed.). New York: McGraw-Hill Medical. p. 375. ISBN 978-0-07-148127-4.
8. Betzler F, Viohl L, Romanczuk-Seiferth N (January 2017). "Decision-making in chronic ecstasy users: a systematic review" (<https://doi.org/10.1111/2Fejn.13480>) . *The European Journal of Neuroscience*. **45** (1): 34–44. doi:10.1111/ejn.13480 (<https://doi.org/10.1111/2Fejn.13480>) . PMID 27859780 (<https://pubmed.ncbi.nlm.nih.gov/27859780>) . S2CID 31694072 (<https://api.semanticscholar.org/CorpusID:31694072>) . "...the addictive potential of MDMA itself is relatively small."

9. Jerome L, Schuster S, Yazar-Klosinski BB (March 2013). "Can MDMA play a role in the treatment of substance abuse?" (<https://web.archive.org/web/20200803194522/http://pdfs.semanticscholar.org/33fd/4f5decd405dee8d4f280a9158bfb16ae6e27.pdf>) (PDF). *Current Drug Abuse Reviews*. **6** (1): 54–62. doi:10.2174/18744737112059990005 (<https://doi.org/10.2174%2F18744737112059990005>) . PMID 23627786 (<https://pubmed.ncbi.nlm.nih.gov/23627786>) . S2CID 9327169 (<https://api.semanticscholar.org/CorpusID:9327169>) . Archived from the original (<http://pdfs.semanticscholar.org/33fd/4f5decd405dee8d4f280a9158bfb16ae6e27.pdf>) (PDF) on 3 August 2020. "Animal and human studies demonstrate moderate abuse liability for MDMA, and this effect may be of most concern to those treating substance abuse disorders."
10. "Methylenedioxymethamphetamine (MDMA or 'Ecstasy')" (<http://www.emcdda.europa.eu/publications/drug-profiles/mdma>) . EMCDDA. European Monitoring Centre for Drugs and Drug Addiction. Archived (<https://web.archive.org/web/20160101211851/http://www.emcdda.europa.eu/publications/drug-profiles/mdma>) from the original on 1 January 2016. Retrieved 17 October 2014.
11. "Methylenedioxymethamphetamine (MDMA, ecstasy)" (<https://web.archive.org/web/20120503102427/http://www.nhtsa.gov/people/injury/research/job185drugs/methylenedioxymethamphetamine.htm>) . *Drugs and Human Performance Fact Sheets*. National Highway Traffic Safety Administration. Archived from the original (<http://www.nhtsa.dot.gov/people/injury/research/job185drugs/methylenedioxymethamphetamine.htm>) on 3 May 2012.
12. Anvisa (24 July 2023). "RDC Nº 804 - Listas de Substâncias Entorpecentes, Psicotrópicas, Precursoras e Outras sob Controle Especial" (<https://www.in.gov.br/en/web/dou/-/resolucao-rdc-n-804-de-24-de-julho-de-2023-498447451>) [Collegiate Board Resolution No. 804 - Lists of Narcotic, Psychotropic, Precursor, and Other Substances under Special Control] (in Brazilian Portuguese). *Diário Oficial da União* (published 25 July 2023). Archived (<https://web.archive.org/web/20230827163149/https://www.in.gov.br/en/web/dou/-/resolucao-rdc-n-804-de-24-de-julho-de-2023-498447451>) from the original on 27 August 2023. Retrieved 27 August 2023.
13. Freye E (28 July 2009). "Pharmacological Effects of MDMA in Man". *Pharmacology and Abuse of Cocaine, Amphetamines, Ecstasy and Related Designer Drugs*. Springer Netherlands. pp. 151–160. doi:10.1007/978-90-481-2448-0_24 (https://doi.org/10.1007%2F978-90-481-2448-0_24) . ISBN 978-90-481-2448-0.
14. "Midomafetamine: Uses, Interactions, Mechanism of Action" (<https://go.drugbank.com/drugs/DB01454>) . *DrugBank Online*. 31 July 2007. Retrieved 11 December 2024.

15. Carvalho M, Carmo H, Costa VM, Capela JP, Pontes H, Remião F, et al. (August 2012). "Toxicity of amphetamines: an update". *Archives of Toxicology*. **86** (8): 1167–1231. Bibcode:2012ArTox..86.1167C (<https://ui.adsabs.harvard.edu/abs/2012ArTox..86.1167C>) . doi:10.1007/s00204-012-0815-5 (<https://doi.org/10.1007%2Fs00204-012-0815-5>) . PMID 22392347 (<https://pubmed.ncbi.nlm.nih.gov/22392347>) . S2CID 2873101 (<https://api.semanticscholar.org/CorpusID:2873101>) .
16. Nakagawa H, Makino Y, Yoshida Y, Ohta S, Hirobe M (August 1993). "A novel metabolite of 3,4-methylenedioxyamphetamine (MDA): formation of 3-methyl-6,7-methylenedioxy-1,2,3,4-tetrahydroisoquinoline and its pharmacological effect". *Biological & Pharmaceutical Bulletin*. **16** (8): 761–764. doi:10.1248/bpb.16.579 (<https://doi.org/10.1248%2Fbpb.16.579>) . PMID 8364510 (<https://pubmed.ncbi.nlm.nih.gov/8364510>) . S2CID 28148811 (<https://api.semanticscholar.org/CorpusID:28148811>) .
17. Straumann I, Avedisian I, Klaiber A, Varghese N, Eckert A, Rudin D, et al. (August 2024). "Acute effects of R-MDMA, S-MDMA, and racemic MDMA in a randomized double-blind cross-over trial in healthy participants" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11631982>) . *Neuropsychopharmacology*. **50** (2): 362–371. doi:10.1038/s41386-024-01972-6 (<https://doi.org/10.1038%2Fs41386-024-01972-6>) . PMC 11631982 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11631982>) . PMID 39179638 (<https://pubmed.ncbi.nlm.nih.gov/39179638>) .
18. Dunlap LE, Andrews AM, Olson DE (October 2018). "Dark Classics in Chemical Neuroscience: 3,4-Methylenedioxymethamphetamine" (<https://shaunlacob.com/wp-content/uploads/2020/12/DC-MDMA.pdf>) (PDF). *ACS Chemical Neuroscience*. **9** (10): 2408–2427. doi:10.1021/acscchemneuro.8b00155 (<https://doi.org/10.1021%2Facschemneuro.8b00155>) . PMC 6197894 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6197894>) . PMID 30001118 (<https://pubmed.ncbi.nlm.nih.gov/30001118>) .
19. Oeri HE (May 2021). "Beyond ecstasy: Alternative entactogens to 3,4-methylenedioxymethamphetamine with potential applications in psychotherapy" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8155739>) . *Journal of Psychopharmacology*. **35** (5). Oxford, England: 512–536. doi:10.1177/0269881120920420 (<https://doi.org/10.1177%2F0269881120920420>) . PMC 8155739 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8155739>) . PMID 32909493 (<https://pubmed.ncbi.nlm.nih.gov/32909493>) .
20. Palamar JJ (7 December 2016). "There's something about Molly: The underresearched yet popular powder form of ecstasy in the United States" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5578728>) . *Substance Abuse*. **38** (1): 15–17. doi:10.1080/08897077.2016.1267070 (<https://doi.org/10.1080%2F08897077.2016.1267070>) . PMC 5578728 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5578728>) . PMID 27925866 (<https://pubmed.ncbi.nlm.nih.gov/27925866>) .

21. Skaug HA, ed. (14 December 2020). "Hva er tryggest av molly og ecstasy?" (<https://www.ung.no/oss/rusmidler/462397.html>) [What is safer: molly or ecstasy?]. *Ung.no* (in Norwegian). Norwegian Directorate for Children, Youth and Family Affairs. Archived (<https://web.archive.org/web/20220811145918/https://www.ung.no/oss/rusmidler/462397.html>) from the original on 11 August 2022. Retrieved 20 June 2022. "MDMA er virkestoffet i både Molly-krystaller og Ecstasy-tabletter. (MDMA is the active substance in both Molly crystals and Ecstasy tablets)"
22. Green AR, Mehan AO, Elliott JM, O'Shea E, Colado MI (September 2003). "The pharmacology and clinical pharmacology of 3,4-methylenedioxymethamphetamine (MDMA, "ecstasy")". *Pharmacological Reviews*. **55** (3): 463–508. doi:10.1124/pr.55.3.3 (<https://doi.org/10.1124/pr.55.3.3>) . PMID 12869661 (<https://pubmed.ncbi.nlm.nih.gov/12869661>) .
23. Meyer JS (2013). "3,4-methylenedioxymethamphetamine (MDMA): current perspectives" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3931692>) . *Substance Abuse and Rehabilitation*. **4**: 83–99. doi:10.2147/SAR.S37258 (<https://doi.org/10.2147/SAR.S37258>) . PMC 3931692 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3931692>) . PMID 24648791 (<https://pubmed.ncbi.nlm.nih.gov/24648791>) .
24. Freudenmann RW, Öxler F, Bernschneider-Reif S (August 2006). "The origin of MDMA (ecstasy) revisited: the true story reconstructed from the original documents" (http://www.thedea.org/docs/2006_Freudenmann_22846_1.pdf) (PDF). *Addiction*. **101** (9). Abingdon, England: 1241–1245. doi:10.1111/j.1360-0443.2006.01511.x (<https://doi.org/10.1111/j.1360-0443.2006.01511.x>) . PMID 16911722 (<https://pubmed.ncbi.nlm.nih.gov/16911722>) . Archived (https://web.archive.org/web/20200922140222/http://www.thedea.org/docs/2006_Freudenmann_22846_1.pdf) (PDF) from the original on 22 September 2020. Retrieved 23 May 2019. "Although MDMA was, in fact, first synthesized at Merck in 1912, it was not tested pharmacologically because it was only an unimportant precursor in a new synthesis for haemostatic substances."
25. Anderson L, ed. (18 May 2014). "MDMA" (<https://www.drugs.com/illicit/mdma.html>) . *Drugs.com*. Drugsite Trust. Archived (<https://web.archive.org/web/20160323161153/http://www.drugs.com/illicit/mdma.html>) from the original on 23 March 2016. Retrieved 30 March 2016.
26. "DrugFacts: MDMA (Ecstasy/Molly)" (<https://www.drugabuse.gov/publications/drugfacts/mdma-ecstasymolly>) . *National Institute on Drug Abuse*. February 2016. Archived (<https://web.archive.org/web/20160323164024/https://www.drugabuse.gov/publications/drugfacts/mdma-ecstasymolly>) from the original on 23 March 2016. Retrieved 30 March 2016.

27. *Neuroscience of Psychoactive Substance Use and Dependence* (<https://books.google.com/books?id=G9OhG-dZdAwC&pg=PA97>) . World Health Organization. 2004. pp. 97–. ISBN 978-92-4-156235-5. Archived (<https://web.archive.org/web/20160428231233/https://books.google.com/books?id=G9OhG-dZdAwC&pg=PA97>) from the original on 28 April 2016.
28. *World Drug Report 2018* (https://www.unodc.org/wdr2018/prelaunch/WDR18_Booklet_1_EXSUM.pdf) (PDF). United Nations. June 2018. p. 7. ISBN 978-92-1-148304-8. Archived (https://web.archive.org/web/20180727030037/http://www.unodc.org/wdr2018/prelaunch/WDR18_Booklet_1_EXSUM.pdf) (PDF) from the original on 27 July 2018. Retrieved 14 July 2018.
29. "MDMA (Ecstasy/Molly)" (<https://www.drugabuse.gov/drugs-abuse/mdma-ecstasymolly>) . National Institute on Drug Abuse. Archived (<https://web.archive.org/web/20180715011009/https://www.drugabuse.gov/drugs-abuse/mdma-ecstasymolly>) from the original on 15 July 2018. Retrieved 14 July 2018.
30. White CM (March 2014). "How MDMA's pharmacology and pharmacokinetics drive desired effects and harms". *Journal of Clinical Pharmacology*. **54** (3): 245–252. doi:10.1002/jcph.266 (<https://doi.org/10.1002%2Fjcph.266>) . PMID 24431106 (<https://pubmed.ncbi.nlm.nih.gov/24431106>) . S2CID 6223741 (<https://api.semanticscholar.org/CorpusID:6223741>) .
31. Freye E (2009). *Pharmacology and Abuse of Cocaine, Amphetamines, Ecstasy and Related Designer Drugs: A comprehensive review on their mode of action, treatment of abuse and intoxication* (<https://books.google.com/books?id=OTAlolM3XlwC&pg=PA147>) . Springer Science & Business Media. p. 147. ISBN 978-90-481-2448-0. Archived (<https://web.archive.org/web/20230113000534/https://books.google.com/books?id=OTAlolM3XlwC&pg=PA147>) from the original on 13 January 2023. Retrieved 12 May 2020.
32. Lyles J, Cadet JL (May 2003). "Methylenedioxymethamphetamine (MDMA, Ecstasy) neurotoxicity: cellular and molecular mechanisms". *Brain Research. Brain Research Reviews*. **42** (2): 155–168. doi:10.1016/S0165-0173(03)00173-5 (<https://doi.org/10.1016%2FS0165-0173%2803%2900173-5>) . PMID 12738056 (<https://pubmed.ncbi.nlm.nih.gov/12738056>) . S2CID 45330713 (<https://api.semanticscholar.org/CorpusID:45330713>) .
33. Philipps D (1 May 2018). "Ecstasy as a Remedy for PTSD? You Probably Have Some Questions" (<https://ghostarchive.org/archive/20220101/https://www.nytimes.com/2018/05/01/us/ecstasy-molly-ptsd-mdma.html>) . *The New York Times*. Archived from the original (<https://www.nytimes.com/2018/05/01/us/ecstasy-molly-ptsd-mdma.html>) on 1 January 2022. Retrieved 14 July 2018.

34. Patel V (2010). *Mental and neurological public health a global perspective* (<https://books.google.com/books?id=8fihf3VWbclC&pg=PA57>) (1st ed.). San Diego, CA: Academic Press/Elsevier. p. 57. ISBN 978-0-12-381527-9. Archived (<https://web.archive.org/web/20170910234523/https://books.google.com/books?id=8fihf3VWbclC&pg=PA57>) from the original on 10 September 2017.
35. Yong AS, Bratuskins S, Sultani MS, Blakeley B, Davey CG, Bell JS (April 2025). "Safety and efficacy of methylenedioxymethamphetamine (MDMA)-assisted psychotherapy in post-traumatic stress disorder: An overview of systematic reviews and meta-analyses" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11924292>) . *The Australian and New Zealand Journal of Psychiatry*. **59** (4): 339–360. doi:10.1177/00048674251315642 (<https://doi.org/10.1177%2F00048674251315642>) . PMC 11924292 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11924292>) . PMID 39979849 (<https://pubmed.ncbi.nlm.nih.gov/39979849>) .
36. Nuwer R (3 May 2021). "A Psychedelic Drug Passes a Big Test for PTSD Treatment" (<https://www.nytimes.com/2021/05/03/health/mdma-approval.html>) . *The New York Times*. Archived (<https://web.archive.org/web/20210605190721/https://www.nytimes.com/2021/05/03/health/mdma-approval.html>) from the original on 5 June 2021. Retrieved 5 June 2021.
37. "Subsection 56(1) class exemption for practitioners, agents, pharmacists, persons in charge of a hospital, hospital employees, and licensed dealers to conduct activities with psilocybin and MDMA in relation to a special access program authorization" (<https://www.canada.ca/en/health-canada/services/health-concerns/controlled-substances-precursor-chemicals/policy-regulations/policy-documents/subsection-56-1-class-exemption-conducting-activities-psilocybin-mdma-special-access-program-authorization.html>) . *Health Canada*. 5 January 2022. Archived (<https://web.archive.org/web/20220211203137/https://www.canada.ca/en/health-canada/services/health-concerns/controlled-substances-precursor-chemicals/policy-regulations/policy-documents/subsection-56-1-class-exemption-conducting-activities-psilocybin-mdma-special-access-program-authorization.html>) from the original on 11 February 2022. Retrieved 20 February 2022.
38. "Change to classification of psilocybin and MDMA to enable prescribing by authorised psychiatrists" (<https://www.tga.gov.au/news/media-releases/change-classification-psilocybin-and-mdma-enable-prescribing-authorised-psychiatrists>) . 3 February 2023. Archived (<https://web.archive.org/web/20240204024230/https://www.tga.gov.au/news/media-releases/change-classification-psilocybin-and-mdma-enable-prescribing-authorised-psychiatrists>) from the original on 4 February 2024. Retrieved 4 February 2024.

39. Reynolds S (1999). *Generation Ecstasy: Into the World of Techno and Rave Culture* (<https://books.google.com/books?id=tGaRJiXe74UC&q=Generation+Ecstasy+MDMA&pg=PA81>) . Routledge. p. 81. ISBN 978-0-415-92373-6. Archived (<https://web.archive.org/web/20231108064150/https://books.google.com/books?id=tGaRJiXe74UC&q=Generation+Ecstasy+MDMA&pg=PA81#v=snippet&q=Generation%20Ecstasy%20MDMA&f=false>) from the original on 8 November 2023. Retrieved 13 October 2020.
40. McCrady BS, Epstein EE, eds. (2013). *Addictions: a comprehensive guidebook* (<https://books.google.com/books?id=MUYfAQAAQBAJ&pg=PA299>) (Second ed.). Oxford: Oxford University Press. p. 299. ISBN 978-0-19-975366-6. Archived (<https://web.archive.org/web/20230113000537/https://books.google.com/books?id=MUYfAQAAQBAJ&pg=PA299>) from the original on 13 January 2023. Retrieved 11 January 2017.
41. Zeifman RJ, Kettner H, Pagni BA, Mallard A, Roberts DE, Erritzoe D, et al. (August 2023). "Co-use of MDMA with psilocybin/LSD may buffer against challenging experiences and enhance positive experiences" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10444769>) . *Scientific Reports*. **13** (1) 13645. Bibcode:2023NatSR..1313645Z (<https://ui.adsabs.harvard.edu/abs/2023NatSR..1313645Z>) . doi:10.1038/s41598-023-40856-5 (<https://doi.org/10.1038/s41598-023-40856-5>) . PMC 10444769 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10444769>) . PMID 37608057 (<https://pubmed.ncbi.nlm.nih.gov/37608057>) .
42. Weiss S (16 February 2024). "Nexus Flipping: What Happens When You Combine MDMA and 2C-B" (<https://doubleblindmag.com/nexus-flipping/>) . *DoubleBlind Mag*. Retrieved 10 July 2024.
43. Hysek CM, Simmler LD, Nicola VG, Vischer N, Donzelli M, Krähenbühl S, et al. (4 May 2012). Laks J (ed.). "Duloxetine inhibits effects of MDMA ("ecstasy") in vitro and in humans in a randomized placebo-controlled laboratory study" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3344887>) . *PLOS ONE*. **7** (5) e36476. Bibcode:2012PLoSO...736476H (<https://ui.adsabs.harvard.edu/abs/2012PLoSO...736476H>) . doi:10.1371/journal.pone.0036476 (<https://doi.org/10.1371/journal.pone.0036476>) . PMC 3344887 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3344887>) . PMID 22574166 (<https://pubmed.ncbi.nlm.nih.gov/22574166>) . "Fig. 7 shows the mean PD effects of MDMA plotted against simultaneous plasma concentrations at the different time points (hysteresis loops). The increases in "any drug effect" (Fig. 7a) and MAP (Fig. 7b) returned to baseline within 6 h when MDMA concentrations were still high. This clockwise hysteresis indicates that a smaller MDMA effect was seen at a given plasma concentration later in time, indicating rapid acute pharmacodynamic tolerance, which was similarly described for cocaine [33]. [...] Figure 7. Pharmacokinetic-pharmacodynamic (PK-PD) relationship. MDMA effects are plotted against simultaneous MDMA plasma concentrations (a, b). The time of sampling is noted next to each point in minutes or hours after MDMA administration. The clockwise hysteresis indicates acute tolerance to the effects of MDMA."

44. Yuki F, Rie I, Miki K, Mitsuhiro W, Naotaka K, Kenichiro N (April 2013). "Warning against co-administration of 3,4-methylenedioxymethamphetamine (MDMA) with methamphetamine from the perspective of pharmacokinetic and pharmacodynamic evaluations in rat brain". *European Journal of Pharmaceutical Sciences*. **49** (1): 57–64. doi:10.1016/j.ejps.2013.01.014 (<https://doi.org/10.1016%2Fj.ejps.2013.01.014>) . PMID 23395913 (<https://pubmed.ncbi.nlm.nih.gov/23395913>) .
45. Hamida SB, Tracqui A, de Vasconcelos AP, Szwarc E, Lazarus C, Kelche C, et al. (July 2009). "Ethanol increases the distribution of MDMA to the rat brain: possible implications in the ethanol-induced potentiation of the psychostimulant effects of MDMA" (<https://doi.org/10.1017%2Fs1461145708009693>) . *The International Journal of Neuropsychopharmacology*. **12** (6): 749–759. doi:10.1017/s1461145708009693 (<https://doi.org/10.1017%2Fs1461145708009693>) . PMID 19046482 (<https://pubmed.ncbi.nlm.nih.gov/19046482>) . S2CID 24230367 (<https://api.semanticscholar.org/CorpusID:24230367>) .
46. Oesterheld JR, Armstrong SC, Cozza KL (1 March 2004). "Ecstasy: pharmacodynamic and pharmacokinetic interactions" (<https://doi.org/10.1176%2Fappi.psy.45.1.84>) . *Psychosomatics*. **45** (1): 84–87. doi:10.1176/appi.psy.45.1.84 (<https://doi.org/10.1176%2Fappi.psy.45.1.84>) . PMID 14709765 (<https://pubmed.ncbi.nlm.nih.gov/14709765>) .
47. Carpenter M, Berry H, Pelletier AL (May 2019). "Clinically Relevant Drug-Drug Interactions in Primary Care". *American Family Physician*. **99** (9): 558–564. PMID 31038898 (<https://pubmed.ncbi.nlm.nih.gov/31038898>) .
48. Climko RP, Roehrich H, Sweeney DR, Al-Razi J (1986). "Ecstasy: a review of MDMA and MDA". *International Journal of Psychiatry in Medicine*. **16** (4): 359–372. doi:10.2190/dcrp-u22m-aumd-d84h (<https://doi.org/10.2190%2Fdcrp-u22m-aumd-d84h>) . PMID 2881902 (<https://pubmed.ncbi.nlm.nih.gov/2881902>) . S2CID 31902958 (<https://api.semanticscholar.org/CorpusID:31902958>) .
49. Wan W (6 August 2017). "Ecstasy could be 'breakthrough' therapy for soldiers, others suffering from PTSD" (https://www.washingtonpost.com/national/health-science/ecstasy-could-be-breakthrough-therapy-for-soldiers-others-suffering-from-ptsd/2017/08/26/009314ca-842f-11e7-b359-15a3617c767b_story.html) . *The Washington Post*. Archived (https://web.archive.org/web/20170828230759/https://www.washingtonpost.com/national/health-science/ecstasy-could-be-breakthrough-therapy-for-soldiers-others-suffering-from-ptsd/2017/08/26/009314ca-842f-11e7-b359-15a3617c767b_story.html) from the original on 28 August 2017. Retrieved 3 April 2021.

50. Kupferschmidt K (26 August 2017). "All clear for the decisive trial of ecstasy in PTSD patients" (<https://www.science.org/content/article/all-clear-decisive-trial-ecstasy-ptsd-patients>) . *Science (magazine)*. Archived (<https://web.archive.org/web/20211228220713/http://www.science.org/content/article/all-clear-decisive-trial-ecstasy-ptsd-patients>) from the original on 28 December 2021. Retrieved 3 April 2021.
51. Philipps D (1 May 2018). "Ecstasy as a Remedy for PTSD? You Probably Have Some Questions" (<https://ghostarchive.org/archive/20220101/https://www.nytimes.com/2018/05/01/us/ecstasy-molly-ptsd-mdma.html>) . *The New York Times*. Archived from the original (<https://www.nytimes.com/2018/05/01/us/ecstasy-molly-ptsd-mdma.html>) on 1 January 2022. Retrieved 14 July 2018.
52. Patel V (2010). *Mental and neurological public health a global perspective* (<https://books.google.com/books?id=8fihf3VWbclC&pg=PA57>) (1st ed.). San Diego, CA: Academic Press/Elsevier. p. 57. ISBN 978-0-12-381527-9. Archived (<https://web.archive.org/web/20170910234523/https://books.google.com/books?id=8fihf3VWbclC&pg=PA57>) from the original on 10 September 2017.
53. "Subsection 56(1) class exemption for practitioners, agents, pharmacists, persons in charge of a hospital, hospital employees, and licensed dealers to conduct activities with psilocybin and MDMA in relation to a special access program authorization" (<https://www.canada.ca/en/health-canada/services/health-concerns/controlled-substances-precursor-chemicals/policy-regulations/policy-documents/subsection-56-1-class-exemption-conducting-activities-psilocybin-mdma-special-access-program-authorization.html>) . *Health Canada*. 5 January 2022. Archived (<https://web.archive.org/web/20220211203137/https://www.canada.ca/en/health-canada/services/health-concerns/controlled-substances-precursor-chemicals/policy-regulations/policy-documents/subsection-56-1-class-exemption-conducting-activities-psilocybin-mdma-special-access-program-authorization.html>) from the original on 11 February 2022. Retrieved 20 February 2022.
54. "Change to classification of psilocybin and MDMA to enable prescribing by authorised psychiatrists" (<https://www.tga.gov.au/news/media-releases/change-classification-psilocybin-and-mdma-enable-prescribing-authorised-psychiatrists>) . 3 February 2023. Archived (<https://web.archive.org/web/20240204024230/https://www.tga.gov.au/news/media-releases/change-classification-psilocybin-and-mdma-enable-prescribing-authorised-psychiatrists>) from the original on 4 February 2024. Retrieved 4 February 2024.
55. Dupuis D, Veissière S (October 2022). "Culture, context, and ethics in the therapeutic use of hallucinogens: Psychedelics as active super-placebos?". *Transcultural Psychiatry*. **59** (5): 571–578. doi:10.1177/13634615221131465 (<https://doi.org/10.1177%2F13634615221131465>) . PMID 36263513 (<https://pubmed.ncbi.nlm.nih.gov/36263513>) .

56. van Elk M, Yaden DB (September 2022). "Pharmacological, neural, and psychological mechanisms underlying psychedelics: A critical review". *Neuroscience and Biobehavioral Reviews*. **140** 104793. doi:10.1016/j.neubiorev.2022.104793 (<https://doi.org/10.1016%2Fj.neubiorev.2022.104793>) . hdl:1887/3515020 (<https://hdl.handle.net/1887%2F3515020>) . PMID 35878791 (<https://pubmed.ncbi.nlm.nih.gov/35878791>) . "In addition, the strong prior expectations that many people have about psychedelics directly contribute to the psychedelic experience and as a consequence it has been suggested that psychedelics may act as a 'super-placebo' (Hartogsohn, 2016). Specifically, strong prior expectations (e.g., that a specific intervention will likely trigger a mystical experience) will increase the likelihood of having e.g., a mystical-type experience (Maij et al., 2019), and this placebo-effect is further boosted by the psychedelic-induced suggestibility."
57. Saunders N (29 July 1995). "The Agony and Ecstasy of God's path" (https://web.archive.org/web/20130424024232/http://csp.org/practices/entheogens/docs/saunders-ecstasy_rel.html) . Council on Spiritual Practices (CSP). Archived from the original (http://csp.org/practices/entheogens/docs/saunders-ecstasy_rel.html) on 24 April 2013. Retrieved 11 June 2011.
58. Watson L, Beck J (1991). "New age seekers: MDMA use as an adjunct to spiritual pursuit" (https://web.archive.org/web/20041122213135/http://research.lycaeum.org/researchpdfs/1991_watson_1.pdf) (PDF). *Journal of Psychoactive Drugs*. **23** (3): 261–270. doi:10.1080/02791072.1991.10471587 (<https://doi.org/10.1080%2F02791072.1991.10471587>) . PMID 1685513 (<https://pubmed.ncbi.nlm.nih.gov/1685513>) . Archived from the original (http://research.lycaeum.org/researchpdfs/1991_watson_1.pdf) (PDF) on 22 November 2004. Retrieved 28 April 2024.
59. "MDMA (3,4-Methylenedioxymethamphetamine)" (<https://web.archive.org/web/20160410100038/http://www.dea.gov/docs/2015%20NDTA%20Report.pdf>) (PDF). *2015 National Drug Threat Assessment Summary*. United States Department of Justice, Drug Enforcement Administration. October 2015. pp. 85–88. Archived from the original (<https://www.dea.gov/docs/2015%20NDTA%20Report.pdf>) (PDF) on 10 April 2016. Retrieved 10 April 2016.
60. *Molly Madness*. *Drugs, Inc.* (TV documentary). National Geographic Channel. 13 August 2014. ASIN B00LIC368M (<https://www.amazon.com/dp/B00LIC368M>) .
61. *Manic Molly*. *Drugs, Inc.* (TV documentary). National Geographic Channel. 10 December 2014. ASIN B00LIC368M (<https://www.amazon.com/dp/B00LIC368M>) .
62. Kelly M (20 June 2019). "Man arrested for possession of ecstasy tablets shaped like Wario" (<https://web.archive.org/web/20210624195901/https://www.nintendoenthusiast.com/man-arrested-for-possession-of-ecstasy-tablets-shaped-like-wario/>) . *Nintendo Enthusiast*. Archived from the original (<https://www.nintendoenthusiast.com/man-arrested-for-possession-of-ecstasy-tablets-shaped-like-wario/>) on 24 June 2021. Retrieved 17 June 2021.

63. "Groesbeck: Students caught with deceptively shaped Ecstasy pills" (<https://www.kwtx.com/content/news/Students-caught-with-deceptively-shaped-Ecstasy-pills-at-local-school-564188221.html>) . KWTX. 31 October 2019. Archived (<https://web.archive.org/web/20210624195735/https://www.kwtx.com/content/news/Students-caught-with-deceptively-shaped-Ecstasy-pills-at-local-school-564188221.html>) from the original on 24 June 2021. Retrieved 17 June 2021.
64. "3,4-Methylenedioxymethamphetamine" (<https://www.nlm.nih.gov/toxnet/index.html>) . *Hazardous Substances Data Bank*. National Library of Medicine. 28 August 2008. Archived (<https://web.archive.org/web/20190404201340/https://toxnet.nlm.nih.gov/cgi-bin/sis/search2/r?dbs%20hsdb%3A%40term%20%40rn%20%40rel%2042542-10-9>) from the original on 4 April 2019. Retrieved 22 August 2014.
65. Liechti ME, Gamma A, Vollenweider FX (March 2001). "Gender differences in the subjective effects of MDMA". *Psychopharmacology*. **154** (2): 161–168. doi:10.1007/s002130000648 (<https://doi.org/10.1007/s002130000648>) . PMID 11314678 (<https://pubmed.ncbi.nlm.nih.gov/11314678>) . S2CID 20251888 (<https://api.semanticscholar.org/CorpusID:20251888>) .
66. Greene SL, Kerr F, Braitberg G (October 2008). "Review article: amphetamines and related drugs of abuse". *Emergency Medicine Australasia*. **20** (5): 391–402. doi:10.1111/j.1742-6723.2008.01114.x (<https://doi.org/10.1111/j.1742-6723.2008.01114.x>) . PMID 18973636 (<https://pubmed.ncbi.nlm.nih.gov/18973636>) . S2CID 20755466 (<https://api.semanticscholar.org/CorpusID:20755466>) .
67. Landriscina F (1995). "MDMA and the states of Consciousness". *Eleusis*. **2**: 3–9.
68. Baggott MJ, Kirkpatrick MG, Bedi G, de Wit H (June 2015). "Intimate insight: MDMA changes how people talk about significant others" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4698152>) . *Journal of Psychopharmacology*. **29** (6). Oxford, England: 669–677. doi:10.1177/0269881115581962 (<https://doi.org/10.1177/0269881115581962>) . PMC 4698152 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4698152>) . PMID 25922420 (<https://pubmed.ncbi.nlm.nih.gov/25922420>) .
69. Schmid Y, Hysek CM, Simmler LD, Crockett MJ, Quednow BB, Liechti ME (September 2014). "Differential effects of MDMA and methylphenidate on social cognition" (https://edoc.unibas.ch/42235/1/20160316152928_56e96dc8bdaad.pdf) (PDF). *Journal of Psychopharmacology*. **28** (9). Oxford, England: 847–856. doi:10.1177/0269881114542454 (<https://doi.org/10.1177/0269881114542454>) . PMID 25052243 (<https://pubmed.ncbi.nlm.nih.gov/25052243>) . S2CID 25713943 (<https://api.semanticscholar.org/CorpusID:25713943>) . Archived (https://web.archive.org/web/20200917151735/https://edoc.unibas.ch/42235/1/20160316152928_56e96dc8bdaad.pdf) (PDF) from the original on 17 September 2020. Retrieved 29 June 2019.

70. Wardle MC, de Wit H (October 2014). "MDMA alters emotional processing and facilitates positive social interaction" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4194242>) . *Psychopharmacology*. **231** (21): 4219–4229. doi:10.1007/s00213-014-3570-x (<https://doi.org/10.1007/s00213-014-3570-x>) . PMC 4194242 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4194242>) . PMID 24728603 (<https://pubmed.ncbi.nlm.nih.gov/24728603>) .
71. Bravo GL (2001). "What does MDMA feel like?". In Holland J (ed.). *Ecstasy: The complete guide. A comprehensive look at the risks and benefits of MDMA*. Rochester: Park Street Press.
72. Metzner R (2005). "Psychedelic, Psychoactive, and Addictive Drugs and States of Consciousness" (<https://web.archive.org/web/20171009092337/http://www.greenearthfound.org/write/psychedelic.html>) . In Earleywine M (ed.). *Mind-Altering Drugs: The Science of Subjective Experience*. New York: Oxford University. Archived from the original (<http://www.greenearthfound.org/write/psychedelic.html>) on 9 October 2017. Retrieved 8 October 2017.
73. Kamilar-Britt P, Bedi G (October 2015). "The prosocial effects of 3,4-methylenedioxymethamphetamine (MDMA): Controlled studies in humans and laboratory animals" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4678620>) . *Neuroscience and Biobehavioral Reviews*. **57**: 433–446. doi:10.1016/j.neubiorev.2015.08.016 (<https://doi.org/10.1016/j.neubiorev.2015.08.016>) . PMC 4678620 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4678620>) . PMID 26408071 (<https://pubmed.ncbi.nlm.nih.gov/26408071>) .
74. Keane M (February 2014). "Recognising and managing acute hyponatraemia". *Emergency Nurse*. **21** (9): 32–6, quiz 37. doi:10.7748/en2014.02.21.9.32.e1128 (<https://doi.org/10.7748/en2014.02.21.9.32.e1128>) . PMID 24494770 (<https://pubmed.ncbi.nlm.nih.gov/24494770>) .
75. Spauwen LW, Niekamp AM, Hoebe CJ, Dukers-Muijters NH (February 2015). "Drug use, sexual risk behaviour and sexually transmitted infections among swingers: a cross-sectional study in The Netherlands" (<https://doi.org/10.1136/sextrans-2014-051626>) . *Sexually Transmitted Infections*. **91** (1): 31–36. doi:10.1136/sextrans-2014-051626 (<https://doi.org/10.1136/sextrans-2014-051626>) . PMID 25342812 (<https://pubmed.ncbi.nlm.nih.gov/25342812>) . "It is known that some recreational drugs (eg, MDMA or GHB) may hamper the potential to ejaculate or maintain an erection."
76. Hahn IH (25 March 2015). "MDMA Toxicity: Background, Pathophysiology, Epidemiology" (<http://emedicine.medscape.com/article/821572-overview#showall>) . *Medscape*. Archived (<https://web.archive.org/web/20160511222711/http://emedicine.medscape.com/article/821572-overview#showall>) from the original on 11 May 2016. Retrieved 14 May 2016.

77. Parrott AC (2012). "13. MDMA and LSD" (<https://books.google.com/books?id=yGeBj9U6Za8C>) . In Verster J, Brady K, Galanter M, Conrod P (eds.). *Drug Abuse and Addiction in Medical Illness: Causes, Consequences and Treatment*. Springer Science & Business Media. p. 179. ISBN 978-1-4614-3375-0.
78. Garg A, Kapoor S, Goel M, Chopra S, Chopra M, Kapoor A, et al. (2015). "Functional Magnetic Resonance Imaging in Abstinent MDMA Users: A Review". *Current Drug Abuse Reviews*. **8** (1): 15–25. doi:10.2174/1874473708666150303115833 (<https://doi.org/10.2174%2F1874473708666150303115833>) . PMID 25731754 (<https://pubmed.ncbi.nlm.nih.gov/25731754>) .
79. Mueller F, Lenz C, Steiner M, Dolder PC, Walter M, Lang UE, et al. (March 2016). "Neuroimaging in moderate MDMA use: A systematic review" (<https://doi.org/10.1016%2Fj.neubiorev.2015.12.010>) . *Neuroscience and Biobehavioral Reviews*. **62**: 21–34. doi:10.1016/j.neubiorev.2015.12.010 (<https://doi.org/10.1016%2Fj.neubiorev.2015.12.010>) . PMID 26746590 (<https://pubmed.ncbi.nlm.nih.gov/26746590>) .
80. Gouzoulis-Mayfrank E, Daumann J (2009). "Neurotoxicity of drugs of abuse—the case of methylenedioxyamphetamines (MDMA, ecstasy), and amphetamines" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3181923>) . *Dialogues in Clinical Neuroscience*. **11** (3): 305–317. doi:10.31887/DCNS.2009.11.3/egmayfrank (<https://doi.org/10.31887%2FDCNS.2009.11.3%2Fegmayfrank>) . PMC 3181923 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3181923>) . PMID 19877498 (<https://pubmed.ncbi.nlm.nih.gov/19877498>) .
81. Halpin LE, Collins SA, Yamamoto BK (February 2014). "Neurotoxicity of methamphetamine and 3,4-methylenedioxymethamphetamine" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3870191>) . *Life Sciences*. **97** (1): 37–44. doi:10.1016/j.lfs.2013.07.014 (<https://doi.org/10.1016%2Fj.lfs.2013.07.014>) . PMC 3870191 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3870191>) . PMID 23892199 (<https://pubmed.ncbi.nlm.nih.gov/23892199>) . "In contrast, MDMA produces damage to serotonergic, but not dopaminergic axon terminals in the striatum, hippocampus, and prefrontal cortex (Battaglia et al., 1987, O'Hearn et al., 1988). The damage associated with Meth and MDMA has been shown to persist for at least 2 years in rodents, non-human primates and humans (Seiden et al., 1988, Woolverton et al., 1989, McCann et al., 1998, Volkow et al., 2001a, McCann et al., 2005)"

82. Szigeti B, Winstock AR, Erritzoe D, Maier LJ (July 2018). "Are ecstasy induced serotonergic alterations overestimated for the majority of users?". *Journal of Psychopharmacology*. **32** (7). Oxford, England: 741–748. doi:10.1177/0269881118767646 (<https://doi.org/10.1177%2F0269881118767646>) . PMID 29733742 (<https://pubmed.ncbi.nlm.nih.gov/29733742>) . S2CID 13660975 (<https://api.semanticscholar.org/CorpusID:13660975>) . "Given the dose-response relationship between MDMA exposure and SERT reductions and the statistically non-significant SERT binding differences for users with use levels similar to the majority of real-life users, it can be speculated that SERT levels may not be significantly affected for most recreational ecstasy users."
83. Roberts CA, Jones A, Montgomery C (April 2016). "Meta-analysis of molecular imaging of serotonin transporters in ecstasy/polydrug users" (<https://doi.org/10.1016%2Fj.neubiorev.2016.02.003>) . *Neuroscience and Biobehavioral Reviews*. **63**: 158–167. doi:10.1016/j.neubiorev.2016.02.003 (<https://doi.org/10.1016%2Fj.neubiorev.2016.02.003>) . PMID 26855234 (<https://pubmed.ncbi.nlm.nih.gov/26855234>) .
84. Parrott AC (2014). "The potential dangers of using MDMA for psychotherapy" (<https://www.researchgate.net/publication/262381558>) . *Journal of Psychoactive Drugs*. **46** (1): 37–43. doi:10.1080/02791072.2014.873690 (<https://doi.org/10.1080%2F02791072.2014.873690>) . PMID 24830184 (<https://pubmed.ncbi.nlm.nih.gov/24830184>) . S2CID 23485480 (<https://api.semanticscholar.org/CorpusID:23485480>) .
85. Rogers G, Elston J, Garside R, Roome C, Taylor R, Younger P, et al. (January 2009). "The harmful health effects of recreational ecstasy: a systematic review of observational evidence" (<https://doi.org/10.3310%2Fhta13050>) . *Health Technology Assessment*. **13** (6). Winchester, England: iii–iv, ix–xii, 1–315. doi:10.3310/hta13050 (<https://doi.org/10.3310%2Fhta13050>) . hdl:10871/11534 (<https://hdl.handle.net/10871%2F11534>) . PMID 19195429 (<https://pubmed.ncbi.nlm.nih.gov/19195429>) .
86. Kuypers KP, Theunissen EL, van Wel JH, de Sousa Fernandes Perna EB, Linssen A, Sambeth A, et al. (2016). "Verbal Memory Impairment in Polydrug Ecstasy Users: A Clinical Perspective" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4764468>) . *PLOS ONE*. **11** (2) e0149438. Bibcode:2016PLoSO..1149438K (<https://ui.adsabs.harvard.edu/abs/2016PLoSO..1149438K>) . doi:10.1371/journal.pone.0149438 (<https://doi.org/10.1371%2Fjournal.pone.0149438>) . PMC 4764468 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4764468>) . PMID 26907605 (<https://pubmed.ncbi.nlm.nih.gov/26907605>) .
87. Laws KR, Kokkalis J (August 2007). "Ecstasy (MDMA) and memory function: a meta-analytic update". *Human Psychopharmacology*. **22** (6): 381–388. doi:10.1002/hup.857 (<https://doi.org/10.1002%2Fhup.857>) . PMID 17621368 (<https://pubmed.ncbi.nlm.nih.gov/17621368>) . S2CID 25353240 (<https://api.semanticscholar.org/CorpusID:25353240>) .

88. Kousik SM, Napier TC, Carvey PM (2012). "The effects of psychostimulant drugs on blood brain barrier function and neuroinflammation" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3386512>) . *Frontiers in Pharmacology*. **3**: 121. doi:10.3389/fphar.2012.00121 (<https://doi.org/10.3389%2Ffphar.2012.00121>) . PMC 3386512 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3386512>) . PMID 22754527 (<https://pubmed.ncbi.nlm.nih.gov/22754527>) .
89. McMillan B, Starr C (2014). *Human biology* (10th ed.). Belmont, CA: Brooks/Cole Cengage Learning. ISBN 978-1-133-59916-6.
90. Boyle NT, Connor TJ (September 2010). "Methylenedioxymethamphetamine ('Ecstasy')-induced immunosuppression: a cause for concern?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2962814>) . *British Journal of Pharmacology*. **161** (1): 17–32. doi:10.1111/J.1476-5381.2010.00899.X (<https://doi.org/10.1111%2FJ.1476-5381.2010.00899.X>) . PMC 2962814 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2962814>) . PMID 20718737 (<https://pubmed.ncbi.nlm.nih.gov/20718737>) .
91. Cavero I, Guillon JM (2014). "Safety Pharmacology assessment of drugs with biased 5-HT(2B) receptor agonism mediating cardiac valvulopathy". *Journal of Pharmacological and Toxicological Methods*. **69** (2): 150–161. doi:10.1016/j.vascn.2013.12.004 (<https://doi.org/10.1016%2Fj.vascn.2013.12.004>) . PMID 24361689 (<https://pubmed.ncbi.nlm.nih.gov/24361689>) .
92. Padhariya K, Bhandare R, Canney D, Velingkar V (2017). "Cardiovascular Concern of 5-HT2B Receptor and Recent Vistas in the Development of Its Antagonists". *Cardiovascular & Hematological Disorders Drug Targets*. **17** (2): 86–104. doi:10.2174/1871529X17666170703115111 (<https://doi.org/10.2174%2F1871529X17666170703115111>) . PMID 28676029 (<https://pubmed.ncbi.nlm.nih.gov/28676029>) .
93. Koczor CA, Ludlow I, Hight RS, Jiao Z, Fields E, Ludaway T, et al. (November 2015). "Ecstasy (MDMA) Alters Cardiac Gene Expression and DNA Methylation: Implications for Circadian Rhythm Dysfunction in the Heart" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4731408>) . *Toxicological Sciences*. **148** (1): 183–191. doi:10.1093/toxsci/kfv170 (<https://doi.org/10.1093%2Ftoxsci%2Fkfv170>) . PMC 4731408 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4731408>) . PMID 26251327 (<https://pubmed.ncbi.nlm.nih.gov/26251327>) .

94. Nutt D, King LA, Saulsbury W, Blakemore C (March 2007). "Development of a rational scale to assess the harm of drugs of potential misuse". *Lancet*. **369** (9566). London, England: 1047–1053. Bibcode:2007Lanc..369.1047N (<https://ui.adsabs.harvard.edu/abs/2007Lanc..369.1047N>) . doi:10.1016/S0140-6736(07)60464-4 (<https://doi.org/10.1016%2FS0140-6736%2807%2960464-4>) . PMID 17382831 (<https://pubmed.ncbi.nlm.nih.gov/17382831>) . S2CID 5903121 (<https://api.semanticscholar.org/CorpusID:5903121>) . Lay summary: "Scientists want new drug rankings" (<https://news.bbc.co.uk/2/hi/health/6474053.stm>) . *BBC News*. 23 March 2007. Archived (<https://web.archive.org/web/20071202233524/http://news.bbc.co.uk/2/hi/health/6474053.stm>) from the original on 2 December 2007. Retrieved 4 April 2008.
95. Olausson P, Jentsch JD, Tronson N, Neve RL, Nestler EJ, Taylor JR (September 2006). "DeltaFosB in the nucleus accumbens regulates food-reinforced instrumental behavior and motivation" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6674495>) . *The Journal of Neuroscience*. **26** (36): 9196–9204. doi:10.1523/JNEUROSCI.1124-06.2006 (<https://doi.org/10.1523%2FJNEUROSCI.1124-06.2006>) . PMC 6674495 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6674495>) . PMID 16957076 (<https://pubmed.ncbi.nlm.nih.gov/16957076>) .
96. Robison AJ, Nestler EJ (October 2011). "Transcriptional and epigenetic mechanisms of addiction" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272277>) . *Nature Reviews. Neuroscience*. **12** (11): 623–637. doi:10.1038/nrn3111 (<https://doi.org/10.1038%2Fnrn3111>) . PMC 3272277 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3272277>) . PMID 21989194 (<https://pubmed.ncbi.nlm.nih.gov/21989194>) .
97. Mack AH, Brady KT, Miller SI, Frances RJ (12 May 2016). *Clinical Textbook of Addictive Disorders* (https://books.google.com/books?id=88W_CwAAQBAJ&pg=PA171) . Guilford Publications. p. 169. ISBN 978-1-4625-2169-2. "MDMA's addictive liability appears to be lower than that of other drugs of abuse...."
98. Favrod-Coune T, Broers B (July 2010). "The Health Effect of Psychostimulants: A Literature Review" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4036656>) . *Pharmaceuticals*. **3** (7). Basel, Switzerland: 2333–2361. doi:10.3390/ph3072333 (<https://doi.org/10.3390%2Fph3072333>) . PMC 4036656 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4036656>) . PMID 27713356 (<https://pubmed.ncbi.nlm.nih.gov/27713356>) . "It seems to present a smaller addiction potential than cocaine or methamphetamine."

99. Ries R, Miller SC, Fiellin DA (2009). *Principles of addiction medicine* (<https://books.google.com/books?id=j6GGBud8DXcC&pg=PA226>) (4th ed.). Philadelphia: Wolters Kluwer/Lippincott Williams & Wilkins. p. 226. ISBN 978-0-7817-7477-2. Archived (<https://web.archive.org/web/20230113000536/https://books.google.com/books?id=j6GGBud8DXcC&pg=PA226>) from the original on 13 January 2023. Retrieved 11 January 2017. "MDA and MDMA are less reinforcing than amphetamine..."
100. Steinkellner T, Freissmuth M, Sitte HH, Montgomery T (January 2011). "The ugly side of amphetamines: short- and long-term toxicity of 3,4-methylenedioxymethamphetamine (MDMA, 'Ecstasy'), methamphetamine and D-amphetamine" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4497800>) . *Biological Chemistry*. **392** (1–2): 103–115. doi:10.1515/BC.2011.016 (<https://doi.org/10.1515%2FBC.2011.016>) . PMC 4497800 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4497800>) . PMID 21194370 (<https://pubmed.ncbi.nlm.nih.gov/21194370>) . "...approximately 15% of routine MDMA users recently fit the diagnostic criteria for MDMA dependence according to the Diagnostic and Statistical Manual, fourth edition/DSMIV."
101. Mack AH, Brady KT, Miller SI, Frances RJ (12 May 2016). *Clinical Textbook of Addictive Disorders* (https://books.google.com/books?id=88W_CwAAQBAJ&q=mdma+addiction&pg=PA171) . Guilford Publications. p. 171. ISBN 978-1-4625-2169-2. Archived (https://web.archive.org/web/20230119130056/https://books.google.com/books?id=88W_CwAAQBAJ&q=mdma+addiction&pg=PA171) from the original on 19 January 2023. Retrieved 13 October 2020. "There are no known pharmacological treatments for MDMA addiction."
102. Vorhees CV (November 1997). "Methods for detecting long-term CNS dysfunction after prenatal exposure to neurotoxins". *Drug and Chemical Toxicology*. **20** (4): 387–399. doi:10.3109/01480549709003895 (<https://doi.org/10.3109%2F01480549709003895>) . PMID 9433666 (<https://pubmed.ncbi.nlm.nih.gov/9433666>) .
103. Meamar R, Karamali F, Sadeghi HM, Etebari M, Nasr-Esfahani MH, Baharvand H (June 2010). "Toxicity of ecstasy (MDMA) towards embryonic stem cell-derived cardiac and neural cells". *Toxicology in Vitro*. **24** (4): 1133–1138. Bibcode:2010ToxVi..24.1133M (<https://ui.adsabs.harvard.edu/abs/2010ToxVi..24.1133M>) . doi:10.1016/j.tiv.2010.03.005 (<https://doi.org/10.1016%2Fj.tiv.2010.03.005>) . PMID 20230888 (<https://pubmed.ncbi.nlm.nih.gov/20230888>) . "In summary, MDMA is a moderate teratogen that could influence cardiac and neuronal differentiation in the ESC model and these results are in concordance with previous in vivo and in vitro models."

104. Singer LT, Moore DG, Fulton S, Goodwin J, Turner JJ, Min MO, et al. (2012). "Neurobehavioral outcomes of infants exposed to MDMA (Ecstasy) and other recreational drugs during pregnancy" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3367027>) . *Neurotoxicology and Teratology*. **34** (3): 303–310. Bibcode:2012NTxT...34..303S (<https://ui.adsabs.harvard.edu/abs/2012NTxT...34..303S>) . doi:10.1016/j.ntt.2012.02.001 (<https://doi.org/10.1016%2Fj.ntt.2012.02.001>) . PMC 3367027 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3367027>) . PMID 22387807 (<https://pubmed.ncbi.nlm.nih.gov/22387807>) .
105. Lipton JW, Tolod EG, Thompson VB, Pei L, Paumier KL, Terpstra BT, et al. (October 2008). "3,4-Methylenedioxy-N-methamphetamine (ecstasy) promotes the survival of fetal dopamine neurons in culture" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2572681>) . *Neuropharmacology*. **55** (5): 851–859. doi:10.1016/j.neuropharm.2008.06.062 (<https://doi.org/10.1016%2Fj.neuropharm.2008.06.062>) . PMC 2572681 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2572681>) . PMID 18655796 (<https://pubmed.ncbi.nlm.nih.gov/18655796>) .
106. Vanattou-Saïfoudine N, McNamara R, Harkin A (November 2012). "Caffeine provokes adverse interactions with 3,4-methylenedioxymethamphetamine (MDMA, 'ecstasy') and related psychostimulants: mechanisms and mediators" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3492978>) . *British Journal of Pharmacology*. **167** (5): 946–959. doi:10.1111/j.1476-5381.2012.02065.x (<https://doi.org/10.1111%2Fj.1476-5381.2012.02065.x>) . PMC 3492978 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3492978>) . PMID 22671762 (<https://pubmed.ncbi.nlm.nih.gov/22671762>) .
107. Henríquez-Hernández LA, Rojas-Hernández J, Quintana-Hernández DJ, Borkel LF (February 2023). "Hofmann vs. Paracelsus: Do Psychedelics Defy the Basics of Toxicology?-A Systematic Review of the Main Ergolamines, Simple Tryptamines, and Phenylethylamines" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9963058>) . *Toxics*. **11** (2): 148. Bibcode:2023Toxic..11..148H (<https://ui.adsabs.harvard.edu/abs/2023Toxic..11..148H>) . doi:10.3390/toxics11020148 (<https://doi.org/10.3390%2Ftoxics11020148>) . PMC 9963058 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9963058>) . PMID 36851023 (<https://pubmed.ncbi.nlm.nih.gov/36851023>) .
108. Richards JR, Albertson TE, Derlet RW, Lange RA, Olson KR, Horowitz BZ (May 2015). "Treatment of toxicity from amphetamines, related derivatives, and analogues: a systematic clinical review". *Drug and Alcohol Dependence*. **150**: 1–13. doi:10.1016/j.drugalcdep.2015.01.040 (<https://doi.org/10.1016%2Fj.drugalcdep.2015.01.040>) . PMID 25724076 (<https://pubmed.ncbi.nlm.nih.gov/25724076>) .

109. Hysek C, Schmid Y, Rickli A, Simmler LD, Donzelli M, Grouzmann E, et al. (August 2012). "Carvedilol inhibits the cardiostimulant and thermogenic effects of MDMA in humans" (<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3448893>) . *British Journal of Pharmacology*. **166** (8): 2277–2288. doi:10.1111/j.1476-5381.2012.01936.x (<https://doi.org/10.1111%2Fj.1476-5381.2012.01936.x>) . PMC 3448893 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3448893>) . PMID 22404145 (<https://pubmed.ncbi.nlm.nih.gov/22404145>) .
110. Hall AP, Henry JA (June 2006). "Acute toxic effects of 'Ecstasy' (MDMA) and related compounds: overview of pathophysiology and clinical management" (<https://doi.org/10.1093%2Fbja%2Fael078>) . *British Journal of Anaesthesia*. **96** (6): 678–685. doi:10.1093/bja/ael078 (<https://doi.org/10.1093%2Fbja%2Fael078>) . PMID 16595612 (<http://pubmed.ncbi.nlm.nih.gov/16595612>) .
111. de la Torre R, Farré M, Roset PN, Pizarro N, Abanades S, Segura M, et al. (April 2004). "Human pharmacology of MDMA: pharmacokinetics, metabolism, and disposition" (https://web.archive.org/web/20140305194315/http://www.maps.org/w3pb/new/2004/2004_de_20593_2.pdf) (PDF). *Therapeutic Drug Monitoring*. **26** (2): 137–144. doi:10.1097/00007691-200404000-00009 (<https://doi.org/10.1097%2F00007691-200404000-00009>) . PMID 15228154 (<https://pubmed.ncbi.nlm.nih.gov/15228154>) . Archived from the original (http://www.maps.org/w3pb/new/2004/2004_de_20593_2.pdf) (PDF) on 5 March 2014. Retrieved 25 October 2009.
112. Kellum JA, Gunn SR, Singer M (2008). *Oxford American Handbook of Critical Care*. Oxford University Press. p. 464. ISBN 978-0-19-530528-9. OCLC 1003197730 (<https://search.worldcat.org/oclc/1003197730>) .
113. Chummun H, Tilley V, Ibe J (2010). "3,4-methylenedioxyamphetamine (ecstasy) use reduces cognition". *British Journal of Nursing*. **19** (2). Mark Allen Publishing: 94–100. PMID 20235382 (<https://pubmed.ncbi.nlm.nih.gov/20235382>) .
114. Pendergraft WF, Herlitz LC, Thornley-Brown D, Rosner M, Niles JL (November 2014). "Nephrotoxic effects of common and emerging drugs of abuse" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4220747>) . *Clinical Journal of the American Society of Nephrology*. **9** (11): 1996–2005. doi:10.2215/CJN.00360114 (<https://doi.org/10.2215%2FCJN.00360114>) . PMC 4220747 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4220747>) . PMID 25035273 (<https://pubmed.ncbi.nlm.nih.gov/25035273>) .
115. Silins E, Copeland J, Dillon P (August 2007). "Qualitative review of serotonin syndrome, ecstasy (MDMA) and the use of other serotonergic substances: hierarchy of risk". *The Australian and New Zealand Journal of Psychiatry*. **41** (8): 649–655. doi:10.1080/00048670701449237 (<https://doi.org/10.1080%2F00048670701449237>) . PMID 17620161 (<https://pubmed.ncbi.nlm.nih.gov/17620161>) . S2CID 25832516 (<https://api.semanticscholar.org/CorpusID:25832516>) .

116. Papaseit E, Pérez-Mañá C, Torrens M, Farré A, Poyatos L, Hladun O, et al. (May 2020). "MDMA interactions with pharmaceuticals and drugs of abuse". *Expert Opinion on Drug Metabolism & Toxicology*. **16** (5): 357–369. doi:10.1080/17425255.2020.1749262 (<https://doi.org/10.1080/17425255.2020.1749262>) . PMID 32228243 (<https://pubmed.ncbi.nlm.nih.gov/32228243>) . S2CID 214750903 (<https://api.semanticscholar.org/CorpusID:214750903>) .
117. Fonseca DA, Ribeiro DM, Tapadas M, Cotrim MD (July 2021). "Ecstasy (3,4-methylenedioxymethamphetamine): Cardiovascular effects and mechanisms". *European Journal of Pharmacology*. **903** 174156. doi:10.1016/j.ejphar.2021.174156 (<https://doi.org/10.1016/j.ejphar.2021.174156>) . PMID 33971177 (<https://pubmed.ncbi.nlm.nih.gov/33971177>) .
118. Schmid Y, Rickli A, Schaffner A, Duthaler U, Grouzmann E, Hysek CM, et al. (April 2015). "Interactions between bupropion and 3,4-methylenedioxymethamphetamine in healthy subjects". *The Journal of Pharmacology and Experimental Therapeutics*. **353** (1): 102–111. doi:10.1124/jpet.114.222356 (<https://doi.org/10.1124/jpet.114.222356>) . PMID 25655950 (<https://pubmed.ncbi.nlm.nih.gov/25655950>) .
119. Vuori E, Henry JA, Ojanperä I, Nieminen R, Savolainen T, Wahlsten P, et al. (March 2003). "Death following ingestion of MDMA (ecstasy) and moclobemide". *Addiction*. **98** (3). Abingdon, England: 365–368. doi:10.1046/j.1360-0443.2003.00292.x (<https://doi.org/10.1046/j.1360-0443.2003.00292.x>) . PMID 12603236 (<https://pubmed.ncbi.nlm.nih.gov/12603236>) .
120. Halberstadt AL, Nichols DE (2020). "Serotonin and serotonin receptors in hallucinogen action". *Handbook of the Behavioral Neurobiology of Serotonin*. Handbook of Behavioral Neuroscience. Vol. 31. pp. 843–863. doi:10.1016/B978-0-444-64125-0.00043-8 (<https://doi.org/10.1016/B978-0-444-64125-0.00043-8>) . ISBN 978-0-444-64125-0. ISSN 1569-7339 (<https://search.worldcat.org/issn/1569-7339>) . S2CID 241134396 (<https://api.semanticscholar.org/CorpusID:241134396>) .

121. Mendelson J, Baggott MJ, Li L, Coyle J, Galloway GP (2012). "Poster Session II (PII 1-111): PII-41. MDMA-Induced Increases in Blood Pressure Are Not Mediated by α -Adrenergic Mechanisms and Are Not Due To Elevated Peripheral Vascular Resistance". *Clinical Pharmacology & Therapeutics*. **91** (S1 [American Society for Clinical Pharmacology and Therapeutics Abstract of papers, 2012 Annual Meeting Gaylord National Hotel and Convention Center National Harbor, Maryland March 14–17, 2012]): S51–S93 (S66–S66). doi:10.1038/clpt.2011.361 (<https://doi.org/10.1038%2Fclpt.2011.361>) . ISSN 0009-9236 (<https://search.worldcat.org/issn/0009-9236>) . "MDMA increased heart rate (HR) by 25 bpm ($p < .001$), [cardiac output (CO)] by 1.75 L/min ($p < 0.01$) but did not alter [stroke volume (SV)] or [systemic vascular resistance (SVR)]. Compared to MDMA alone the combination of MDMA + prazosin further increased HR by 24 bpm ($p < 0.001$) and CO by 3.3L/min ($p < 0.02$). MDMA increased systolic and diastolic blood pressure (SBP, DBP) by 26 mmHg ($p < 0.001$ each); prazosin attenuated MDMA effects on DBP by 9.3 mmHg ($p < 0.001$) but did not alter SBP. [...] MDMA increases HR, producing elevations in CO. The hypertensive effects of MDMA are not due to elevated peripheral vascular resistance and the blood pressure effects of MDMA are not attenuated by α -adrenergic blockade, suggesting that MDMA may produce CV effects through non- α -adrenergic mechanisms."
122. Liechti ME, Saur MR, Gamma A, Hell D, Vollenweider FX (October 2000). "Psychological and physiological effects of MDMA ("Ecstasy") after pretreatment with the 5-HT(2) antagonist ketanserin in healthy humans". *Neuropsychopharmacology*. **23** (4): 396–404. doi:10.1016/S0893-133X(00)00126-3 (<https://doi.org/10.1016%2FS0893-133X%2800%2900126-3>) . PMID 10989266 (<https://pubmed.ncbi.nlm.nih.gov/10989266>) .
123. Wsół A (December 2023). "Cardiovascular safety of psychedelic medicine: current status and future directions" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10661823>) . *Pharmacological Reports*. **75** (6): 1362–1380. doi:10.1007/s43440-023-00539-4 (<https://doi.org/10.1007%2FS43440-023-00539-4>) . PMC 10661823 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10661823>) . PMID 37874530 (<https://pubmed.ncbi.nlm.nih.gov/37874530>) .
124. Neumann J, Dhein S, Kirchhefer U, Hofmann B, Gergs U (2024). "Effects of hallucinogenic drugs on the human heart" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10869618>) . *Frontiers in Pharmacology*. **15** 1334218. doi:10.3389/fphar.2024.1334218 (<https://doi.org/10.3389%2Ffphar.2024.1334218>) . PMC 10869618 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10869618>) . PMID 38370480 (<https://pubmed.ncbi.nlm.nih.gov/38370480>) .

125. Ley L, Holze F, Arikci D, Becker AM, Straumann I, Klaiber A, et al. (October 2023). "Comparative acute effects of mescaline, lysergic acid diethylamide, and psilocybin in a randomized, double-blind, placebo-controlled cross-over study in healthy participants" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10517157>) . *Neuropsychopharmacology*. **48** (11): 1659–1667. doi:10.1038/s41386-023-01607-2 (<https://doi.org/10.1038%2Fs41386-023-01607-2>) . PMC 10517157 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10517157>) . PMID 37231080 (<https://pubmed.ncbi.nlm.nih.gov/37231080>) .
126. Hart XM, Spangemacher M, Defert J, Uchida H, Gründer G (April 2024). "Update Lessons from PET Imaging Part II: A Systematic Critical Review on Therapeutic Plasma Concentrations of Antidepressants". *Therapeutic Drug Monitoring*. **46** (2): 155–169. doi:10.1097/FTD.0000000000001142 (<https://doi.org/10.1097%2FFTD.0000000000001142>) . PMID 38287888 (<https://pubmed.ncbi.nlm.nih.gov/38287888>) .
127. Eap CB, Gründer G, Baumann P, Ansermot N, Conca A, Corruble E, et al. (October 2021). "Tools for optimising pharmacotherapy in psychiatry (therapeutic drug monitoring, molecular brain imaging and pharmacogenetic tests): focus on antidepressants" (https://serval.unil.ch/resource/serval:BIB_6FD14CC75A02.P001/REF.pdf) (PDF). *The World Journal of Biological Psychiatry*. **22** (8): 561–628. doi:10.1080/15622975.2021.1878427 (<https://doi.org/10.1080%2F15622975.2021.1878427>) . PMID 33977870 (<https://pubmed.ncbi.nlm.nih.gov/33977870>) . S2CID 234472488 (<https://api.semanticscholar.org/CorpusID:234472488>) . Archived (https://web.archive.org/web/20220505232003/https://serval.unil.ch/resource/serval:BIB_6FD14CC75A02.P001/REF.pdf) (PDF) from the original on 5 May 2022. Retrieved 10 April 2022.
128. Hysek CM, Simmler LD, Schillinger N, Meyer N, Schmid Y, Donzelli M, et al. (March 2014). "Pharmacokinetic and pharmacodynamic effects of methylphenidate and MDMA administered alone or in combination" (<https://edoc.unibas.ch/35606/2/371.full.pdf>) (PDF). *The International Journal of Neuropsychopharmacology*. **17** (3): 371–381. doi:10.1017/S1461145713001132 (<https://doi.org/10.1017%2FS1461145713001132>) . PMID 24103254 (<https://pubmed.ncbi.nlm.nih.gov/24103254>) .
129. Hysek CM, Vollenweider FX, Liechti ME (August 2010). "Effects of a beta-blocker on the cardiovascular response to MDMA (Ecstasy)" (<https://www.zora.uzh.ch/id/eprint/40820/1/Hysek.pdf>) (PDF). *Emergency Medicine Journal*. **27** (8): 586–589. doi:10.1136/emj.2009.079905 (<https://doi.org/10.1136%2Femj.2009.079905>) . PMID 20378736 (<https://pubmed.ncbi.nlm.nih.gov/20378736>) .

130. Hysek CM, Brugger R, Simmler LD, Bruggisser M, Donzelli M, Grouzmann E, et al. (February 2012). "Effects of the α_2 -adrenergic agonist clonidine on the pharmacodynamics and pharmacokinetics of 3,4-methylenedioxymethamphetamine in healthy volunteers". *The Journal of Pharmacology and Experimental Therapeutics*. **340** (2): 286–294. doi:10.1124/jpet.111.188425 (<https://doi.org/10.1124%2Fjpet.111.188425>) . PMID 22034656 (<https://pubmed.ncbi.nlm.nih.gov/22034656>) .
131. Hysek CM, Fink AE, Simmler LD, Donzelli M, Grouzmann E, Liechti ME (October 2013). " α_1 -Adrenergic receptors contribute to the acute effects of 3,4-methylenedioxymethamphetamine in humans". *Journal of Clinical Psychopharmacology*. **33** (5): 658–666. doi:10.1097/JCP.0b013e3182979d32 (<https://doi.org/10.1097%2FJCP.0b013e3182979d32>) . PMID 23857311 (<https://pubmed.ncbi.nlm.nih.gov/23857311>) .
132. Liechti ME, Vollenweider FX (July 2000). "Acute psychological and physiological effects of MDMA ("Ecstasy") after haloperidol pretreatment in healthy humans". *European Neuropsychopharmacology*. **10** (4): 289–295. doi:10.1016/s0924-977x(00)00086-9 (<https://doi.org/10.1016%2Fs0924-977x%2800%2900086-9>) . PMID 10871712 (<https://pubmed.ncbi.nlm.nih.gov/10871712>) .
133. Richards JR, Hollander JE, Ramoska EA, Fareed FN, Sand IC, Izquierdo Gómez MM, et al. (May 2017). " β -Blockers, Cocaine, and the Unopposed α -Stimulation Phenomenon". *Journal of Cardiovascular Pharmacology and Therapeutics*. **22** (3): 239–249. doi:10.1177/1074248416681644 (<https://doi.org/10.1177%2F1074248416681644>) . PMID 28399647 (<https://pubmed.ncbi.nlm.nih.gov/28399647>) .
134. "PDSP Database" (<https://pdsp.unc.edu/databases/pdsp.php?testFreeRadio=testFreeRadio&testLigand=MDMA&kiAllRadio=all&doQuery=Submit+Query>) . UNC (in Zulu). Retrieved 11 December 2024.
135. Liu T. "BindingDB BDBM50010588 (RS)-3,4-(methylenedioxy)methamphetamine::1-(1,3-Benzodioxol-5-yl)-N-methyl-2-propanamine::1-(1,3-benzodioxol-5-yl)-N-methylpropan-2-amine::3,4-methylenedioxymethamphetamine::CHEMBL43048::DL-(3,4-Methylenedioxy)methamphetamine::MDMA::N,alpha-dimethyl-1,3-benzodioxole-5-ethanamine::N-Methyl-3,4-methylenedioxyamphetamine::US11767305, Compound MDMA" (<https://www.bindingdb.org/rwd/bind/chemsearch/marvin/MolStructure.jsp?monomerid=50010588>) . BindingDB. Retrieved 11 December 2024.
136. Jain MK, Gumpfer RH, Slocum ST, Schmitz GP, Madsen JS, Tummino TA, et al. (July 2025). "The polypharmacology of psychedelics reveals multiple targets for potential therapeutics" (<https://www.cell.com/cms/10.1016/j.neuron.2025.06.012/attachment/7d8365fe-51f3-4a28-bf40-9999bec837f6/mmc11.pdf>) (PDF). *Neuron*. **113** (19): 3129–3142.e9. doi:10.1016/j.neuron.2025.06.012 (<https://doi.org/10.1016%2Fj.neuron.2025.06.012>) . PMID 40683247 (<https://pubmed.ncbi.nlm.nih.gov/40683247>) .

137. Ray TS (February 2010). "Psychedelics and the human receptorome" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2814854>) . *PLOS ONE*. **5** (2) e9019. Bibcode:2010PLoSO...5.9019R (<https://ui.adsabs.harvard.edu/abs/2010PLoSO...5.9019R>) . doi:10.1371/journal.pone.0009019 (<https://doi.org/10.1371%2Fjournal.pone.0009019>) . PMC 2814854 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2814854>) . PMID 20126400 (<https://pubmed.ncbi.nlm.nih.gov/20126400>) .
138. Simmler LD, Buser TA, Donzelli M, Schramm Y, Dieu LH, Huwyler J, et al. (January 2013). "Pharmacological characterization of designer cathinones in vitro" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3572571>) . *British Journal of Pharmacology*. **168** (2): 458–470. doi:10.1111/j.1476-5381.2012.02145.x (<https://doi.org/10.1111%2Fj.1476-5381.2012.02145.x>) . PMC 3572571 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3572571>) . PMID 22897747 (<https://pubmed.ncbi.nlm.nih.gov/22897747>) . "β-Keto-analogue cathinones also exhibited approximately 10-fold lower affinity for the TA1 receptor compared with their respective non-β-keto amphetamines. [...] Activation of TA1 receptors negatively modulates dopaminergic neurotransmission. Importantly, methamphetamine decreased DAT surface expression via a TA1 receptor-mediated mechanism and thereby reduced the presence of its own pharmacological target (Xie and Miller, 2009). MDMA and amphetamine have been shown to produce enhanced DA and 5-HT release and locomotor activity in TA1 receptor knockout mice compared with wild-type mice (Lindemann et al., 2008; Di Cara et al., 2011). Because methamphetamine and MDMA auto-inhibit their neurochemical and functional effects via TA1 receptors, low affinity for these receptors may result in stronger effects on monoamine systems by cathinones compared with the classic amphetamines."
139. Simmler LD, Rickli A, Hoener MC, Liechti ME (April 2014). "Monoamine transporter and receptor interaction profiles of a new series of designer cathinones". *Neuropharmacology*. **79**: 152–160. doi:10.1016/j.neuropharm.2013.11.008 (<https://doi.org/10.1016%2Fj.neuropharm.2013.11.008>) . PMID 24275046 (<https://pubmed.ncbi.nlm.nih.gov/24275046>) .
140. Rickli A, Kopf S, Hoener MC, Liechti ME (July 2015). "Pharmacological profile of novel psychoactive benzofurans" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4500375>) . *British Journal of Pharmacology*. **172** (13): 3412–3425. doi:10.1111/bph.13128 (<https://doi.org/10.1111%2Fbph.13128>) . PMC 4500375 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4500375>) . PMID 25765500 (<https://pubmed.ncbi.nlm.nih.gov/25765500>) .

141. Luethi D, Kolaczynska KE, Walter M, Suzuki M, Rice KC, Blough BE, et al. (July 2019). "Metabolites of the ring-substituted stimulants MDMA, methylenedioxymethamphetamine and MDPV differentially affect human monoaminergic systems" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8269116>) . *Journal of Psychopharmacology*. **33** (7). Oxford, England: 831–841. doi:10.1177/0269881119844185 (<https://doi.org/10.1177%2F0269881119844185>) . PMC 8269116 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8269116>) . PMID 31038382 (<https://pubmed.ncbi.nlm.nih.gov/31038382>) .
142. Eshleman AJ, Forster MJ, Wolfrum KM, Johnson RA, Janowsky A, Gatch MB (March 2014). "Behavioral and neurochemical pharmacology of six psychoactive substituted phenethylamines: mouse locomotion, rat drug discrimination and in vitro receptor and transporter binding and function" (<https://www.researchgate.net/publication/258061356>) . *Psychopharmacology*. **231** (5): 875–888. doi:10.1007/s00213-013-3303-6 (<https://doi.org/10.1007%2Fs00213-013-3303-6>) . PMC 3945162 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3945162>) . PMID 24142203 (<https://pubmed.ncbi.nlm.nih.gov/24142203>) .
143. Gainetdinov RR, Hoener MC, Berry MD (July 2018). "Trace Amines and Their Receptors" (<https://www.researchgate.net/publication/325975689>) . *Pharmacological Reviews*. **70** (3): 549–620. doi:10.1124/pr.117.015305 (<https://doi.org/10.1124%2Fpr.117.015305>) . PMID 29941461 (<https://pubmed.ncbi.nlm.nih.gov/29941461>) .

144. Simmler LD, Buchy D, Chaboz S, Hoener MC, Liechti ME (April 2016). "In Vitro Characterization of Psychoactive Substances at Rat, Mouse, and Human Trace Amine-Associated Receptor 1" (https://web.archive.org/web/20250509235235/https://d1wqtxts1xzle7.cloudfront.net/74120533/eae6c6e62565b82d46b4d111bbea0f77b9c2-libre.pdf?1635931703=&response-content-disposition=inline%3B+filename%3DIn_Vitro_Characterization_of_Psychoactiv.pdf&Expires=1746838268&Signature=Sy4fJ90yUhxs68314NxYsW5PAaNRBGePRu35WRR4PIF-3YC7Z~sLdnCn5wfqqbLg9bDEGdt~oW55ugMP3D3jgA0BoRI~~GOB0NQOwrtfUEQK1PQs1uuN9qg5Y1ct8z5NsABm44RgtukkwRMdU6f07OIfsQ68hOiFk129LI7UYqldxD2f1xhE2fTTfsxSpb8cMCJzHn7-ItqLdwnAUPFK7WggDljmY1kCnaHLwlxMwdJCAq8L6DYzSTg7pZkbR8qlou~GXbTPQt~gYpyZTJp5hgW-7V6K5wLIQ7Z2xE7B0f9wEfuc1W1QNafg125Tr-vvAe4LEGKXV58bnn1bpfWKw__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA) (PDF). *The Journal of Pharmacology and Experimental Therapeutics*. **357** (1): 134–144. doi:10.1124/jpet.115.229765 (<https://doi.org/10.1124%2Fjpet.115.229765>) . PMID 26791601 (<https://pubmed.ncbi.nlm.nih.gov/26791601>) . Archived from the original (https://d1wqtxts1xzle7.cloudfront.net/74120533/eae6c6e62565b82d46b4d111bbea0f77b9c2-libre.pdf?1635931703=&response-content-disposition=inline%3B+filename%3DIn_Vitro_Characterization_of_Psychoactiv.pdf&Expires=1746838268&Signature=Sy4fJ90yUhxs68314NxYsW5PAaNRBGePRu35WRR4PIF-3YC7Z~sLdnCn5wfqqbLg9bDEGdt~oW55ugMP3D3jgA0BoRI~~GOB0NQOwrtfUEQK1PQs1uuN9qg5Y1ct8z5NsABm44RgtukkwRMdU6f07OIfsQ68hOiFk129LI7UYqldxD2f1xhE2fTTfsxSpb8cMCJzHn7-ItqLdwnAUPFK7WggDljmY1kCnaHLwlxMwdJCAq8L6DYzSTg7pZkbR8qlou~GXbTPQt~gYpyZTJp5hgW-7V6K5wLIQ7Z2xE7B0f9wEfuc1W1QNafg125Tr-vvAe4LEGKXV58bnn1bpfWKw__&Key-Pair-Id=APKAJLOHF5GGSLRBV4ZA) (PDF) on 9 May 2025.
145. Sotnikova TD, Caron MG, Gainetdinov RR (August 2009). "Trace amine-associated receptors as emerging therapeutic targets" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2713119>) . *Molecular Pharmacology*. **76** (2): 229–235. doi:10.1124/mol.109.055970 (<https://doi.org/10.1124%2Fmol.109.055970>) . PMC 2713119 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2713119>) . PMID 19389919 (<https://pubmed.ncbi.nlm.nih.gov/19389919>) .
146. Nichols DE (2022). "Entactogens: How the Name for a Novel Class of Psychoactive Agents Originated" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8990025>) . *Frontiers in Psychiatry*. **13** 863088. doi:10.3389/fpsyt.2022.863088 (<https://doi.org/10.3389%2Ffpsyt.2022.863088>) . PMC 8990025 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8990025>) . PMID 35401275 (<https://pubmed.ncbi.nlm.nih.gov/35401275>) .

147. Docherty JR, Alsufyani HA (August 2021). "Pharmacology of Drugs Used as Stimulants" (<https://doi.org/10.1002%2Fjcph.1918>) . *Journal of Clinical Pharmacology*. **61** (Suppl 2): S53–S69. doi:10.1002/jcph.1918 (<https://doi.org/10.1002%2Fjcph.1918>) . PMID 34396557 (<https://pubmed.ncbi.nlm.nih.gov/34396557>) . "Receptor-mediated actions of amphetamine and other amphetamine derivatives [...] may involve trace amine-associated receptors (TAARs) at which amphetamine and MDMA also have significant potency.^{85–87} Many stimulants have potency at the rat TAAR1 in the micromolar range but tend to be about 5 to 10 times less potent at the human TAAR1, [...] Activation of the TAAR1 receptor causes inhibition of dopaminergic transmission in the mesocorticolimbic system, and TAAR1 agonists attenuated psychostimulant abuse-related behaviors.⁸⁹ It is likely that TAARs contribute to the actions of specific stimulants to modulate dopaminergic, serotonergic, and glutamate signaling,⁹⁰ and drugs acting on the TAAR1 may have therapeutic potential.⁹¹ In the periphery, stimulants such as MDMA and cathinone produce vasoconstriction, part of which may involve TAARs, although only relatively high concentrations produced vascular contractions resistant to a cocktail of monoamine antagonist drugs.⁸⁶"
148. Rothman RB, Baumann MH (October 2003). "Monoamine transporters and psychostimulant drugs". *European Journal of Pharmacology*. **479** (1–3): 23–40. doi:10.1016/j.ejphar.2003.08.054 (<https://doi.org/10.1016%2Fj.ejphar.2003.08.054>) . PMID 14612135 (<https://pubmed.ncbi.nlm.nih.gov/14612135>) .
149. Rothman RB, Baumann MH (2006). "Therapeutic potential of monoamine transporter substrates". *Current Topics in Medicinal Chemistry*. **6** (17): 1845–1859. doi:10.2174/156802606778249766 (<https://doi.org/10.2174%2F156802606778249766>) . PMID 17017961 (<https://pubmed.ncbi.nlm.nih.gov/17017961>) .
150. Sulzer D, Sonders MS, Poulsen NW, Galli A (April 2005). "Mechanisms of neurotransmitter release by amphetamines: a review". *Progress in Neurobiology*. **75** (6): 406–433. doi:10.1016/j.pneurobio.2005.04.003 (<https://doi.org/10.1016%2Fj.pneurobio.2005.04.003>) . PMID 15955613 (<https://pubmed.ncbi.nlm.nih.gov/15955613>) .
151. Reith ME, Gnegy ME (2020). "Molecular Mechanisms of Amphetamines". *Substance Use Disorders. Handbook of Experimental Pharmacology*. Vol. 258. pp. 265–297. doi:10.1007/164_2019_251 (https://doi.org/10.1007%2F164_2019_251) . ISBN 978-3-030-33678-3. PMID 31286212 (<https://pubmed.ncbi.nlm.nih.gov/31286212>) .

152. Vaughan RA, Henry LK, Foster JD, Brown CR (2024). "Post-translational mechanisms in psychostimulant-induced neurotransmitter efflux" (<https://books.google.com/books?id=2Sr6EAAQBAJ&pg=PA1>) . *Pharmacological Advances in Central Nervous System Stimulants*. Advances in Pharmacology. Vol. 99. pp. 1–33. doi:10.1016/bs.apha.2023.10.003 (<https://doi.org/10.1016%2Fbs.apha.2023.10.003>) . ISBN 978-0-443-21933-7. PMID 38467478 (<https://pubmed.ncbi.nlm.nih.gov/38467478>) .
153. Casey AB, Heifets BD (October 2025). "History, pharmacology and therapeutic mechanisms of 3,4-methylenedioxymethamphetamine (MDMA)" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12873472>) . *Br J Pharmacol*. doi:10.1111/bph.70225 (<https://doi.org/10.1111%2Fbph.70225>) . PMC 12873472 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12873472>) . PMID 41121647 (<https://pubmed.ncbi.nlm.nih.gov/41121647>) . "Unlike structurally related psychostimulants such as amphetamine and methamphetamine (Figure 1), which primarily release noradrenaline and dopamine over 5-HT via an identical mechanism, MDMA preferentially releases 5-HT and noradrenaline (Rothman et al., 2001) (Table 1). However, the selectivity by which MDMA releases monoamines is species-dependent. For example, in rat brain synaptosomes, MDMA preferentially releases 5-HT with modest selectivity (<10-fold) over noradrenaline and dopamine (Rothman et al., 2001; Setola et al., 2003). In contrast, studies using cloned human transporters in heterologous cells indicate that MDMA preferentially inhibits the release of noradrenaline with modest selectivity over 5-HT and dopamine (Han & Gu, 2006; Verrico et al., 2007). These species differences are most prominent for SERT, as MDMA more potently inhibits rodent over human SERT, with no species differences at NET and DAT (Han & Gu, 2006)."
154. Pitts EG, Curry DW, Hampshire KN, Young MB, Howell LL (February 2018). "(±)-MDMA and its enantiomers: potential therapeutic advantages of R(-)-MDMA". *Psychopharmacology*. **235** (2): 377–392. doi:10.1007/s00213-017-4812-5 (<https://doi.org/10.1007%2Fs00213-017-4812-5>) . PMID 29248945 (<https://pubmed.ncbi.nlm.nih.gov/29248945>) .
155. Setola V, Hufeisen SJ, Grande-Allen KJ, Vesely I, Glennon RA, Blough B, et al. (June 2003). "3,4-methylenedioxymethamphetamine (MDMA, "Ecstasy") induces fenfluramine-like proliferative actions on human cardiac valvular interstitial cells in vitro". *Molecular Pharmacology*. **63** (6): 1223–1229. doi:10.1124/mol.63.6.1223 (<https://doi.org/10.1124%2Fmol.63.6.1223>) . PMID 12761331 (<https://pubmed.ncbi.nlm.nih.gov/12761331>) . S2CID 839426 (<https://api.semanticscholar.org/CorpusID:839426>) .
156. Nash JF, Roth BL, Brodtkin JD, Nichols DE, Gudelsky GA (August 1994). "Effect of the R(-) and S(+) isomers of MDA and MDMA on phosphatidyl inositol turnover in cultured cells expressing 5-HT_{2A} or 5-HT_{2C} receptors". *Neuroscience Letters*. **177** (1–2): 111–115. doi:10.1016/0304-3940(94)90057-4 (<https://doi.org/10.1016%2F0304-3940%2894%2990057-4>) . PMID 7824160 (<https://pubmed.ncbi.nlm.nih.gov/7824160>) .

157. Lewin AH, Miller GM, Gilmour B (December 2011). "Trace amine-associated receptor 1 is a stereoselective binding site for compounds in the amphetamine class" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3236098>) . *Bioorganic & Medicinal Chemistry*. **19** (23): 7044–7048. doi:10.1016/j.bmc.2011.10.007 (<https://doi.org/10.1016%2Fj.bmc.2011.10.007>) . PMC 3236098 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3236098>) . PMID 22037049 (<https://pubmed.ncbi.nlm.nih.gov/22037049>) .
158. Espinoza S, Gainetdinov RR (2014). "Neuronal Functions and Emerging Pharmacology of TAAR1". *Taste and Smell*. Topics in Medicinal Chemistry. Vol. 23. Cham: Springer International Publishing. pp. 175–194. doi:10.1007/7355_2014_78 (https://doi.org/10.1007%2F7355_2014_78) . ISBN 978-3-319-48925-4. "Interestingly, the concentrations of amphetamine found to be necessary to activate TAAR1 are in line with what was found in drug abusers [3, 51, 52]. Thus, it is likely that some of the effects produced by amphetamines could be mediated by TAAR1. Indeed, in a study in mice, MDMA effects were found to be mediated in part by TAAR1, in a sense that MDMA auto-inhibits its neurochemical and functional actions [46]. Based on this and other studies (see other section), it has been suggested that TAAR1 could play a role in reward mechanisms and that amphetamine activity on TAAR1 counteracts their known behavioral and neurochemical effects mediated via dopamine neurotransmission."
159. Kuroпка P, Zawadzki M, Szpot P (May 2023). "A narrative review of the neuropharmacology of synthetic cathinones-Popular alternatives to classical drugs of abuse". *Human Psychopharmacology*. **38** (3) e2866. doi:10.1002/hup.2866 (<https://doi.org/10.1002%2Fhup.2866>) . PMID 36866677 (<https://pubmed.ncbi.nlm.nih.gov/36866677>) . "Another feature that distinguishes [synthetic cathinones (SCs)] from amphetamines is their negligible interaction with the trace amine associated receptor 1 (TAAR1). Activation of this receptor reduces the activity of dopaminergic neurones, thereby reducing psychostimulatory effects and addictive potential (Miller, 2011; Simmler et al., 2016). Amphetamines are potent agonists of this receptor, making them likely to self-inhibit their stimulating effects. In contrast, SCs show negligible activity towards TAAR1 (Kolaczynska et al., 2021; Rickli et al., 2015; Simmler et al., 2014, 2016). [...] It is worth noting, however, that for TAAR1 there is considerable species variability in its interaction with ligands, and it is possible that the in vitro activity of [rodent TAAR1 agonists] may not translate into activity in the human body (Simmler et al., 2016). The lack of self-regulation by TAAR1 may partly explain the higher addictive potential of SCs compared to amphetamines (Miller, 2011; Simmler et al., 2013)."

160. Di Cara B, Maggio R, Aloisi G, Rivet JM, Lundius EG, Yoshitake T, et al. (November 2011). "Genetic deletion of trace amine 1 receptors reveals their role in auto-inhibiting the actions of ecstasy (MDMA)" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6623861>) . *The Journal of Neuroscience*. **31** (47): 16928–16940. doi:10.1523/JNEUROSCI.2502-11.2011 (<https://doi.org/10.1523%2FJNEUROSCI.2502-11.2011>) . PMC 6623861 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6623861>) . PMID 22114263 (<https://pubmed.ncbi.nlm.nih.gov/22114263>) .
161. Rein B, Raymond K, Boustani C, Tuy S, Zhang J, St Laurent R, et al. (April 2024). "MDMA enhances empathy-like behaviors in mice via 5-HT release in the nucleus accumbens" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11042730>) . *Science Advances*. **10** (17) eadl6554. Bibcode:2024SciA...10L6554R (<https://ui.adsabs.harvard.edu/abs/2024SciA...10L6554R>) . doi:10.1126/sciadv.adl6554 (<https://doi.org/10.1126%2Fsciadv.adl6554>) . PMC 11042730 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11042730>) . PMID 38657057 (<https://pubmed.ncbi.nlm.nih.gov/38657057>) .
162. Esaki H, Sasaki Y, Nishitani N, Kamada H, Mukai S, Ohshima Y, et al. (May 2023). "Role of 5-HT1A receptors in the basolateral amygdala on 3,4-methylenedioxymethamphetamine-induced prosocial effects in mice". *European Journal of Pharmacology*. **946** 175653. doi:10.1016/j.ejphar.2023.175653 (<https://doi.org/10.1016%2Fj.ejphar.2023.175653>) . PMID 36907260 (<https://pubmed.ncbi.nlm.nih.gov/36907260>) .
163. Heifets BD, Salgado JS, Taylor MD, Hoerbelt P, Cardozo Pinto DF, Steinberg EE, et al. (December 2019). "Distinct neural mechanisms for the prosocial and rewarding properties of MDMA" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7123941>) . *Science Translational Medicine*. **11** (522) eaaw6435. doi:10.1126/scitranslmed.aaw6435 (<https://doi.org/10.1126%2Fscitranslmed.aaw6435>) . PMC 7123941 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7123941>) . PMID 31826983 (<https://pubmed.ncbi.nlm.nih.gov/31826983>) .
164. Pitts EG, Minerva AR, Chandler EB, Kohn JN, Logun MT, Sulima A, et al. (September 2017). "3,4-Methylenedioxymethamphetamine Increases Affiliative Behaviors in Squirrel Monkeys in a Serotonin 2A Receptor-Dependent Manner" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5561347>) . *Neuropsychopharmacology*. **42** (10): 1962–1971. doi:10.1038/npp.2017.80 (<https://doi.org/10.1038%2Fnpp.2017.80>) . PMC 5561347 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5561347>) . PMID 28425496 (<https://pubmed.ncbi.nlm.nih.gov/28425496>) .

165. Blanco-Gandía MC, Mateos-García A, García-Pardo MP, Montagud-Romero S, Rodríguez-Arias M, Miñarro J, et al. (September 2015). "Effect of drugs of abuse on social behaviour: a review of animal models". *Behavioural Pharmacology*. **26** (6): 541–570. doi:10.1097/FBP.0000000000000162 (<https://doi.org/10.1097%2FFBP.0000000000000162>) . PMID 26221831 (<https://pubmed.ncbi.nlm.nih.gov/26221831>) .
166. Heifets BD, Olson DE (January 2024). "Therapeutic mechanisms of psychedelics and entactogens" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10700553>) . *Neuropsychopharmacology*. **49** (1): 104–118. doi:10.1038/s41386-023-01666-5 (<https://doi.org/10.1038%2Fs41386-023-01666-5>) . PMC 10700553 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10700553>) . PMID 37488282 (<https://pubmed.ncbi.nlm.nih.gov/37488282>) .
167. Kaur H, Karabulut S, Gauld JW, Fagot SA, Holloway KN, Shaw HE, et al. (2023). "Balancing Therapeutic Efficacy and Safety of MDMA and Novel MDXX Analogues as Novel Treatments for Autism Spectrum Disorder" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11661495>) . *Psychedelic Medicine*. **1** (3). New Rochelle, N.Y.: 166–185. doi:10.1089/psymed.2023.0023 (<https://doi.org/10.1089%2Fpsymed.2023.0023>) . PMC 11661495 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11661495>) . PMID 40046567 (<https://pubmed.ncbi.nlm.nih.gov/40046567>) . "It is postulated that MDMA-induced neuronal apoptosis arises from directly stimulating the 5HT2A receptor. However, it is unclear whether MDMA binds here directly or whether one of its active metabolites (for example, MDA exhibits a 5-HT2A affinity almost 10-fold better than MDMA) is responsible.^{70,80,81} In addition, R-MDMA more potently activates 5-HT2A second messenger signaling, with S-MDMA having a minimal effect and racemic MDMA acting as a weak partial agonist."
168. Bankson MG, Cunningham KA (June 2001). "3,4-Methylenedioxymethamphetamine (MDMA) as a unique model of serotonin receptor function and serotonin-dopamine interactions" (<https://web.archive.org/web/20241209141749/https://bibliography.maps.org/resources/download/983>) . *The Journal of Pharmacology and Experimental Therapeutics*. **297** (3): 846–852. doi:10.1016/S0022-3565(24)29607-5 (<https://doi.org/10.1016%2FS0022-3565%2824%2929607-5>) . PMID 11356903 (<https://pubmed.ncbi.nlm.nih.gov/11356903>) . Archived from the original (<https://bibliography.maps.org/resources/download/983>) on 9 December 2024.

169. Baumann MH, Clark RD, Rothman RB (August 2008). "Locomotor stimulation produced by 3,4-methylenedioxymethamphetamine (MDMA) is correlated with dialysate levels of serotonin and dopamine in rat brain" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2491560>) . *Pharmacology, Biochemistry, and Behavior*. **90** (2): 208–217.
doi:10.1016/j.pbb.2008.02.018 (<https://doi.org/10.1016%2Fj.pbb.2008.02.018>) .
PMC 2491560 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2491560>) .
PMID 18403002 (<https://pubmed.ncbi.nlm.nih.gov/18403002>) . "Determining the role of specific 5-HT receptors in MDMA's effects in vivo is complicated by the presence of 14 different 5-HT receptor subtypes (Barnes and Sharp, 1999; Hoyer et al., 2002), many of which affect DA function (Alex and Pehek, 2007; Bubar and Cunningham, 2006). Nonetheless, there is general agreement that 5-HT1B, 5-HT2A and 5-HT2C receptor subtypes influence locomotor effects of MDMA. Hyperactivity produced by MDMA is mimicked by administration of the 5-HT1B agonist RU-24969 (Rempel et al., 1993), and 5-HT1B antagonists inhibit MDMA-induced ambulation (Callaway and Geyer, 1992b; Fletcher et al., 2002; McCreary et al., 1999). 5-HT2A antagonists reduce ambulation produced by MDMA (Fletcher et al., 2002; Kehne et al., 1996), whereas 5-HT2C antagonists markedly enhance it (Bankson and Cunningham, 2002; Fletcher et al., 2006). Taken together, these data reveal that 5-HT1B and 5-HT2A receptors facilitate, while 5-HT2C receptors suppress, forward locomotion evoked by MDMA administration. Importantly, the neural circuits underlying serotonergic modulation of MDMA-induced activity are largely unexplored."
170. Bankson MG, Cunningham KA (January 2002). "Pharmacological studies of the acute effects of (+)-3,4-methylenedioxymethamphetamine on locomotor activity: role of 5-HT(1B/1D) and 5-HT(2) receptors". *Neuropsychopharmacology*. **26** (1): 40–52.
doi:10.1016/S0893-133X(01)00345-1 (<https://doi.org/10.1016%2FS0893-133X%2801%2900345-1>) . PMID 11751031 (<https://pubmed.ncbi.nlm.nih.gov/11751031>) .

171. Cole JC, Sumnall HR (May 2003). "The pre-clinical behavioural pharmacology of 3,4-methylenedioxymethamphetamine (MDMA)". *Neuroscience and Biobehavioral Reviews*. **27** (3): 199–217. doi:10.1016/s0149-7634(03)00031-9 (<https://doi.org/10.1016%2Fs0149-7634%2803%2900031-9>) . PMID 12788333 (<https://pubmed.ncbi.nlm.nih.gov/12788333>) . "[...] 5-HT1B receptor agonists elicit locomotor hyperactivity similar to MDMA [247] and transgenic mice lacking the SERT or 5-HT1B receptor show no or reduced MDMA-induced locomotion and a behavioural pattern qualitatively reminiscent of other amphetamines [262,27]. Prior 5-HT depletion or pre-treatment of animals with SSRIs or 5-HT1B receptor antagonists block low dose MDMA hyperkinesis [42,44]. Activation of this receptor subtype may be fundamental in defining MDMA-specific locomotor activity. The 5-HT2A antagonist MDL 100,907 blocked high dose (20 mg/kg) but not low dose (3 mg/kg) MDMA-induced locomotion [157]. 5-HT2C receptor antagonists potentiated locomotor activation after low dose MDMA [20,115]. It has been proposed that unmasking of 5-HT1B receptor-mediated hyperactivity via 5-HT2C antagonism is only possible in the presence of the elevated DA and 5-HT concentrations seen after MDMA [19]."
172. Pomrenze MB, Vaillancourt S, Salgado JS, Raymond KB, Llorach P, Sacai H, et al. (July 2025). "5-HT2C receptors in the nucleus accumbens constrain the rewarding effects of MDMA" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12532699>) . *Molecular Psychiatry*. **30** (11): 5405–5416. doi:10.1038/s41380-025-03128-4 (<https://doi.org/10.1038%2Fs41380-025-03128-4>) . PMC 12532699 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12532699>) . PMID 40707786 (<https://pubmed.ncbi.nlm.nih.gov/40707786>) .
173. Canal CE, Murnane KS (January 2017). "The serotonin 5-HT2C receptor and the non-addictive nature of classic hallucinogens" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5445387>) . *Journal of Psychopharmacology*. **31** (1). Oxford, England: 127–143. doi:10.1177/0269881116677104 (<https://doi.org/10.1177%2F0269881116677104>) . PMC 5445387 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5445387>) . PMID 27903793 (<https://pubmed.ncbi.nlm.nih.gov/27903793>) .
174. Martinez-Price D, Krebs-Thomson K, Geyer M (1 January 2002). "Behavioral Psychopharmacology of MDMA and MDMA-Like Drugs: A Review of Human and Animal Studies". *Addiction Research & Theory*. **10** (1). Informa UK Limited: 43–67. doi:10.1080/16066350290001704 (<https://doi.org/10.1080%2F16066350290001704>) . ISSN 1606-6359 (<https://search.worldcat.org/issn/1606-6359>) .
175. Stove CP, De Letter EA, Piette MH, Lambert WE (August 2010). "Mice in ecstasy: advanced animal models in the study of MDMA". *Current Pharmaceutical Biotechnology*. **11** (5): 421–433. doi:10.2174/138920110791591508 (<https://doi.org/10.2174%2F138920110791591508>) . PMID 20420576 (<https://pubmed.ncbi.nlm.nih.gov/20420576>) .

176. Doly S, Valjent E, Setola V, Callebert J, Hervé D, Launay JM, et al. (March 2008). "Serotonin 5-HT_{2B} receptors are required for 3,4-methylenedioxymethamphetamine-induced hyperlocomotion and 5-HT release in vivo and in vitro" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6670669>) . *The Journal of Neuroscience*. **28** (11): 2933–2940. doi:10.1523/JNEUROSCI.5723-07.2008 (<https://doi.org/10.1523%2FJNEUROSCI.5723-07.2008>) . PMC 6670669 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6670669>) . PMID 18337424 (<https://pubmed.ncbi.nlm.nih.gov/18337424>) . "Here we show that acute pharmacological inhibition or genetic ablation of the 5-HT_{2B} receptor in mice completely abolishes MDMA-induced hyperlocomotion and 5-HT release in nucleus accumbens and ventral tegmental area. Furthermore, the 5-HT_{2B} receptor dependence of MDMA-stimulated release of endogenous 5-HT from superfused midbrain synaptosomes suggests that 5-HT_{2B} receptors act, unlike any other 5-HT receptor, presynaptically to affect MDMA-stimulated 5-HT release."
177. Doly S, Bertran-Gonzalez J, Callebert J, Bruneau A, Banas SM, Belmer A, et al. (23 November 2009). "Role of Serotonin via 5-HT_{2B} Receptors in the Reinforcing Effects of MDMA in Mice" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2775951>) . *PLOS ONE*. **4** (11) e7952. Bibcode:2009PLoSO...4.7952D (<https://ui.adsabs.harvard.edu/abs/2009PLoSO...4.7952D>) . doi:10.1371/journal.pone.0007952 (<https://doi.org/10.1371%2Fjournal.pone.0007952>) . ISSN 1932-6203 (<https://search.worldcat.org/issn/1932-6203>) . PMC 2775951 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2775951>) . PMID 19956756 (<https://pubmed.ncbi.nlm.nih.gov/19956756>) .
178. Belmer A, Quentin E, Diaz SL, Guiard BP, Fernandez SP, Doly S, et al. (2018). "Positive regulation of raphe serotonin neurons by serotonin 2B receptors" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5983540>) . *Neuropsychopharmacology*. **43** (7): 1623–1632. doi:10.1038/s41386-018-0013-0 (<https://doi.org/10.1038%2Fs41386-018-0013-0>) . ISSN 0893-133X (<https://search.worldcat.org/issn/0893-133X>) . PMC 5983540 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5983540>) . PMID 29453444 (<https://pubmed.ncbi.nlm.nih.gov/29453444>) .
179. Baggott M (23 June 2023). *Beyond Ecstasy: Progress in Developing and Understanding a Novel Class of Therapeutic Medicine* (<https://web.archive.org/web/20250109061208/https://2023.psychedelicscience.org/sessions/beyond-ecstasy-progress-in-developing-and-understanding-a-novel-class-of-therapeutic-medicine/>) . PS2023 [Psychedelic Science 2023, June 19–23, 2023, Denver, Colorado]. Denver, CO: Multidisciplinary Association for Psychedelic Studies. Archived from the original (<https://2023.psychedelicscience.org/sessions/beyond-ecstasy-progress-in-developing-and-understanding-a-novel-class-of-therapeutic-medicine/>) on 9 January 2025. Retrieved 9 January 2025.

180. "Better Than Ecstasy: Progress in Developing a Novel Class of Therapeutic with Matthew Baggott, PhD" (<https://www.youtube.com/watch?v=OnhJvKxwfZI&t=1048>) . YouTube. 6 March 2024. Retrieved 20 November 2024.
181. Rothman RB, Baumann MH (July 2002). "Therapeutic and adverse actions of serotonin transporter substrates". *Pharmacology & Therapeutics*. **95** (1): 73–88. doi:10.1016/s0163-7258(02)00234-6 (<https://doi.org/10.1016%2Fs0163-7258%2802%2900234-6>) . PMID 12163129 (<https://pubmed.ncbi.nlm.nih.gov/12163129>) .
182. Luethi D, Liechti ME (April 2020). "Designer drugs: mechanism of action and adverse effects" (<https://repositorium.meduniwien.ac.at/obvumwoa/content/titleinfo/5270457/full.pdf>) (PDF). *Archives of Toxicology*. **94** (4): 1085–1133. Bibcode:2020ArTox..94.1085L (<https://ui.adsabs.harvard.edu/abs/2020ArTox..94.1085L>) . doi:10.1007/s00204-020-02693-7 (<https://doi.org/10.1007%2Fs00204-020-02693-7>) . PMC 7225206 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7225206>) . PMID 32249347 (<https://pubmed.ncbi.nlm.nih.gov/32249347>) .
183. Liechti ME, Vollenweider FX (December 2001). "Which neuroreceptors mediate the subjective effects of MDMA in humans? A summary of mechanistic studies". *Human Psychopharmacology*. **16** (8): 589–598. doi:10.1002/hup.348 (<https://doi.org/10.1002%2Fhup.348>) . PMID 12404538 (<https://pubmed.ncbi.nlm.nih.gov/12404538>) .
184. Schenk S, Newcombe D (December 2018). "Methylenedioxymethamphetamine (MDMA) in Psychiatry: Pros, Cons, and Suggestions". *Journal of Clinical Psychopharmacology*. **38** (6): 632–638. doi:10.1097/JCP.0000000000000962 (<https://doi.org/10.1097%2FJCP.0000000000000962>) . PMID 30303861 (<https://pubmed.ncbi.nlm.nih.gov/30303861>) .
185. Puxty DJ, Ramaekers JG, de la Torre R, Farré M, Pizarro N, Pujadas M, et al. (2017). "MDMA-Induced Dissociative State not Mediated by the 5-HT_{2A} Receptor" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5504523>) . *Frontiers in Pharmacology*. **8** 455. doi:10.3389/fphar.2017.00455 (<https://doi.org/10.3389%2Ffphar.2017.00455>) . PMC 5504523 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5504523>) . PMID 28744219 (<https://pubmed.ncbi.nlm.nih.gov/28744219>) .
186. Meyer JS (2013). "3,4-methylenedioxymethamphetamine (MDMA): current perspectives" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3931692>) . *Substance Abuse and Rehabilitation*. **4**: 83–99. doi:10.2147/SAR.S37258 (<https://doi.org/10.2147%2FSAR.S37258>) . PMC 3931692 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3931692>) . PMID 24648791 (<https://pubmed.ncbi.nlm.nih.gov/24648791>) .

187. Liechti ME, Baumann C, Gamma A, Vollenweider FX (May 2000). "Acute psychological effects of 3,4-methylenedioxymethamphetamine (MDMA, "Ecstasy") are attenuated by the serotonin uptake inhibitor citalopram". *Neuropsychopharmacology*. **22** (5): 513–521. doi:10.1016/S0893-133X(99)00148-7 (<https://doi.org/10.1016%2FS0893-133X%2899%2900148-7>) . PMID 10731626 (<https://pubmed.ncbi.nlm.nih.gov/10731626>) .
188. Simmler LD, Liechti ME (2018). "Pharmacology of MDMA- and Amphetamine-Like New Psychoactive Substances". *New Psychoactive Substances*. Handbook of Experimental Pharmacology. Vol. 252. pp. 143–164. doi:10.1007/164_2018_113 (https://doi.org/10.1007%2F164_2018_113) . ISBN 978-3-030-10560-0. PMID 29633178 (<https://pubmed.ncbi.nlm.nih.gov/29633178>) . "MDMA is also a low-potency partial agonist of the 5-HT_{2A} receptor. Although not frequent, mild hallucinogen-like effects of MDMA have been reported, which may be attributable to 5-HT_{2A} agonism (Nichols 2004; Liechti et al. 2000). MDA, the active metabolite of MDMA (Hysek et al. 2011), shows a tenfold higher potency for 5-HT_{2A} agonism than MDMA (Rickli et al. 2015c). MDA likely contributes to the mode of action of MDMA and might contribute to the mild hallucinogenic effects of MDMA."
189. Nichols DE (2018). *Chemistry and Structure-Activity Relationships of Psychedelics* (https://bitnest.netfirms.com/external/10.1007/7854_2017_475) . Current Topics in Behavioral Neurosciences. Vol. 36. pp. 1–43. doi:10.1007/7854_2017_475 (https://doi.org/10.1007%2F7854_2017_475) . ISBN 978-3-662-55878-2. PMID 28401524 (<https://pubmed.ncbi.nlm.nih.gov/28401524>) . "Although the most active tryptamine hallucinogens are N,N-dialkylated, the phenethylamines generally cannot tolerate even a single N-substitution. Even small groups such as methyl or ethyl (see Table 2) abolish their hallucinogenic activity."
190. Jacob P, Shulgin AT (1994). "Structure-Activity Relationships of the Classic Hallucinogens and Their Analogs" (<https://web.archive.org/web/20250713011914/https://bibliography.maps.org/resources/download/11534>) . In Lin GC, Glennon RA (eds.). *Hallucinogens: An Update* (<https://archives.nida.nih.gov/sites/default/files/monograph146.pdf>) (PDF). National Institute on Drug Abuse Research Monograph Series. Vol. 146. National Institute on Drug Abuse. pp. 74–91. PMID 8742795 (<https://pubmed.ncbi.nlm.nih.gov/8742795>) . Archived from the original (<https://bibliography.maps.org/resources/download/11534>) on 13 July 2025. "[MDA] is also remarkable because the N-methyl homolog 3,4 (MDMA) has biological activity, although the nature of its action places it outside of this review. No other phenethylamine hallucinogen retains central activity on N-methylation."

191. Halberstadt AL, Geyer MA (2018). "Effect of Hallucinogens on Unconditioned Behavior". *Behavioral Neurobiology of Psychedelic Drugs*. Current Topics in Behavioral Neurosciences. Vol. 36. pp. 159–199. doi:10.1007/7854_2016_466 (https://doi.org/10.1007%2F7854_2016_466) . ISBN 978-3-662-55878-2. PMC 5787039 (<https://www.ncbi.nlm.nih.gov/pmc/article/s/PMC5787039>) . PMID 28224459 (<https://pubmed.ncbi.nlm.nih.gov/28224459>) . " [MDxx] have been assessed in head twitch studies. Racemic [MDA] and S-(+)-MDA reportedly induce WDS in monkeys and rats, respectively (Schlemmer and Davis 1986; Hiramatsu et al. 1989). Although [MDMA] does not induce the HTR in mice, both of the stereoisomers of MDMA have been shown to elicit the response (Fantegrossi et al. 2004, 2005b). 5-HT depletion inhibits the response to S-(+)-MDMA but does not alter the response to R-(–)-MDMA, suggesting the isomers act through different mechanisms (Fantegrossi et al. 2005b). This suggestion is consistent with the fact that S-(+)- and R-(–)-MDMA exhibit qualitatively distinct pharmacological profiles, with the S-(+)isomer working primarily as a monoamine releaser (Johnson et al. 1986; Baumann et al. 2008; Murnane et al. 2010) and the R-(–)-enantiomer acting directly through 5-HT_{2A} receptors (Lyon et al. 1986; Nash et al. 1994). In contrast to their effects in mice, Hiramatsu reported that S-(+)- and R-(–)-MDMA fail to produce WDS in rats (Hiramatsu et al. 1989). The discrepant findings with MDMA in mice and rats may reflect species differences in sensitivity to the HTR (see below for further discussion)."
192. Dunlap LE (2022). *Development of Non-Hallucinogenic Psychoplastogens* (<https://escholarship.org/uc/item/5qr3w0gm>) (Thesis). University of California, Davis. Retrieved 18 November 2024. "Finally, since R-MDMA is known to partially substitute for LSD in animal models we decided to test both compounds in the head twitch response assay (HTR) (FIG 3.3C).3 The HTR is a well-validated mouse model for predicting the hallucinogenic potential of test drugs. Serotonergic psychedelics will cause a rapid back and forth head movement in mice. The potency measured in the HTR assay has been shown to correlate very well with the human potencies of psychedelics.18 Neither R-MDMA or [...] produced any head twitches at all doses tested, suggesting that neither has high hallucinogenic potential."
193. Nichols DE (2016). "Psychedelics" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4813425>) . *Pharmacological Reviews*. **68** (2): 264–355. doi:10.1124/pr.115.011478 (<https://doi.org/10.1124%2Fpr.115.011478>) . ISSN 0031-6997 (<https://search.worldcat.org/issn/0031-6997>) . PMC 4813425 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4813425>) . PMID 26841800 (<https://pubmed.ncbi.nlm.nih.gov/26841800>) .
194. Nichols DE, Oberlender R (1989). "Structure-activity relationships of MDMA-like substances". *NIDA Research Monograph*. **94**: 1–29. PMID 2575223 (<https://pubmed.ncbi.nlm.nih.gov/2575223>) .

195. Nichols DE, Oberlender R (1990). "Structure-activity relationships of MDMA and related compounds: a new class of psychoactive drugs?". *Annals of the New York Academy of Sciences*. **600**: 613–23, discussion 623–5. doi:10.1007/978-1-4613-1485-1_7 (https://doi.org/10.1007/978-1-4613-1485-1_7) . PMID 1979214 (<https://pubmed.ncbi.nlm.nih.gov/1979214>) .
196. McIntyre RS (2023). "Serotonin 5-HT_{2B} receptor agonism and valvular heart disease: implications for the development of psilocybin and related agents". *Expert Opinion on Drug Safety*. **22** (10): 881–883. doi:10.1080/14740338.2023.2248883 (<https://doi.org/10.1080/14740338.2023.2248883>) . PMID 37581427 (<https://pubmed.ncbi.nlm.nih.gov/37581427>) .
197. Tagen M, Mantuani D, van Heerden L, Holstein A, Klumpers LE, Knowles R (September 2023). "The risk of chronic psychedelic and MDMA microdosing for valvular heart disease" (<https://unlimitedsciences.org/wp-content/uploads/2024/01/tagen-et-al-2023-the-risk-of-chronic-psychedelic-and-mdma-microdosing-for-valvular-heart-disease.pdf>) (PDF). *Journal of Psychopharmacology*. **37** (9). Oxford, England: 876–890. doi:10.1177/02698811231190865 (<https://doi.org/10.1177/02698811231190865>) . PMID 37572027 (<https://pubmed.ncbi.nlm.nih.gov/37572027>) . "[...] Both [MDMA and MDA] bind to the human 5-HT_{2B} receptor, although with a 5-fold lower K_i value for MDA compared to MDMA (Ray, 2010; Setola et al., 2003). Both compounds were agonists in an assay of PI hydrolysis, with MDA (EC₅₀=190nM) 10-fold more potent than MDMA (EC₅₀=2000 nM) in addition to greater intrinsic efficacy (90% vs 32%) (Setola et al., 2003). [...] A 50mg dose of MDMA resulted in a mean plasma C_{max} 266nM for MDMA and 28.5nM for MDA (de la Torre et al., 2000)."
198. Rothman RB, Baumann MH (May 2009). "Serotonergic drugs and valvular heart disease" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2695569>) . *Expert Opinion on Drug Safety*. **8** (3): 317–329. doi:10.1517/14740330902931524 (<https://doi.org/10.1517/14740330902931524>) . PMC 2695569 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2695569>) . PMID 19505264 (<https://pubmed.ncbi.nlm.nih.gov/19505264>) .
199. Rothman RB, Baumann MH (April 2002). "Serotonin releasing agents. Neurochemical, therapeutic and adverse effects". *Pharmacology, Biochemistry, and Behavior*. **71** (4): 825–836. doi:10.1016/s0091-3057(01)00669-4 ([https://doi.org/10.1016/s0091-3057\(01\)00669-4](https://doi.org/10.1016/s0091-3057(01)00669-4)) . PMID 11888573 (<https://pubmed.ncbi.nlm.nih.gov/11888573>) .

200. Costa G, Gołembowska K (January 2022). "Neurotoxicity of MDMA: Main effects and mechanisms" (<https://www.didyouno.fr/wp-content/uploads/2023/03/1-s2.0-S0014488621003022-main.pdf>) (PDF). *Experimental Neurology*. **347** 113894. doi:10.1016/j.expneurol.2021.113894 (<https://doi.org/10.1016%2Fj.expneurol.2021.113894>) . hdl:11584/325355 (<https://hdl.handle.net/11584%2F325355>) . PMID 34655576 (<https://pubmed.ncbi.nlm.nih.gov/34655576>) .
201. Sprague JE, Everman SL, Nichols DE (June 1998). "An integrated hypothesis for the serotonergic axonal loss induced by 3,4-methylenedioxymethamphetamine" (<https://www.researchgate.net/publication/13663847>) . *Neurotoxicology*. **19** (3): 427–441. PMID 9621349 (<https://pubmed.ncbi.nlm.nih.gov/9621349>) .
202. Green AR, Mehan AO, Elliott JM, O'Shea E, Colado MI (September 2003). "The pharmacology and clinical pharmacology of 3,4-methylenedioxymethamphetamine (MDMA, "ecstasy")". *Pharmacol Rev*. **55** (3): 463–508. doi:10.1124/pr.55.3.3 (<https://doi.org/10.1124%2Fpr.55.3.3>) . PMID 12869661 (<https://pubmed.ncbi.nlm.nih.gov/12869661>) .
203. Gudelsky GA, Yamamoto BK, Nash JF (November 1994). "Potentiation of 3,4-methylenedioxymethamphetamine-induced dopamine release and serotonin neurotoxicity by 5-HT₂ receptor agonists". *Eur J Pharmacol*. **264** (3): 325–330. doi:10.1016/0014-2999(94)90669-6 (<https://doi.org/10.1016%2F0014-2999%2894%2990669-6>) . PMID 7698172 (<https://pubmed.ncbi.nlm.nih.gov/7698172>) .
204. Schmidt CJ, Black CK, Abbate GM, Taylor VL (October 1990). "Methylenedioxymethamphetamine-induced hyperthermia and neurotoxicity are independently mediated by 5-HT₂ receptors". *Brain Res*. **529** (1–2): 85–90. doi:10.1016/0006-8993(90)90813-q (<https://doi.org/10.1016%2F0006-8993%2890%2990813-q>) . PMID 1980848 (<https://pubmed.ncbi.nlm.nih.gov/1980848>) .
205. Armstrong BD, Paik E, Chhith S, Lelievre V, Waschek JA, Howard SG (2004). "Potentiation of (DL)-3,4-methylenedioxymethamphetamine (MDMA)-induced toxicity by the serotonin 2A receptor partial agonist d-lysergic acid diethylamide (LSD), and the protection of same by the serotonin 2A/2C receptor antagonist MDL 11,939". *Neuroscience Research Communications*. **35** (2): 83–95. doi:10.1002/nrc.20023 (<https://doi.org/10.1002%2Fnrc.20023>) .
206. Anderson GM, Braun G, Braun U, Nichols DE, Shulgin AT (1978). "Absolute configuration and psychotomimetic activity" (<https://citeseerx.ist.psu.edu/document?repid=rep1&type=pdf&doi=2ab674b010611df18c029a78f6d17e52dba5f82f>) . *NIDA Research Monograph* (22): 8–15. PMID 101890 (<https://pubmed.ncbi.nlm.nih.gov/101890>) .

207. Acquas E, Pisanu A, Spiga S, Plumitallo A, Zernig G, Di Chiara G (July 2007). "Differential effects of intravenous R,S-(+/-)-3,4-methylenedioxymethamphetamine (MDMA, Ecstasy) and its S(+)- and R(-)-enantiomers on dopamine transmission and extracellular signal regulated kinase phosphorylation (pERK) in the rat nucleus accumbens shell and core". *Journal of Neurochemistry*. **102** (1): 121–132. doi:10.1111/j.1471-4159.2007.04451.x (<https://doi.org/10.1111%2Fj.1471-4159.2007.04451.x>) . PMID 17564678 (<https://pubmed.ncbi.nlm.nih.gov/17564678>) .
208. Curry DW, Young MB, Tran AN, Daoud GE, Howell LL (January 2018). "Separating the agony from ecstasy: R(-)-3,4-methylenedioxymethamphetamine has prosocial and therapeutic-like effects without signs of neurotoxicity in mice" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5714650>) . *Neuropharmacology*. **128**: 196–206. doi:10.1016/j.neuropharm.2017.10.003 (<https://doi.org/10.1016%2Fj.neuropharm.2017.10.003>) . PMC 5714650 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5714650>) . PMID 28993129 (<https://pubmed.ncbi.nlm.nih.gov/28993129>) .
209. Bedi G (October 2024). "Is the stereoisomer R-MDMA a safer version of MDMA?" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11631934>) . *Neuropsychopharmacology*. **50** (2): 360–361. doi:10.1038/s41386-024-02009-8 (<https://doi.org/10.1038%2Fs41386-024-02009-8>) . PMC 11631934 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11631934>) . PMID 39448866 (<https://pubmed.ncbi.nlm.nih.gov/39448866>) .
210. Luethi D, Liechti ME (2021). "Drugs of Abuse Affecting 5-HT_{2B} Receptors". *5-HT_{2B} Receptors. The Receptors*. Vol. 35. Cham: Springer International Publishing. pp. 277–289. doi:10.1007/978-3-030-55920-5_16 (https://doi.org/10.1007%2F978-3-030-55920-5_16) . ISBN 978-3-030-55919-9. "Notably, in a study by Rickli and colleagues, MDMA did not activate the 5-HT_{2B} receptor in the functional assay at investigated concentrations (EC₅₀ > 20 µM); however, [MDA], the main psychoactive N-demethylated phase I metabolite of MDMA, potently activated the receptor at submicromolar concentrations [14]. This suggests that the metabolite MDA rather than MDMA itself may lead to valvulopathy and that there could be a significant metabolic contribution to MDMA-induced effects and adverse effect."
211. Mead J, Parrott A (May 2020). "Mephedrone and MDMA: A comparative review". *Brain Research*. **1735** 146740. doi:10.1016/j.brainres.2020.146740 (<https://doi.org/10.1016%2Fj.brainres.2020.146740>) . PMID 32087112 (<https://pubmed.ncbi.nlm.nih.gov/32087112>) . "A controlled study on eight experienced MDMA users reported that 1.5 mg/kg (comparable to what was deemed a typical dosage amount) consumed orally resulted in the subjective effects peaking within 2 h of ingestion (Harris et al., 2002). Other research indicates effects to emerge between 20 and 60 min, with them peaking between 60 and 90 min and lasting up to 5 h (Green et al., 2003). A dose of 100 mg has a half-life of 8–9h(De la Torre et al., 2004), although as mentioned above, users are unaware of the dose they ingest."

212. Ermer JC, Pennick M, Frick G (May 2016). "Lisdexamfetamine Dimesylate: Prodrug Delivery, Amphetamine Exposure and Duration of Efficacy" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4823324>) . *Clinical Drug Investigation*. **36** (5): 341–356. doi:10.1007/s40261-015-0354-y (<https://doi.org/10.1007/s40261-015-0354-y>) . PMC 4823324 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4823324>) . PMID 27021968 (<https://pubmed.ncbi.nlm.nih.gov/27021968>) .
213. Cruickshank CC, Dyer KR (July 2009). "A review of the clinical pharmacology of methamphetamine". *Addiction*. **104** (7). Abingdon, England: 1085–1099. doi:10.1111/j.1360-0443.2009.02564.x (<https://doi.org/10.1111/j.1360-0443.2009.02564.x>) . PMID 19426289 (<https://pubmed.ncbi.nlm.nih.gov/19426289>) . "Metabolism does not appear to be altered by chronic exposure, thus dose escalation appears to arise from pharmacodynamic rather than pharmacokinetic tolerance [24]. [...] The terminal plasma half-life of methamphetamine of approximately 10 hours is similar across administration routes, but with substantial inter-individual variability. Acute effects persist for up to 8 hours following a single moderate dose of 30 mg [30]. [...] peak plasma methamphetamine concentration occurs after 4 hours [35]. Nevertheless, peak cardiovascular and subjective effects occur rapidly (within 5–15 minutes). The dissociation between peak plasma concentration and clinical effects indicates acute tolerance, which may reflect rapid molecular processes such as redistribution of vesicular monoamines and internalization of monoamine receptors and transporters [6,36]. Acute subjective effects diminish over 4 hours, while cardiovascular effects tend to remain elevated. This is important, as the marked acute tachyphylaxis to subjective effects may drive repeated use within intervals of 4 hours, while cardiovascular risks may increase [11,35]."

214. Abbas K, Barnhardt EW, Nash PL, Streng M, Coury DL (April 2024). "A review of amphetamine extended release once-daily options for the management of attention-deficit hyperactivity disorder" (<https://doi.org/10.1080%2F14737175.2024.2321921>) . *Expert Review of Neurotherapeutics*. **24** (4): 421–432. doi:10.1080/14737175.2024.2321921 (<http://doi.org/10.1080%2F14737175.2024.2321921>) . PMID 38391788 (<https://pubmed.ncbi.nlm.nih.gov/38391788>) . "For several decades, clinical benefits of amphetamines have been limited by the pharmacologic half-life of around 4 hours. Although higher doses can produce higher maximum concentrations, they do not affect the half-life of the dose. Therefore, to achieve longer durations of effect, stimulants had to be dosed at least twice daily. Further, these immediate-release doses were found to have their greatest effect shortly after administration, with a rapid decline in effect after reaching peak blood concentrations. The clinical correlation of this was found in comparing math problems attempted and solved between a mixed amphetamine salts preparation (MAS) 10 mg once at 8 am vs 8 am followed by 12 pm [14]. The study also demonstrated the phenomenon of acute tolerance, where even if blood concentrations were maintained over the course of the day, clinical efficacy in the form of math problems attempted and solved would diminish over the course of the day. These findings eventually led to the development of a once daily preparation (MAS XR) [15], which is a composition of 50% immediate-release beads and 50% delayed release beads intended to mimic this twice-daily dosing with only a single administration."
215. van Gaalen MM, Schlumbohm C, Folgering JH, Adhikari S, Bhattacharya C, Steinbach D, et al. (April 2019). "Development of a Semimechanistic Pharmacokinetic-Pharmacodynamic Model Describing Dextroamphetamine Exposure and Striatal Dopamine Response in Rats and Nonhuman Primates following a Single Dose of Dextroamphetamine" (<https://doi.org/10.1124%2Fjpet.118.254508>) . *The Journal of Pharmacology and Experimental Therapeutics*. **369** (1): 107–120. doi:10.1124/jpet.118.254508 (<https://doi.org/10.1124%2Fjpet.118.254508>) . PMID 30733244 (<https://pubmed.ncbi.nlm.nih.gov/30733244>) . "Acute tolerance has been demonstrated for methamphetamine in rats (Segal and Kuczenski, 2006), and for D-amphetamine in rats (Lewander, 1971), [non-human primates (NHPs)] (Jedema et al., 2014) and humans (Angrist et al., 1987; Brauer et al., 1996; Dolder et al., 2017). In vivo measurement of dopamine by microdialysis was used in rats and NHPs to evaluate these time-dependent effects. In humans, various subjective measures of mood related to the drug's euphoric effects were observed to decline more rapidly than plasma concentrations following D-amphetamine oral doses ranging from 20 to 40 mg (Angrist et al., 1987; Brauer et al., 1996; Dolder et al., 2017). Whereas peak plasma concentrations and subjective effects occurred between 2 and 4 hours following administration, drug effect measures had largely returned to baseline values by 8 hours despite continued exposure to the drug (mean half-life = 8 hours following a 40 mg dose (Dolder et al., 2017))."

216. Bisagno V, Cadet JL (2021). "Methamphetamine and MDMA Neurotoxicity: Biochemical and Molecular Mechanisms". *Handbook of Neurotoxicity*. Cham: Springer International Publishing. pp. 1–24. doi:10.1007/978-3-030-71519-9_80-1 (https://doi.org/10.1007%2F978-3-030-71519-9_80-1) . ISBN 978-3-030-71519-9. "Injections of large doses of MDMA cause massive release of 5-HT from presynaptic vesicles, followed by a rapid decrease in 5-HT and 5-hydroxyindoleacetic acid (5-HIAA) levels and decreased TPH activity (Górska et al., 2018; Lyles & Cadet, 2003). There do not appear to be losses of 5-HT uptake sites at early time points after MDMA administration (Lyles & Cadet, 2003). [...] MDMA also perturbs the function of SERT (Green et al., 2003), a marker of the integrity of serotonin neurons (Blakely et al., 1994). By virtue of its moderating synaptic 5-HT levels, SERT is crucial for the process of 5-HT neurotransmission (Green et al., 2003). MDMA downregulates SERT function without altering SERT mRNA or protein expression, and this rapid downregulation is sustained for at least 90 min and is dose-dependent (Kivell et al., 2010)."
217. Kivell B, Day D, Bosch P, Schenk S, Miller J (June 2010). "MDMA causes a redistribution of serotonin transporter from the cell surface to the intracellular compartment by a mechanism independent of phospho-p38-mitogen activated protein kinase activation". *Neuroscience*. **168** (1): 82–95. doi:10.1016/j.neuroscience.2010.03.018 (<https://doi.org/10.1016%2Fj.neuroscience.2010.03.018>) . PMID 20298763 (<https://pubmed.ncbi.nlm.nih.gov/20298763>) .
218. Holley A, Simonson B, Kivell BM (April 2013). "MDMA regulates serotonin transporter function via a Protein kinase C dependent mechanism" (<https://www.researchgate.net/publication/256328051>) . *Journal of Addiction & Prevention*. **1** (1): 5. ISSN 2330-2178 (<https://search.worldcat.org/issn/2330-2178>) .
219. Underhill S, Amara S (2020). "MDMA and TAAR1-mediated RhoA Activation in Serotonin Neurons" (<https://doi.org/10.1096%2Ffasebj.2020.34.s1.05856>) . *The FASEB Journal*. **34** (S1): 1. doi:10.1096/fasebj.2020.34.s1.05856 (<https://doi.org/10.1096%2Ffasebj.2020.34.s1.05856>) . ISSN 0892-6638 (<https://search.worldcat.org/issn/0892-6638>) .
220. Underhill S, Amara S (2022). "3,4-methylenedioxymethamphetamine (MDMA) stimulates activation of TAAR1 and subsequent neurotransmitter transporter internalization in serotonin neurons" (<https://doi.org/10.1096%2Ffasebj.2022.36.S1.R5394>) . *The FASEB Journal*. **36** (S1) fasebj.2022.36.S1.R5394. doi:10.1096/fasebj.2022.36.S1.R5394 (<https://doi.org/10.1096%2Ffasebj.2022.36.S1.R5394>) . ISSN 0892-6638 (<https://search.worldcat.org/issn/0892-6638>) .

221. Kittler K, Lau T, Schloss P (March 2010). "Antagonists and substrates differentially regulate serotonin transporter cell surface expression in serotonergic neurons". *European Journal of Pharmacology*. **629** (1–3): 63–67. doi:10.1016/j.ejphar.2009.12.010 (<https://doi.org/10.1016%2Fj.ejphar.2009.12.010>) . PMID 20006597 (<https://pubmed.ncbi.nlm.nih.gov/20006597>) . "Our results show that exposure to the SSRIs citalopram, fluoxetine, sertraline and paroxetine all induced SERT internalization, but with different efficacies. The substrates 5-HT and MDMA also induced SERT internalization, while cocaine elevated SERT cell surface expression."
222. Rothman RB, Baumann MH, Dersch CM, Romero DV, Rice KC, Carroll FI, et al. (January 2001). "Amphetamine-type central nervous system stimulants release norepinephrine more potently than they release dopamine and serotonin". *Synapse*. **39** (1). New York, N.Y.: 32–41. doi:10.1002/1098-2396(20010101)39:1<32::AID-SYN5>3.0.CO;2-3 (<https://doi.org/10.1002%2F1098-2396%2820010101%2939%3A1%3C32%3A%3AAID-SYN5%3E3.0.CO%3B2-3>) . PMID 11071707 (<https://pubmed.ncbi.nlm.nih.gov/11071707>) . S2CID 15573624 (<https://api.semanticscholar.org/CorpusID:15573624>) .
223. Rothman RB, Partilla JS, Baumann MH, Lightfoot-Siordia C, Blough BE (April 2012). "Studies of the biogenic amine transporters. 14. Identification of low-efficacy "partial" substrates for the biogenic amine transporters" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3364510>) . *The Journal of Pharmacology and Experimental Therapeutics*. **341** (1): 251–262. doi:10.1124/jpet.111.188946 (<https://doi.org/10.1124%2Fjpet.111.188946>) . PMC 3364510 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3364510>) . PMID 22271821 (<https://pubmed.ncbi.nlm.nih.gov/22271821>) .
224. Marusich JA, Antonazzo KR, Blough BE, Brandt SD, Kavanagh PV, Partilla JS, et al. (February 2016). "The new psychoactive substances 5-(2-aminopropyl)indole (5-IT) and 6-(2-aminopropyl)indole (6-IT) interact with monoamine transporters in brain tissue" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4681602>) . *Neuropharmacology*. **101**: 68–75. doi:10.1016/j.neuropharm.2015.09.004 (<https://doi.org/10.1016%2Fj.neuropharm.2015.09.004>) . PMC 4681602 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4681602>) . PMID 26362361 (<https://pubmed.ncbi.nlm.nih.gov/26362361>) .
225. Nagai F, Nonaka R, Satoh Hisashi Kamimura K (March 2007). "The effects of non-medically used psychoactive drugs on monoamine neurotransmission in rat brain". *European Journal of Pharmacology*. **559** (2–3): 132–137. doi:10.1016/j.ejphar.2006.11.075 (<https://doi.org/10.1016%2Fj.ejphar.2006.11.075>) . PMID 17223101 (<https://pubmed.ncbi.nlm.nih.gov/17223101>) .

226. Halberstadt AL, Brandt SD, Walther D, Baumann MH (March 2019). "2-Aminoindan and its ring-substituted derivatives interact with plasma membrane monoamine transporters and α 2-adrenergic receptors" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6848746>) . *Psychopharmacology*. **236** (3): 989–999. doi:10.1007/s00213-019-05207-1 (<https://doi.org/10.1007/s00213-019-05207-1>) . PMC 6848746 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6848746>) . PMID 30904940 (<https://pubmed.ncbi.nlm.nih.gov/30904940>) .
227. Blough B (July 2008). "Dopamine-releasing agents" (https://bitnest.netfirms.com/external/Books/Dopamine-releasing-agents_c11.pdf) (PDF). In Trudell ML, Izenwasser S (eds.). *Dopamine Transporters: Chemistry, Biology and Pharmacology* (<https://books.google.com/books?id=QCagLAAACAAJ>) . Hoboken [NJ]: Wiley. pp. 305–320. ISBN 978-0-470-11790-3. OCLC 181862653 (<https://search.worldcat.org/oclc/181862653>) . OL 18589888W (<https://openlibrary.org/works/OL18589888W>) .
228. Kolaczynska KE, Ducret P, Trachsel D, Hoener MC, Liechti ME, Luethi D (June 2022). "Pharmacological characterization of 3,4-methylenedioxymphetamine (MDA) analogs and two amphetamine-based compounds: N, α -DEPEA and DPIA" (<https://www.researchgate.net/publication/359686098>) . *European Neuropsychopharmacology*. **59**: 9–22. doi:10.1016/j.euroneuro.2022.03.006 (<https://doi.org/10.1016/j.euroneuro.2022.03.006>) . PMID 35378384 (<https://pubmed.ncbi.nlm.nih.gov/35378384>) .
229. Mas M, Farré M, de la Torre R, Roset PN, Ortuño J, Segura J, et al. (July 1999). "Cardiovascular and neuroendocrine effects and pharmacokinetics of 3, 4-methylenedioxymphetamine in humans". *The Journal of Pharmacology and Experimental Therapeutics*. **290** (1): 136–145. doi:10.1016/S0022-3565(24)34877-3 ([https://doi.org/10.1016/S0022-3565\(24\)34877-3](https://doi.org/10.1016/S0022-3565(24)34877-3)) . PMID 10381769 (<https://pubmed.ncbi.nlm.nih.gov/10381769>) .
230. de la Torre R, Farré M, Ortuño J, Mas M, Brenneisen R, Roset PN, et al. (February 2000). "Non-linear pharmacokinetics of MDMA ('ecstasy') in humans" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2014905>) . *British Journal of Clinical Pharmacology*. **49** (2): 104–109. doi:10.1046/j.1365-2125.2000.00121.x (<https://doi.org/10.1046/j.1365-2125.2000.00121.x>) . PMC 2014905 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2014905>) . PMID 10671903 (<https://pubmed.ncbi.nlm.nih.gov/10671903>) .
231. Farré M, Roset PN, Lopez CH, Mas M, Ortuño J, Menoyo E, et al. (September 2000). "Pharmacology of MDMA in humans" (<https://doi.org/10.1111/2Fj.1749-6632.2000.tb05199.x>) . *Annals of the New York Academy of Sciences*. **914** (1): 225–237. Bibcode:2000NYASA.914..225D (<https://ui.adsabs.harvard.edu/abs/2000NYASA.914..225D>) . doi:10.1111/j.1749-6632.2000.tb05199.x (<https://doi.org/10.1111/j.1749-6632.2000.tb05199.x>) . PMID 11085324 (<https://pubmed.ncbi.nlm.nih.gov/11085324>) . S2CID 29247621 (<https://api.semanticscholar.org/CorpusID:29247621>) .

232. Kolbrich EA, Goodwin RS, Gorelick DA, Hayes RJ, Stein EA, Huestis MA (June 2008). "Plasma pharmacokinetics of 3,4-methylenedioxymethamphetamine after controlled oral administration to young adults" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2663855>) . *Therapeutic Drug Monitoring*. **30** (3): 320–332. doi:10.1097/FTD.0b013e3181684fa0 (<https://doi.org/10.1097%2FFTD.0b013e3181684fa0>) . PMC 2663855 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2663855>) . PMID 18520604 (<https://pubmed.ncbi.nlm.nih.gov/18520604>) .
233. Shima N, Kamata H, Katagi M, Tsuchihashi H, Sakuma T, Nemoto N (September 2007). "Direct determination of glucuronide and sulfate of 4-hydroxy-3-methoxymethamphetamine, the main metabolite of MDMA, in human urine". *Journal of Chromatography. B, Analytical Technologies in the Biomedical and Life Sciences*. **857** (1): 123–129. doi:10.1016/j.jchromb.2007.07.003 (<https://doi.org/10.1016%2Fj.jchromb.2007.07.003>) . PMID 17643356 (<https://pubmed.ncbi.nlm.nih.gov/17643356>) .
234. Fallon JK, Kicman AT, Henry JA, Milligan PJ, Cowan DA, Hutt AJ (July 1999). "Stereospecific analysis and enantiomeric disposition of 3, 4-methylenedioxymethamphetamine (Ecstasy) in humans" (<https://doi.org/10.1093%2Fclinchem%2F45.7.1058>) . *Clinical Chemistry*. **45** (7): 1058–1069. doi:10.1093/clinchem/45.7.1058 (<https://doi.org/10.1093%2Fclinchem%2F45.7.1058>) . PMID 10388483 (<https://pubmed.ncbi.nlm.nih.gov/10388483>) .
235. Mueller M, Peters FT, Maurer HH, McCann UD, Ricaurte GA (October 2008). "Nonlinear pharmacokinetics of (+/-)3,4-methylenedioxymethamphetamine (MDMA, "Ecstasy") and its major metabolites in squirrel monkeys at plasma concentrations of MDMA that develop after typical psychoactive doses". *The Journal of Pharmacology and Experimental Therapeutics*. **327** (1): 38–44. doi:10.1124/jpet.108.141366 (<https://doi.org/10.1124%2Fjpet.108.141366>) . PMID 18591215 (<https://pubmed.ncbi.nlm.nih.gov/18591215>) . S2CID 38043715 (<https://api.semanticscholar.org/CorpusID:38043715>) .
236. Shulgin AT, Shulgin A (1991). "Chapters 12, 22". *PiHKAL: A Chemical Love Story* (7th printing, 1st ed.). Berkeley, CA: Transform Press. ISBN 978-0-9630096-0-9.
237. Milhazes N, Martins P, Uriarte E, Garrido J, Calheiros R, Marques MP, et al. (July 2007). "Electrochemical and spectroscopic characterisation of amphetamine-like drugs: application to the screening of 3,4-methylenedioxymethamphetamine (MDMA) and its synthetic precursors". *Analytica Chimica Acta*. **596** (2): 231–241. Bibcode:2007AcAC..596..231M (<https://ui.adsabs.harvard.edu/abs/2007AcAC..596..231M>) . doi:10.1016/j.aca.2007.06.027 (<https://doi.org/10.1016%2Fj.aca.2007.06.027>) . hdl:10316/45124 (<https://hdl.handle.net/10316%2F45124>) . PMID 17631101 (<https://pubmed.ncbi.nlm.nih.gov/17631101>) .

238. Milhazes N, Cunha-Oliveira T, Martins P, Garrido J, Oliveira C, Rego AC, et al. (October 2006). "Synthesis and cytotoxic profile of 3,4-methylenedioxymethamphetamine ("ecstasy") and its metabolites on undifferentiated PC12 cells: A putative structure-toxicity relationship" (<http://estudogeral.sib.uc.pt/bitstream/10316/12872/1/Synthesis%20and%20Cytotoxic%20Profile.pdf>) (PDF). *Chemical Research in Toxicology*. **19** (10): 1294–1304. doi:10.1021/tx060123i (<https://doi.org/10.1021%2Ftx060123i>) . hdl:10316/12872 (<https://hdl.handle.net/10316%2F12872>) . PMID 17040098 (<https://pubmed.ncbi.nlm.nih.gov/17040098>) . Archived (<https://web.archive.org/web/20210628070418/https://estudogeral.sib.uc.pt/bitstream/10316/12872/1/Synthesis%20and%20Cytotoxic%20Profile.pdf>) (PDF) from the original on 28 June 2021. Retrieved 24 September 2019.
239. Baxter EW, Reitz AB (April 2004). "Reductive aminations of carbonyl compounds with borohydride and borane reducing agents". *Organic Reactions*. **59**. Hoboken, New Jersey, United States: 59. doi:10.1002/0471264180.or059.01 (<https://doi.org/10.1002%2F0471264180.or059.01>) . ISBN 978-0-471-26418-7.
240. Gimeno P, Besacier F, Bottex M, Dujourdy L, Chaudron-Thozet H (December 2005). "A study of impurities in intermediates and 3,4-methylenedioxymethamphetamine (MDMA) samples produced via reductive amination routes". *Forensic Science International*. **155** (2–3): 141–157. doi:10.1016/j.forsciint.2004.11.013 (<https://doi.org/10.1016%2Fj.forsciint.2004.11.013>) . PMID 16226151 (<https://pubmed.ncbi.nlm.nih.gov/16226151>) .
241. Palhol F, Boyer S, Naulet N, Chabrilat M (September 2002). "Impurity profiling of seized MDMA tablets by capillary gas chromatography". *Analytical and Bioanalytical Chemistry*. **374** (2): 274–281. doi:10.1007/s00216-002-1477-6 (<https://doi.org/10.1007%2Fs00216-002-1477-6>) . PMID 12324849 (<https://pubmed.ncbi.nlm.nih.gov/12324849>) . S2CID 42666306 (<https://api.semanticscholar.org/CorpusID:42666306>) .
242. Renton RJ, Cowie JS, Oon MC (August 1993). "A study of the precursors, intermediates and reaction by-products in the synthesis of 3,4-methylenedioxymethylamphetamine and its application to forensic drug analysis". *Forensic Science International*. **60** (3): 189–202. doi:10.1016/0379-0738(93)90238-6 (<https://doi.org/10.1016%2F0379-0738%2893%2990238-6>) . PMID 7901132 (<https://pubmed.ncbi.nlm.nih.gov/7901132>) .
243. Mohan J, ed. (June 2014). *World Drug Report 2014* (http://www.unodc.org/documents/wdr2014/World_Drug_Report_2014_web.pdf) (PDF). Vienna, Austria: United Nations Office on Drugs and Crime. pp. 2, 3, 123–152. ISBN 978-92-1-056752-7. Archived (https://web.archive.org/web/20141113173315/http://www.unodc.org/documents/wdr2014/World_Drug_Report_2014_web.pdf) (PDF) from the original on 13 November 2014. Retrieved 1 December 2014.

244. "Early Warning - MDMA and MDA Producers Using *Ocotea Cymbarum* as a Precursor" (<http://web.archive.org/web/20121018052300/http://www.justice.gov/dea/pr/micrograms/2005/mg1105.pdf>) (PDF). *DEA Microgram Newsletter*. **38** (11). Drug Enforcement Agency, U.S. Department of Justice: 166. 11 November 2005. Archived from the original (<https://www.justice.gov/dea/pr/micrograms/2005/mg1105.pdf>) (PDF) on 18 October 2012.
245. Barnes AJ, De Martinis BS, Gorelick DA, Goodwin RS, Kolbrich EA, Huestis MA (March 2009). "Disposition of MDMA and metabolites in human sweat following controlled MDMA administration" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2669283>) . *Clinical Chemistry*. **55** (3): 454–462. doi:10.1373/clinchem.2008.117093 (<https://doi.org/10.1373%2Fclinchem.2008.117093>) . PMC 2669283 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2669283>) . PMID 19168553 (<https://pubmed.ncbi.nlm.nih.gov/19168553>) .
246. Baselt RC (2011). *Disposition of toxic drugs and chemicals in man* (9th ed.). Seal Beach, Ca.: Biomedical Publications. pp. 1078–1080. ISBN 978-0-9626523-8-7.
247. Fukuto JM, Kumagai Y, Cho AK (September 1991). "Determination of the mechanism of demethylenation of (methylenedioxy)phenyl compounds by cytochrome P450 using deuterium isotope effects". *J Med Chem*. **34** (9): 2871–2876. doi:10.1021/jm00113a028 (<https://doi.org/10.1021%2Fjm00113a028>) . PMID 1680196 (<https://pubmed.ncbi.nlm.nih.gov/1680196>) .
248. Lin DL, Liu HC, Yin RM, Chen DT, Soong SJ, Liu RH (2004). "Effectiveness of multiple internal standards: deuterated analogues of methylenedioxymethamphetamine, methylenedioxyamphetamine, methamphetamine, and amphetamine". *J Anal Toxicol*. **28** (8): 650–654. doi:10.1093/jat/28.8.650 (<https://doi.org/10.1093%2Fjat%2F28.8.650>) . PMID 15538959 (<https://pubmed.ncbi.nlm.nih.gov/15538959>) .
249. Romberg RW, Ntamack AG, Blacik LJ, Kazarian CM, Snyder JJ, Welsh ER (January 2011). "Differences in binding affinities of MDA, MDMA, MDEA, Amphetamine, Methamphetamine, and their deuterated analogues to solid-phase extraction cartridges". *J Anal Toxicol*. **35** (1): 15–22. doi:10.1093/anatox/35.1.15 (<https://doi.org/10.1093%2Fanatox%2F35.1.15>) . PMID 21219698 (<https://pubmed.ncbi.nlm.nih.gov/21219698>) .
250. Berquist MD, Leth-Petersen S, Kristensen JL, Fantegrossi WE (February 2020). "Locomotor effects of 3,4-methylenedioxymethamphetamine (MDMA) and its deuterated form in mice: psychostimulant effects, stereotypy, and sensitization" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7388080>) . *Psychopharmacology (Berl)*. **237** (2): 431–442. doi:10.1007/s00213-019-05380-3 (<https://doi.org/10.1007%2Fs00213-019-05380-3>) . PMC 7388080 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7388080>) . PMID 31729537 (<https://pubmed.ncbi.nlm.nih.gov/31729537>) .

251. Berquist MD, Leth-Petersen S, Kristensen JL, Fantegrossi WE (March 2020). "In vivo effects of 3,4-methylenedioxymethamphetamine (MDMA) and its deuterated form in rodents: Drug discrimination and thermoregulation". *Drug Alcohol Depend.* **208** 107850. doi:10.1016/j.drugalcdep.2020.107850 (<https://doi.org/10.1016%2Fj.drugalcdep.2020.107850>) . PMID 31954950 (<https://pubmed.ncbi.nlm.nih.gov/31954950>) .
252. Passie T (29 June 2023). *The History of MDMA* (<https://books.google.com/books?id=KSvCEAAAQBAJ&pg=PA6>) . Oxford University Press. pp. 6–16, 18, 27, 29, 32, 40. doi:10.1093/oso/9780198867364.001.0001 (<https://doi.org/10.1093%2Foso%2F9780198867364.001.0001>) . ISBN 978-0-19-886736-4.
253. Bernschneider-Reif S, Oxler F, Freudenmann RW (November 2006). "The origin of MDMA ("ecstasy")--separating the facts from the myth". *Die Pharmazie.* **61** (11): 966–972. PMID 17152992 (<https://pubmed.ncbi.nlm.nih.gov/17152992>) .
254. Darmstadt FE (16 May 1914). "German Patent 274350: Verfahren zur Darstellung von Alkyloxyaryl-, Dialkyloxyaryl- und Alkylendioxyarylaminoopropanen bzw. deren am Stickstoff monoalkylierten Derivaten" (<http://v3.espacenet.com/publicationDetails/originalDocument?CC=DE&NR=274350C&FT=D>) . Kaiserliches Patentamt. Archived (https://web.archive.org/web/20210828153545/https://worldwide.espacenet.com/publicationDetails/originalDocument?locale=en_EP&FT=D&CC=DE&NR=274350C) from the original on 28 August 2021. Retrieved 12 April 2009.
255. Darmstadt FE (15 October 1914). "German Patent 279194: Verfahren zur Darstellung von Hydrastinin Derivaten" (<http://v3.espacenet.com/publicationDetails/originalDocument?CC=DE&NR=279194C&FT=D>) . Kaiserliches Patentamt. Archived (https://web.archive.org/web/20210828153659/https://worldwide.espacenet.com/publicationDetails/originalDocument?locale=en_EP&FT=D&CC=DE&NR=279194C) from the original on 28 August 2021. Retrieved 20 July 2009.
256. Shulgin AT (1990). "History of MDMA" (https://link.springer.com/10.1007/978-1-4613-1485-1_1) . In Peroutka SJ (ed.). *Ecstasy: The Clinical, Pharmacological and Neurotoxicological Effects of the Drug MDMA*. Topics in the Neurosciences. Vol. 9. Boston, MA: Springer US. pp. 1–20 (2, 14). doi:10.1007/978-1-4613-1485-1_1 (https://doi.org/10.1007%2F978-1-4613-1485-1_1) . ISBN 978-1-4612-8799-5. Retrieved 15 May 2025.

257. Alles GA (1959). "Some Relations Between Chemical Structure and Physiological Action of Mescaline and Related Compounds / Structure and Action of Phenethylamines" (<https://web.archive.org/web/20250321230359/https://bitnest.netfirms.com/external/Books/NeuropharmacologyTrans.4.181#page=5>) . In Abramson HA (ed.). *Neuropharmacology: Transactions of the Fourth Conference, September 25, 26, and 27, 1957, Princeton, N. J.* (<https://books.google.com/books?id=sDQLAQAAAMAJ&q=%22Some+relations+between+chemical+structure+and+physiological+action+of+mescaline+and+related+compounds%22>) New York: Josiah Macy Foundation. pp. 181–268. OCLC 9802642 (<https://search.worldcat.org/oclc/9802642>) . Archived from the original (<https://bitnest.netfirms.com/external/Books/NeuropharmacologyTrans.4.181#page=5>) on 21 March 2025.
258. Alles GA (1959). "Subjective Reactions to Phenethylamine Hallucinogens" (<https://archive.org/details/pharmacologicapp0000univ/page/238/mode/1up>) . *A Pharmacologic Approach to the Study of the Mind* (<https://books.google.com/books?id=x45rAAAAMAJ>) . Springfield: CC Thomas. pp. 238–250 (241–246).
259. Benzenhöfer U, Passie T (August 2010). "Rediscovering MDMA (ecstasy): the role of the American chemist Alexander T. Shulgin". *Addiction*. **105** (8). Abingdon, England: 1355–1361. doi:10.1111/j.1360-0443.2010.02948.x (<https://doi.org/10.1111%2Fj.1360-0443.2010.02948.x>) . PMID 20653618 (<https://pubmed.ncbi.nlm.nih.gov/20653618>) .
260. Hardman HF, Haavik CO, Seevers MH (June 1973). "Relationship of the structure of mescaline and seven analogs to toxicity and behavior in five species of laboratory animals" (http://www.erowid.org/references/refs_view.php?A=ShowDoc1&ID=639) . *Toxicology and Applied Pharmacology*. **25** (2): 299–309. Bibcode:1973ToxAP..25..299H (<https://ui.adsabs.harvard.edu/abs/1973ToxAP..25..299H>) . doi:10.1016/S0041-008X(73)80016-X (<https://doi.org/10.1016%2FS0041-008X%2873%2980016-X>) . hdl:2027.42/33868 (<https://hdl.handle.net/2027.42%2F33868>) . PMID 4197635 (<https://pubmed.ncbi.nlm.nih.gov/4197635>) . Archived (https://web.archive.org/web/20081021151006/http://www.erowid.org/reference/refs_view.php?A=ShowDoc1&ID=639) from the original on 21 October 2008. Retrieved 19 April 2009.
261. Biniecki S, Krajewski E (1960). "Production of d,1-N-methyl-beta-(3,4-methylenedioxyphenyl)-isopropylamine and d,1-N-methyl-beta-(3,4-dimethoxyphenyl)-isopropylamine". *Acta Polon Pharm* (in Polish). **17**: 421–425.
262. Siegel RK (October 1986). "MDMA. Nonmedical use and intoxication" (http://www.maps.org/images/pdf/1986_siegel_1.pdf) (PDF). *Journal of Psychoactive Drugs*. **18** (4): 349–354. doi:10.1080/02791072.1986.10472368 (<https://doi.org/10.1080%2F02791072.1986.10472368>) . PMID 2880950 (<https://pubmed.ncbi.nlm.nih.gov/2880950>) . Archived (https://web.archive.org/web/20160304090940/http://www.maps.org/images/pdf/1986_siegel_1.pdf) (PDF) from the original on 4 March 2016. Retrieved 11 August 2015.

263. The first confirmed sample was seized and identified by Chicago Police in 1970, see Sreenivasan VR (1972). "Problems in Identification of Methylenedioxy and Methoxy Amphetamines" (http://www.erowid.org/references/refs_view.php?A=ShowDoc1&ID=1149) . *Journal of Criminal Law, Criminology, and Police Science*. **63** (2): 304–312. doi:10.2307/1142315 (<https://doi.org/10.2307%2F1142315>) . JSTOR 1142315 (<https://www.jstor.org/stable/1142315>) . Archived (https://web.archive.org/web/20081021200016/http://www.erowid.org/references/refs_view.php?A=ShowDoc1&ID=1149) from the original on 21 October 2008. Retrieved 19 April 2009.
264. Foderaro LW (11 December 1988). "Psychedelic Drug Called Ecstasy Gains Popularity in Manhattan Nightclubs" (<https://www.nytimes.com/1988/12/11/nyregion/psychedelic-drug-called-ecstasy-gains-popularity-in-manhattan-nightclubs.html?pagewanted=2>) . *The New York Times*. Archived (<https://web.archive.org/web/20151117042408/http://www.nytimes.com/1988/12/11/nyregion/psychedelic-drug-called-ecstasy-gains-popularity-in-manhattan-nightclubs.html?pagewanted=2>) from the original on 17 November 2015. Retrieved 27 August 2015.
265. Brown E (September 2002). "Professor X" (https://web.archive.org/web/20150625015832/http://archive.wired.com/wired/archive/10.09/professorx.html?pg=3&topic=&topic_set=) . *Wired*. No. 10.09. p. 3. Archived from the original (http://archive.wired.com/wired/archive/10.09/professorx.html?pg=3&topic=&topic_set=) on 25 June 2015. Retrieved 4 January 2015.
266. Beck JE (April 1987). "Drug Abuse Series: MDMA" (https://erowid.org/chemicals/mdma/mdma_info6.shtml) . *Erowid*. Drug Abuse Information and Monitoring Project. Archived (https://web.archive.org/web/20150804125927/https://www.erowid.org/chemicals/mdma/mdma_info6.shtml) from the original on 4 August 2015. Retrieved 6 August 2015.
267. Pentney AR (2001). "An exploration of the history and controversies surrounding MDMA and MDA". *Journal of Psychoactive Drugs*. **33** (3): 213–221. doi:10.1080/02791072.2001.10400568 (<https://doi.org/10.1080%2F02791072.2001.10400568>) . PMID 11718314 (<https://pubmed.ncbi.nlm.nih.gov/11718314>) . S2CID 31142434 (<https://api.semanticscholar.org/CorpusID:31142434>) .
268. "Alexander 'Sasha' Shulgin" (<https://web.archive.org/web/20141220131529/http://www.shulginresearch.org/home/about/alexander-sasha-shulgin/>) . Alexander Shulgin Research Institute. Archived from the original (<http://www.shulginresearch.org/home/about/alexander-sasha-shulgin/>) on 20 December 2014. Retrieved 8 January 2015.

269. Shulgin AT, Nichols DE (1978). "Characterization of Three New Psychotomimetics" (<https://bitnest.netfirms.com/external/10.1016/B978-0-08-021938-7.50010-2>) . In Stillman RC, Willette RE (eds.). *The Psychopharmacology of Hallucinogens* (<https://books.google.com/books?id=trkgBQAAQBAJ>) . Elsevier. pp. 74–83. doi:10.1016/b978-0-08-021938-7.50010-2 (<https://doi.org/10.1016%2Fb978-0-08-021938-7.50010-2>) . ISBN 978-0-08-021938-7.
270. Bennett D (30 January 2005). "Dr. Ecstasy" (<https://www.nytimes.com/2005/01/30/magazine/30ECSTASY.html>) . *The New York Times Magazine*. Archived (<https://web.archive.org/web/20111117063356/http://www.nytimes.com/2005/01/30/magazine/30ECSTASY.html>) from the original on 17 November 2011. Retrieved 10 February 2017.
271. Jennings P (1 April 2004). "Ecstasy Rising" (https://web.archive.org/web/20150527163756/http://www.thedocumentarygroup.com/PJP/Transcripts%20Files/Script_Ecstasy.doc) . *Primetime Thursday*. No. Special edition. ABC News. Archived from the original (http://www.thedocumentarygroup.com/PJP/Transcripts%20Files/Script_Ecstasy.doc) on 27 May 2015.
272. Shulgin A (2004). "Tribute to Jacob" (<https://web.archive.org/web/20180916024833/http://www.maps.org/images/pdf/books/scr/scr.pdf>) (PDF). In Doblin R (ed.). *The Secret Chief Revealed* (2nd ed.). Sarasota, FL: Multidisciplinary Association for Psychedelic Studies. pp. 17–18. ISBN 978-0-9660019-6-9. Archived from the original (<http://maps.org/images/pdf/books/scr/scr.pdf>) (PDF) on 16 September 2018. Retrieved 7 January 2015.
273. "Ecstasy on Prescription" (<https://www.bbc.co.uk/programmes/w3cswgvh>) . *BBC Business Daily*. 29 May 2018. Archived (<https://web.archive.org/web/20210710093100/https://www.bbc.co.uk/programmes/w3cswgvh>) from the original on 10 July 2021. Retrieved 10 July 2021.
274. Milroy CM (February 1999). "Ten years of 'ecstasy' " (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1297063>) . *Journal of the Royal Society of Medicine*. **92** (2): 68–72. doi:10.1177/014107689909200206 (<https://doi.org/10.1177%2F014107689909200206>) . PMC 1297063 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1297063>) . PMID 10450215 (<https://pubmed.ncbi.nlm.nih.gov/10450215>) .
275. Eisner B (1994). *Ecstasy: The MDMA Story* (<https://books.google.com/books?id=8aqUu5M6UpwC>) (Expanded 2nd ed.). Berkeley, CA: Ronin Publishing. ISBN 978-0-914171-68-3. Archived (<https://web.archive.org/web/20230113000538/https://books.google.com/books?id=8aqUu5M6UpwC>) from the original on 13 January 2023. Retrieved 1 February 2016.
276. Beck J, Rosenbaum M (1994). "The Distribution of Ecstasy" (<https://books.google.com/books?id=SwdedK36bVMC>) . *Pursuit of Ecstasy: The MDMA Experience* (<https://archive.org/details/pursuitofecstasy0000beck>) . Albany: State Univ. of New York Press. ISBN 978-0-7914-1817-8.

277. Doblin R, Rosenbaum M (1991). "Chapter 6: Why MDMA Should Not Have Been Made Illegal" (<http://www.drugtext.org/pdf/Dance/party-drugs-clubbing/why-mdma-should-not-have-been-made-illegal.pdf>) (PDF). In Inciardi JA (ed.). *The Drug Legalization Debate* (<https://archive.org/details/druglegalization00inci>) (2nd ed.). London: SAGE Publications, Inc. ISBN 978-0-8039-3678-2. Retrieved 10 August 2015.
278. Collin M, Godfrey J (2010). "The Technologies of Pleasure" (<https://books.google.com/books?id=fc8x9qeCekQC>) . *Altered State: The Story of Ecstasy Culture and Acid House* (Updated new ed.). London: Profile Books. ISBN 978-1-84765-641-4.
279. Savlov M (12 June 2000). "Countdown to Ecstasy: A New Drug for a New Millennium" (http://web.archive.org/web/20160121022839/http://weeklywire.com/ww/06-12-00/austin_music_feature.html) . *The Austin Chronicle*. Weekly Wire. Archived from the original (http://www.weeklywire.com/ww/06-12-00/austin_music_feature.html) on 21 January 2016. Retrieved 6 August 2015.
280. Owen F, Gavin L (20 October 2013). "Molly Isn't Who You Think She Is: A Deeper Look at MDMA" (<https://web.archive.org/web/20150727080003/http://www.playboy.com/articles/molly-party-drug-ecstasy>) . *Playboy*. Archived from the original (<http://www.playboy.com/articles/molly-party-drug-ecstasy>) on 27 July 2015. Retrieved 6 August 2015.
281. Sylvan R (2005). "A Brief History of the Rave Scene". *Trance Formation: The Spiritual and Religious Dimensions of Global Rave Culture*. New York, NY: Routledge. pp. 21–22. ISBN 978-0-415-97090-7.
282. Parrott AC (May 2004). "Is ecstasy MDMA? A review of the proportion of ecstasy tablets containing MDMA, their dosage levels, and the changing perceptions of purity" (<http://psy.swansea.ac.uk/staff/parrott/p-IsEcstasyMDMA-Psychopharm-2004.pdf>) (PDF). *Psychopharmacology*. **173** (3–4): 234–241. doi:10.1007/s00213-003-1712-7 (<https://doi.org/10.1007/s00213-003-1712-7>) . PMID 15007594 (<https://pubmed.ncbi.nlm.nih.gov/15007594>) . S2CID 3347303 (<https://api.semanticscholar.org/CorpusID:3347303>) . Archived (<https://web.archive.org/web/20150918181517/http://psy.swansea.ac.uk/staff/parrott/p-IsEcstasyMDMA-Psychopharm-2004.pdf>) (PDF) from the original on 18 September 2015. Retrieved 7 August 2015.
283. Renfro CL (October 1986). "MDMA on the street: Analysis Anonymous". *Journal of Psychoactive Drugs*. **18** (4): 363–369. doi:10.1080/02791072.1986.10472371 (<https://doi.org/10.1080/02791072.1986.10472371>) . PMID 2880953 (<https://pubmed.ncbi.nlm.nih.gov/2880953>) .

284. "Schedules of Controlled Substances Proposed Placement of 3,4-Methylenedioxymethamphetamine in Schedule I" (<https://maps.org/research-archive/dea-mdma/pdf/0194.PDF>) (PDF). *Federal Register*. **49** (146) 30210. 27 July 1984. Archived (<http://web.archive.org/web/20160304002700/http://www.maps.org/research-archive/dea-mdma/pdf/0194.PDF>) (PDF) from the original on 4 March 2016. Retrieved 15 January 2015.
285. Adler J, Abramson P, Katz S, Hager M (15 April 1985). "Getting High on 'Ecstasy' " (<http://www.maps.org/research-archive/hmma/Newsweek4.85.pdf>) (PDF). *Newsweek Magazine*. Life/Style. p. 96. Archived (<https://web.archive.org/web/20160420122637/http://www.maps.org/research-archive/hmma/Newsweek4.85.pdf>) (PDF) from the original on 20 April 2016. Retrieved 1 February 2015.
286. Holland J (2001). "The History of MDMA". In Holland J (ed.). *Ecstasy: the complete guide; a comprehensive look at the risks and benefits of MDMA*. Rochester, VT: Park Street Press. ISBN 978-0-89281-857-0.
287. "U.S. will ban 'ecstasy,' a hallucinogenic drug" (<https://www.nytimes.com/1985/06/01/us/us-will-ban-ecstasy-a-hallucinogenic-drug.html>) . *The New York Times*. The Associated Press. 1 June 1985. Archived (<https://web.archive.org/web/20150524161818/http://www.nytimes.com/1985/06/01/us/us-will-ban-ecstasy-a-hallucinogenic-drug.html>) from the original on 24 May 2015. Retrieved 29 April 2015.
288. "MDMA – FDA REPORT, 1985" (https://www.erowid.org/chemicals/mdma/mdma_law3.shtml) . *Erowid*. Food and Drug Administration. 1985. Archived (https://web.archive.org/web/20150811142554/https://www.erowid.org/chemicals/mdma/mdma_law3.shtml) from the original on 11 August 2015. Retrieved 11 August 2015.
289. Baker K (30 May 1985). "DEA To Ban "Ecstasy" – The Drug MDMA" (<https://apnews.com/26214cd32a8b4c05e361158df14715b0>) . The Associated Press. Archived (<https://web.archive.org/web/20180831174744/https://apnews.com/26214cd32a8b4c05e361158df14715b0>) from the original on 31 August 2018. Retrieved 7 August 2015.
290. Corwin M (31 May 1985). "U.S. to Ban Use of Drug MDMA: Street Abuse Cited; Used by Psychiatrists" (<https://www.latimes.com/archives/la-xpm-1985-05-31-mn-14566-story.html>) . *Los Angeles Times*. Archived (https://web.archive.org/web/20151117032749/http://articles.latimes.com/1985-05-31/news/mn-14566_1_street-drugs) from the original on 17 November 2015. Retrieved 11 August 2015.
291. Weber B (7 June 2014). "Alexander Shulgin, Psychedelia Researcher, Dies at 88" (https://www.nytimes.com/2014/06/08/us/alexander-shulgin-psychedelia-researcher-dies-at-88.html?_r=0) . *The New York Times*. Archived (https://web.archive.org/web/20150905200423/http://www.nytimes.com/2014/06/08/us/alexander-shulgin-psychedelia-researcher-dies-at-88.html?_r=0) from the original on 5 September 2015. Retrieved 28 August 2015.

292. Vastag B (3 June 2014). "Chemist Alexander Shulgin, popularizer of the drug Ecstasy, dies at 88" (https://www.washingtonpost.com/national/health-science/chemist-alexander-shulgin-popularizer-of-the-drug-ecstasy-dies-at-88/2014/06/03/19fd9580-eb34-11e3-b98c-72cef4a00499_story.html) . *The Washington Post*. WP Company LLC. Archived (https://web.archive.org/web/20150805104454/http://www.washingtonpost.com/national/health-science/chemist-alexander-shulgin-popularizer-of-the-drug-ecstasy-dies-at-88/2014/06/03/19fd9580-eb34-11e3-b98c-72cef4a00499_story.html) from the original on 5 August 2015. Retrieved 28 August 2015.
293. "Ecstasy has its pros and cons" (https://newspaperarchive.com/profile/Robert_Thomas/clip-number/34618/) . *Kokomo Tribune*. Kokomo, Indiana. Harper's Bazaar. 23 November 1985. p. 6. Archived (<https://web.archive.org/web/20210828153542/https://newspaperarchive.com/profile/robert-thomas-512557/>) from the original on 28 August 2021. Retrieved 9 September 2017 – via newspaperarchive.com. 
294. "Lester Grinspoon, M.d., Petitioner, v. Drug Enforcement Administration, Respondent, 828 F.2d 881 (1st Cir. 1987)" (<https://law.justia.com/cases/federal/appellate-courts/F2/828/881/368975/>) . *Justia Law*. US Court of Appeals for the First Circuit. Archived (<https://web.archive.org/web/20211009212855/https://law.justia.com/cases/federal/appellate-courts/F2/828/881/368975/>) from the original on 9 October 2021. Retrieved 9 October 2021.
295. Halvorsen JØ, Naudet F, Cristea IA (October 2021). "Challenges with benchmarking of MDMA-assisted psychotherapy" (<https://hal.archives-ouvertes.fr/hal-03414583/file/Halvorsen%20et%20al%20-%202021%20-%20Challenges%20with%20benchmarking%20of%20MDMA-assisted%20psychotherapy.pdf>) (PDF). *Nature Medicine*. **27** (10): 1689–1690. doi:10.1038/s41591-021-01525-0 (<https://doi.org/10.1038/s41591-021-01525-0>) . PMID 34635857 (<https://pubmed.ncbi.nlm.nih.gov/34635857/>) . S2CID 238636360 (<https://api.semanticscholar.org/CorpusID:238636360>) . Archived (<https://web.archive.org/web/20220522022351/https://hal.archives-ouvertes.fr/hal-03414583/file/Halvorsen%20et%20al%20-%202021%20-%20Challenges%20with%20benchmarking%20of%20MDMA-assisted%20psychotherapy.pdf>) (PDF) from the original on 22 May 2022. Retrieved 9 May 2022.
296. *WHO Expert Committee on Drug Dependence: Twenty-second Report* (https://web.archive.org/web/20141019192225/http://whqlibdoc.who.int/trs/WHO_TRS_729.pdf) (PDF). Geneva: World Health Organization. 1985. pp. 24–25. Archived from the original (http://whqlibdoc.who.int/trs/WHO_TRS_729.pdf) (PDF) on 19 October 2014. Retrieved 29 August 2012.

297. "Decision to place MDMA into Schedule I" (http://www.unodc.org/documents/commission/s/CND/Drug_Resolutions/1980-1989/1986/CND_Decision-1986-07_S-IX.pdf) (PDF). UNODC. Commission on Narcotic Drugs. 11 February 1986. Archived (https://web.archive.org/web/20150922000603/http://www.unodc.org/documents/commissions/CND/Drug_Resolutions/1980-1989/1986/CND_Decision-1986-07_S-IX.pdf) (PDF) from the original on 22 September 2015. Retrieved 9 May 2015.
298. McKinley JC (12 September 2013). "Overdoses of 'Molly' Led to Electric Zoo Deaths" (http://artsbeat.blogs.nytimes.com/2013/09/12/overdoses-of-molly-led-to-electric-zoo-deaths/?_r=0) . *The New York Times*. Archived (https://web.archive.org/web/20131204085710/http://artsbeat.blogs.nytimes.com/2013/09/12/overdoses-of-molly-led-to-electric-zoo-deaths/?_r=0) from the original on 4 December 2013. Retrieved 9 December 2013.
299. Nelson LS, Lewin NA, Howland MA, Hoffman RS, Goldfrank LR, Flomenbaum NE (2011). *Goldfrank's toxicologic emergencies* (9th ed.). New York: McGraw-Hill Medical. ISBN 978-0-07-160593-9.
300. "Bibliography of Psychedelic Research Studies" (<https://web.archive.org/web/20131203011733/http://www.maps.org/research/>) . *Multidisciplinary Association for Psychedelic Studies* (MAPS). Santa Cruz, CA. Archived from the original (<http://www.maps.org/research/>) on 3 December 2013.
301. James SD (23 February 2015). "What Is Molly and Why Is It Dangerous?" (<https://www.nbcnews.com/health/health-news/what-molly-why-it-dangerous-n311291>) . NBCNews.com. Archived (<https://web.archive.org/web/20150224063219/http://www.nbcnews.com/health/health-news/what-molly-why-it-dangerous-n311291>) from the original on 24 February 2015. Retrieved 23 February 2015. "Why is it called Molly? That's short for "molecule." "You can put a ribbon and bow on it and call it a cute name like 'Molly' and people are all in," said Paul Doering, professor emeritus of pharmacology at the University of Florida."
302. Aleksander I (21 June 2013). "Molly: Pure, but Not So Simple" (<https://ghostarchive.org/archive/20220101/https://www.nytimes.com/2013/06/23/fashion/molly-pure-but-not-so-simple.html>) . *The New York Times*. Archived from the original (<https://www.nytimes.com/2013/06/23/fashion/molly-pure-but-not-so-simple.html>) on 1 January 2022. Retrieved 24 February 2015.
303. "Mephedrone (4-Methylmethcathinone) appearing in "Ecstasy" in the Netherlands" (<https://web.archive.org/web/20121105125943/http://scientopia.org/blogs/drugmonkey/2010/09/19/mephedrone-4-methylmethcathinone-appearing-in-ecstasy-in-the-netherlands/>) . 19 September 2010. Archived from the original (<http://scientopia.org/blogs/drugmonkey/2010/09/19/mephedrone-4-methylmethcathinone-appearing-in-ecstasy-in-the-netherlands/>) on 5 November 2012. Retrieved 31 December 2012.

304. "Why ecstasy is 'vanishing' from UK nightclubs" (https://news.bbc.co.uk/2/hi/uk_news/england/london/8468372.stm) . *BBC News*. 19 January 2010. Archived (<https://web.archive.org/web/20170907220833/http://news.bbc.co.uk/1/hi/england/london/8468372.stm>) from the original on 7 September 2017. Retrieved 14 February 2010.
305. Bish J (4 August 2017). "Watch Out for Pentylone, the Horrible New MDMA Additive" (<https://www.vice.com/en/article/watch-out-for-pentylone-the-horrible-new-mdma-additive/>) . *Vice*. Archived (https://web.archive.org/web/20200901064627/https://www.vice.com/en_uk/article/pagjxg/watch-out-for-pentylone-the-horrible-new-mdma-additive) from the original on 1 September 2020. Retrieved 31 May 2018.
306. Williamson L (7 March 2025). "MDMA may have protected Nova attack survivors from trauma, study suggests" (<https://www.bbc.com/news/articles/c9wpy14wyd0o>) . *BBC Home*. Retrieved 16 March 2025.
307. Burns B (8 March 2025). "MDMA and LSD may have helped October 7 survivors" (<https://www.news.com.au/world/middle-east/lsd-mdma-party-drugs-may-have-helped-survivors-of-october-7-attack-study-claims/news-story/361e7726844087d6d023f010bccf1c48>) . *news.com.au*. Retrieved 16 March 2025.
308. "Annual prevalence of use of drugs, by region and globally, 2016" (https://dataunodc.un.org/drugs/prevalence_regional) . *World Drug Report 2018* (<https://www.unodc.org/wdr2018/>) . United Nations Office on Drugs and Crime. 2018. Retrieved 7 July 2018.
309. "MDMA and psilocybin: What GPs need to know" (<https://www1.racgp.org.au/newsgp/gp-opinion/mdma-and-psilocybin-what-gps-need-to-know#:~:text=It%20means%20psilocybin%20and%20MDMA,restricts%20supply%20to%20clinical%20trials.>) . *Newsgp*. Archived (<https://web.archive.org/web/20230613073443/https://www1.racgp.org.au/newsgp/gp-opinion/mdma-and-psilocybin-what-gps-need-to-know#:~:text=It%20means%20psilocybin%20and%20MDMA,restricts%20supply%20to%20clinical%20trials.>) from the original on 13 June 2023. Retrieved 13 June 2023.
310. Bedi G (29 March 2018). "Is psychiatry ready for medical MDMA?" (<https://theconversation.com/is-psychiatry-ready-for-medical-mdma-94105>) . *The Conversation*. Archived (<https://web.archive.org/web/20230104131905/https://theconversation.com/is-psychiatry-ready-for-medical-mdma-94105>) from the original on 4 January 2023. Retrieved 12 April 2024.
311. "Misuse of Drugs Act 1981" (https://www.slp.wa.gov.au/legislation/statutes.nsf/main_mrtitle_4607_homepage.html) . The Government of Western Australia. Department of the Premier and Cabinet. 23 October 1981. Archived (https://web.archive.org/web/20160818132249/https://www.slp.wa.gov.au/legislation/statutes.nsf/main_mrtitle_4607_homepage.html) from the original on 18 August 2016. Retrieved 22 July 2016.

312. "ACT government decriminalises small amounts of illicit drugs including speed, heroin and cocaine" (<https://www.abc.net.au/news/2022-10-20/act-decriminalises-small-amounts-of-illicit-drugs-heroin-cocaine/101552008>) . *ABC News*. Australian Broadcasting Corporation. 20 October 2022. Archived (<https://web.archive.org/web/20230613074529/https://www.abc.net.au/news/2022-10-20/act-decriminalises-small-amounts-of-illicit-drugs-heroin-cocaine/101552008>) from the original on 13 June 2023. Retrieved 13 June 2023.
313. Roy T (27 October 2023). "The ACT has today decriminalised small amounts of some illicit drugs. But what does that mean?" (<https://www.abc.net.au/news/2023-10-28/canberra-drug-decriminalisation-laws-begin-today/103032128>) . *ABC News*. Retrieved 7 June 2024.
314. "Schedule I" (<https://web.archive.org/web/20131110200556/http://isomerdesign.com/Cdsa/schedule.php?schedule=1§ion=18.5&structure=C>) . *Controlled Drugs and Substances Act*. Isomer Design. Archived from the original (<http://isomerdesign.com/Cdsa/schedule.php?schedule=1§ion=18.5&structure=C>) on 10 November 2013. Retrieved 9 December 2013.
315. "Definitions and interpretations" (<https://web.archive.org/web/20131110213450/http://isomerdesign.com/Cdsa/definitions.php?structure=C>) . *Controlled Drugs and Substances Act*. Isomer Design. Archived from the original (<http://isomerdesign.com/Cdsa/definitions.php?structure=C>) on 10 November 2013. Retrieved 9 December 2013.
316. "Decriminalizing people who use drugs in B.C." (<https://www2.gov.bc.ca/gov/content/overdose/decriminalization>) *Government of BC*. Government Communications and Public Engagement. Archived (<https://web.archive.org/web/20230309091348/https://www2.gov.bc.ca/gov/content/overdose/decriminalization>) from the original on 9 March 2023. Retrieved 8 March 2023.
317. "B.C. recorded 211 toxic drug deaths — almost 7 a day — in January, coroner reports" (<https://www.cbc.ca/news/canada/british-columbia/bc-toxic-drugs-deaths-january-2023-1.6770643>) . *CBC.ca*. 7 March 2023. Archived (<https://web.archive.org/web/20230308231034/https://www.cbc.ca/news/canada/british-columbia/bc-toxic-drugs-deaths-january-2023-1.6770643>) from the original on 8 March 2023. Retrieved 8 March 2023.
318. "Valtioneuvoston asetus huumausaineina pidettävistä aineista, valmisteista ja kasveista | 543/2008 | Lainsäädäntö | Finlex" (<https://www.finlex.fi/fi/lainsaadanto/2008/543>) .
319. "KKO:2005:56 | 11.5.2005 | Ennakkopäätökset | Korkein oikeus | Finlex" (<https://finlex.fi/fi/oi/keuskaytanto/korkein-oikeus/ennakkopaatokset/2005/56>) .
320. Zaken MV (31 May 2024). "MDMA /// Beyond Ecstasy - Report - Government.nl" (<https://www.government.nl/documents/reports/2024/05/31/mdma-beyond-ecstasy>) . *www.government.nl*. Retrieved 6 April 2025.

321. Hennig AC (2014). "An Examination of Federal Sentencing Guidelines' Treatment of MDMA ('Ecstasy')". *Belmont Law Review*. **1**: 267. SSRN 2481227 (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2481227) .
322. "Rapport Drugs in Lijsten" (<https://web.archive.org/web/20120306112130/http://www.rijksoverheid.nl/documenten-en-publicaties/rapporten/2011/06/27/rapport-drugs-in-lijsten.html>) . *Rijksoverheid.nl*. 27 June 2011. Archived from the original (<http://www.rijksoverheid.nl/documenten-en-publicaties/rapporten/2011/06/27/rapport-drugs-in-lijsten.html>) on 6 March 2012. Retrieved 29 August 2012.
323. "Committee: the current system of the Opium Act does not have to be changed" (<https://web.archive.org/web/20120429071421/http://www.government.nl/documents-and-publications/press-releases/2011/06/24/committee-the-current-system-of-the-opium-act-does-not-have-to-be-changed.html>) . *government.nl*. 24 June 2011. Archived from the original (<http://www.government.nl/documents-and-publications/press-releases/2011/06/24/committee-the-current-system-of-the-opium-act-does-not-have-to-be-changed.html>) on 29 April 2012. Retrieved 29 August 2012.
324. Power M (2013). *Drugs 2.0: the web revolution that's changing how the world gets high* (<http://books.google.com/books?id=gj6zMQEACAAJ&q=Drugs+2.0>) (epub file). London: Portobello. ISBN 978-1-84627-459-6.
325. "Misuse of Drugs Act 1971" (<http://www.statutelaw.gov.uk/content.aspx?LegType=All+Primary&PageNumber=57&NavFrom=2&parentActiveTextDocId=1367412&ActiveTextDocId=1367472&filesize=411>) . *Statutelaw.gov.uk*. 5 January 1998. Archived (<https://web.archive.org/web/20210828153546/https://www.legislation.gov.uk/ukpga/1971/38/schedule/2/part/I/2003-04-01?timeline=true>) from the original on 28 August 2021. Retrieved 11 June 2011.
326. Hope C (7 February 2009). "Ecstasy 'no more dangerous than horse riding' " (<https://www.telegraph.co.uk/news/uknews/law-and-order/4537874/Ecstasy-no-more-dangerous-than-horse-riding.html>) . *Telegraph.co.uk*. Archived (<https://web.archive.org/web/20151210201615/http://www.telegraph.co.uk/news/uknews/law-and-order/4537874/Ecstasy-no-more-dangerous-than-horse-riding.html>) from the original on 10 December 2015. Retrieved 4 December 2015.
327. Nutt DJ (January 2009). "Equasy-- an overlooked addiction with implications for the current debate on drug harms". *Journal of Psychopharmacology*. **23** (1). Oxford, England: 3–5. doi:10.1177/0269881108099672 (<https://doi.org/10.1177/0269881108099672>) . PMID 19158127 (<https://pubmed.ncbi.nlm.nih.gov/19158127>) . S2CID 32034780 (<https://api.semanticscholar.org/CorpusID:32034780>) .

328. Johnson A (2 November 2009). "Why Professor David Nutt was shown the door" (<https://www.theguardian.com/politics/2009/nov/02/drug-policy-alan-johnson-nutt>) . *The Guardian*. London. Archived (<https://web.archive.org/web/20140118223546/http://www.theguardian.com/politics/2009/nov/02/drug-policy-alan-johnson-nutt>) from the original on 18 January 2014. Retrieved 3 November 2009.
329. *Schedules of Controlled Substances; Scheduling of 3,4-Methylenedioxymethamphetamine (MDMA) Into Schedule I of the Controlled Substances Act; Remand*, 53 Fed. Reg. 5,156 (DEA 22 February 1988).
330. "Court Rejects Harsh Federal Drug Sentencing Guideline as Scientifically Unjustified" (<https://www.aclu.org/criminal-law-reform/court-rejects-harsh-federal-drug-sentencing-guideline-scientifically-unjustified>) . *American Civil Liberties Union*. 15 July 2011. Archived (<https://web.archive.org/web/20140314210920/https://www.aclu.org/criminal-law-reform/court-rejects-harsh-federal-drug-sentencing-guideline-scientifically-unjustified>) from the original on 14 March 2014. Retrieved 29 August 2012.
331. "Statistical Bulletin 2018 – prevalence of drug use" (http://www.emcdda.europa.eu/data/stats2018/gps_en) . European Monitoring Centre for Drugs and Drug Addiction (EMCDDA). Archived (https://web.archive.org/web/20181231102708/http://www.emcdda.europa.eu/data/stats2018/gps_en) from the original on 31 December 2018. Retrieved 5 February 2019.
332. Wu P, Liu X, Pham TH, Jin J, Fan B, Jin Z (November 2010). "Ecstasy use among US adolescents from 1999 to 2008" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2967577>) . *Drug and Alcohol Dependence*. **112** (1–2): 33–38. doi:10.1016/j.drugalcdep.2010.05.006 (<https://doi.org/10.1016%2Fj.drugalcdep.2010.05.006>) . PMC 2967577 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2967577>) . PMID 20570447 (<https://pubmed.ncbi.nlm.nih.gov/20570447>) .
333. European Monitoring Centre for Drugs and Drug Addiction (2008). *Annual report: the state of the drugs problem in Europe* (https://web.archive.org/web/20130425191815/http://www.emcdda.europa.eu/attachements.cfm/att_64227_EN_EMCDDA_AR08_en.pdf) (PDF). Luxembourg: Office for Official Publications of the European Communities. p. 49. ISBN 978-92-9168-324-6. Archived from the original (http://www.emcdda.europa.eu/attachements.cfm/att_64227_EN_EMCDDA_AR08_en.pdf) (PDF) on 25 April 2013. Retrieved 1 December 2008.

334. European Monitoring Centre for Drugs Drug Addiction (2014). "Ecstasy: high purity powder available". *European Drug Report* (https://web.archive.org/web/20140907142119/http://www.emcdda.europa.eu/attachements.cfm/att_228272_EN_TDAT14001ENN.pdf) (PDF). European Monitoring Centre for Drugs and Drug Addiction (EMCDDA). p. 26. doi:10.2810/32306 (<https://doi.org/10.2810%2F32306>) . ISBN 978-92-9168-694-0. Archived from the original (http://www.emcdda.europa.eu/attachements.cfm/att_228272_EN_TDAT14001ENN.pdf) (PDF) on 7 September 2014. Retrieved 4 June 2014.
335. "Ecstasy-type substances Retail and wholesale prices* and purity levels, by drug, region and country or territory" (https://web.archive.org/web/20151208142205/http://www.unodc.org/documents/wdr2014/Statistics/Prices_ATS.xls) . United Nations Office on Drugs and Crime. Archived from the original (http://www.unodc.org/documents/wdr2014/Statistics/Prices_ATS.xls) on 8 December 2015. Retrieved 2 January 2015.
336. Camargo J, Esseiva P, González F, Wist J, Patiny L (November 2012). "Monitoring of illicit pill distribution networks using an image collection exploration framework" ([http://www.fsijournal.org/article/S0379-0738\(12\)00458-6/abstract](http://www.fsijournal.org/article/S0379-0738(12)00458-6/abstract)) . *Forensic Science International*. **223** (1–3): 298–305. doi:10.1016/j.forsciint.2012.10.004 (<https://doi.org/10.1016%2Fj.forsciint.2012.10.004>) . PMID 23107059 (<https://pubmed.ncbi.nlm.nih.gov/23107059>) . Archived (<http://web.archive.org/web/20210828153543/http://www.allmallorcahotels.com/fsijournalorg/>) from the original on 28 August 2021. Retrieved 9 December 2013.
337. Dillon P. "10 years of ecstasy and other party drug use in Australia: What have we done and what is there left to do?" (<https://web.archive.org/web/20120209142314/http://www.drugtext.org/library/articles/dillon.htm>) . *Drugtext.org*. Archived from the original (<http://www.drugtext.org/library/articles/dillon.htm>) on 9 February 2012.
338. "Erowid MDMA Vault: Images" (https://www.erowid.org/chemicals/mdma/mdma_images.shtml) . Archived (https://web.archive.org/web/20160306030258/https://www.erowid.org/chemicals/mdma/mdma_images.shtml) from the original on 6 March 2016. Retrieved 3 March 2016.
339. Cork T (31 July 2015). "Now sick dealers peddle Shaun the Sheep Ecstasy tablets" (<https://web.archive.org/web/20150812022920/http://www.westerndailynews.co.uk/sick-dealers-peddle-Shaun-Sheep-Ecstasy-tablets/story-27521142-detail/story.html>) . *Western Daily News*. Archived from the original (<http://www.westerndailynews.co.uk/sick-dealers-peddle-Shaun-Sheep-Ecstasy-tablets/story-27521142-detail/story.html>) on 12 August 2015. Retrieved 3 March 2016.

340. Chaliha D, Mamo JC, Albrecht M, Lam V, Takechi R, Vaccarezza M (1 July 2021). "A Systematic Review of the MDMA Model to Address Social Impairment in Autism" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8686313>) . *Current Neuropharmacology*. **19** (7): 1101–1154. doi:10.2174/1570159X19666210101130258 (<https://doi.org/10.2174%2F1570159X19666210101130258>) . PMC 8686313 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8686313>) . PMID 33388021 (<https://pubmed.ncbi.nlm.nih.gov/33388021>) .
341. Moncrieff J (16 January 2025). "Alternative Approaches: The Good, the Bad and the Worrying: Psychedelics for Depression". *Chemically Imbalanced: The Making and Unmaking of the Serotonin Myth* (<https://books.google.com/books?id=e0MkEQAAQBAJ>) . Flint. ISBN 978-1-80399-680-6. Retrieved 16 October 2025.

External links

-
- MDMA - Isomer Design (<https://isomerdesign.com/pihkal/explore/109>)
 - MDMA - PsychonautWiki (<https://psychonautwiki.org/wiki/MDMA>)
 - MDMA - Erowid (<https://www.erowid.org/chemicals/mdma/mdma.shtml>)
 - MDMA - PiHKAL - Erowid (https://www.erowid.org/library/books_online/pihkal/pihkal109.shtml)
 - MDMA - PiHKAL - Isomer Design (<https://isomerdesign.com/pihkal/read/pk/109>)
 - A Multi-Site Phase 3 Study of MDMA-Assisted Therapy for PTSD (MAPP2) (<https://maps.org/mdma/ptsd/mapp2/>)
 - "MDMA Facts and Statistics" (<https://www.drugabuse.gov/publications/drugfacts/mdma-ecstasy>) . *National Institute on Drug Abuse*. 15 June 2020.